Crackz Documentation

AI Image Analysis

Version: 0.90.0

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Crackz

The Crackz desktop app — project dashboard with model configuration and detection metrics.

	`crackz-gui` `crackz`

1. `crackz-gui`
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3.
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5.

```
crackz init --name demo
crackz import-images --project demo --src ./photos --type raw --size 512 512
crackz detect run --project demo
# Results written to demo/artifacts/detections/
```

Installation

Installation

`uv sync`

`LICENSE`

`crackz-gui`

	`crackz-gui.exe`
	`crackz-gui`
	`crackz-gui`
	`crackz-gui-cu128`
	`crackz-gui-cu118`

`nvidia-smi` `CUDA Version`

```
asset=crackz-gui            # or crackz-gui-cu128 / crackz-gui-cu118
chmod +x "$asset"
sudo install -m 755 "$asset" /usr/local/bin/crackz-gui
crackz-gui
```

`crackz-gui.exe` `PATH`

`gh auth login` `uv`

`gh auth login` `uv`

```
gh auth login
gh release download --repo Anaxiatech-AB/crackz \
  --pattern 'crackz-*.whl' --pattern 'crackz_gui-*.whl'
uv pip install crackz-*.whl crackz_gui-*.whl
crackz-gui
```

```
# Using pip with GitHub authentication
pip install git+https://github.com/OWNER/crackz.git@vVERSION

# Or download the wheel from the private repository's Releases page
# (Requires GitHub authentication - use personal access token)
pip install https://USERNAME:TOKEN@github.com/OWNER/crackz/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl
```

```
pip install git+https://github.com/OWNER/crackz.git
```

```
# Login with GitHub CLI
gh auth login

# Download release asset
gh release download vVERSION --repo OWNER/crackz --pattern '*.whl'

# Install the downloaded wheel
pip install crackz-VERSION-py3-none-any.whl
```

```
# Create a personal access token with 'repo' scope at:
# https://github.com/settings/tokens

# Set your token (keep it secure!)
export GITHUB_TOKEN="ghp_xxxxxxxxxxxx" # gitleaks:allow

# Install using token authentication
pip install git+https://${GITHUB_TOKEN}@github.com/OWNER/crackz.git@vVERSION
```

`https://github.com/OWNER/crackz`
`crackz-VERSION-py3-none-any.whl`

```
pip install crackz-VERSION-py3-none-any.whl
```

```
# Download wheel using GitHub token
curl -L \
  -H "Authorization: token ${GITHUB_TOKEN}" \
  -o crackz.whl \
  https://github.com/OWNER/crackz/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl

# Install the downloaded wheel
pip install crackz.whl
```

```
# If your organization has configured a private index
pip install --index-url https://pypi.company.com/simple/ crackz
```

```
# Using the provided index URL and credentials
pip install crackz --index-url https://INDEX_URL/simple/

# Example with authentication (if required):
pip install crackz --index-url https://USERNAME:TOKEN@pypi.company.com/simple/
```

`~/.config/pip/pip.conf` `%APPDATA%\pip\pip.ini`

```
[global]
index-url = https://USERNAME:TOKEN@pypi.company.com/simple/
```

```
pip install crackz
pip install --upgrade crackz  # For updates
```

```
# Set credentials via environment variables
export PIP_INDEX_URL="https://pypi.company.com/simple/"
export PIP_EXTRA_INDEX_URL="https://pypi.org/simple"  # Fallback to PyPI for dependencies

pip install crackz
```

`https://dl.cloudsmith.io/TOKEN/org/repo/python/simple/`
`https://artifactory.company.com/artifactory/api/pypi/pypi-local/simple/`
`https://pypi.fury.io/TOKEN/USERNAME/`
`https://aws:TOKEN@domain.d.codeartifact.region.amazonaws.com/pypi/repo/simple/`
`https://pkgs.dev.azure.com/org/_packaging/feed/pypi/simple/` `https://devpi.company.com/username/index/+simple/`

```
chmod 600 ~/.config/pip/pip.conf  # Linux/macOS
```

```
pip install --trusted-host pypi.company.com crackz
# Or add CA cert:
pip install --cert /path/to/ca-bundle.crt crackz
```

```
pip install crackz --extra-index-url https://pypi.org/simple/
```

`crackz.exe`
`crackz-gui.exe` `crackz` `crackz-gui` `crackz` `crackz-gui`

```
REM Download the executables from GitHub Releases
REM Move them to a permanent location
mkdir C:\Program Files\Crackz
move crackz.exe "C:\Program Files\Crackz\"
move crackz-gui.exe "C:\Program Files\Crackz\"

REM Add to PATH (optional, for CLI access from anywhere)
setx PATH "%PATH%;C:\Program Files\Crackz"
```

```
# Download the executables from GitHub Releases
chmod +x crackz crackz-gui

# Move to a system directory (optional)
sudo mv crackz /usr/local/bin/
sudo mv crackz-gui /usr/local/bin/

# Or keep in a local directory and add to PATH
mkdir -p ~/Applications/crackz
mv crackz crackz-gui ~/Applications/crackz/
echo 'export PATH="$HOME/Applications/crackz:$PATH"' >> ~/.bashrc
source ~/.bashrc
```

```
# Run CLI directly (no Python needed)
crackz --help
crackz init --name myproject

# Run GUI by double-clicking crackz-gui.exe (Windows)
# Or run from command line:
crackz-gui
```

`crackz-cuda.exe` `crackz-gui-cuda.exe`

```
crackz --help
python -c "import crackz; import importlib.metadata as m; print(m.version('crackz'))"
```

- `crackz-VERSION-py3-none-any.whl`
- `crackz_cu118-VERSION-py3-none-any.whl`
- `crackz_cu128-VERSION-py3-none-any.whl`

```
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl
```

```
# Step 1: Install PyTorch with CUDA 11.8
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu118

# Step 2: Install crackz CUDA variant
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz_cu118-VERSION-py3-none-any.whl
```

```
# Step 1: Install PyTorch with CUDA 12.8
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu128

# Step 2: Install crackz CUDA variant
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz_cu128-VERSION-py3-none-any.whl
```

```
nvidia-smi
```

```
nvcc --version
# Or check driver
nvidia-smi | grep "CUDA Version"
```

`crackz_cu118` `crackz_cu128` `crackz`

```
# Login first
gh auth login

# Download and install CPU version (one step)
gh release download vVERSION --repo OWNER/crackz --pattern 'crackz-*.whl'
pip install crackz-VERSION-py3-none-any.whl
```

```
# Login first
gh auth login

# Step 1: Install PyTorch with CUDA 12.8
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu128

# Step 2: Download and install crackz CUDA variant
gh release download vVERSION --repo OWNER/crackz --pattern 'crackz_cu128-*.whl'
pip install crackz_cu128-VERSION-py3-none-any.whl
```

```
# Check CUDA availability
python -c "import torch; print(f'CUDA available: {torch.cuda.is_available()}')"

# Check PyTorch version (should include +cu118 or +cu128 suffix)
python -c "import torch; print(f'PyTorch version: {torch.__version__}')"

# Check GPU device name
python -c "import torch; print(f'Device: {torch.cuda.get_device_name(0) if torch.cuda.is_available() else \"CPU\"}')"
```

```
CUDA available: True
PyTorch version: 2.5.0+cu128
Device: NVIDIA GeForce RTX 4090
```

```
# Step 1: Install specific PyTorch version
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu121

# Step 2: Install crackz CUDA variant (cu118 or cu128, either works)
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz_cu118-VERSION-py3-none-any.whl
```

`cpu` `cu128`

```
# This installs CPU PyTorch even with the cu128 extra - BROKEN!
pip install "crackz-VERSION-py3-none-any.whl[cu128]"
```

```
# Step 1: Install PyTorch with CUDA
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu128

# Step 2: Install crackz
pip install crackz_cu128-VERSION-py3-none-any.whl
```

`keyring`

```
# With uv
uv add keyring

# With pip
pip install keyring
```

```
from crackz_gui.state.secure_storage import (
    store_api_key,
    retrieve_api_key,
    is_secure_storage_available,
)

# Check availability
if is_secure_storage_available():
    store_api_key("your-api-key")
    api_key = retrieve_api_key()
```

1.

```
mkdir crackz_wheels
cd crackz_wheels

# Download Crackz wheel
wget https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl

# Download all dependencies
pip download -r <(pip show -f crackz | grep Requires) -d .
# Or download from the wheel itself:
pip download crackz-VERSION-py3-none-any.whl -d .
```

2.
3.

```
pip install --no-index --find-links crackz_wheels crackz-VERSION-py3-none-any.whl
```

```
pip install .
```

`pyproject.toml`

```
# Download the new version from GitHub Releases
pip install --upgrade https://github.com/Anaxiatech-AB/crackz/releases/download/vNEW_VERSION/crackz-NEW_VERSION-py3-none-any.whl

# Or if using a private index:
pip install -U --index-url https://pypi.company.com/simple/ crackz
```

```
pip uninstall crackz
```

```
crackz --version
crackz init --name sanity-test
crackz detect run --project sanity-test  # (Will use default config; may download model weights)
```

`torch.__version__`

`No module named crackz`

`CUDA not available` `torch.__version__`

`ModuleNotFoundError: No module named 'torch'`

`CRACKZ_LOG_DIR`

`pip install --force-reinstall crackz-VERSION.whl`

`pyproject.toml`

```
# Check if CUDA is available
python -c "import torch; print(f'CUDA available: {torch.cuda.is_available()}')"
python -c "import torch; print(f'CUDA device: {torch.cuda.get_device_name(0)}')"
```

```
CUDA available: True
CUDA device: NVIDIA GeForce RTX 3080
```

```
# These commands automatically use GPU if available:
crackz detect train --project my_project
crackz detect run --project my_project
crackz-gui  # GUI will use GPU automatically
```

`cuda` `mps` `cpu`

```
# Force GPU usage (will error if CUDA not available)
crackz detect train --project my_project --device cuda

# Force CPU usage (useful for debugging)
crackz detect train --project my_project --device cpu

# Let Crackz auto-detect (default)
crackz detect train --project my_project
```

`<project>/<project>_project.yaml`

```
ai:
  model:
    device: cuda  # or cpu, or mps
```

```
crackz detect train --project my_project
```

```
GPU available: True, used: True
TPU available: False, using: 0 TPU cores
IPU available: False, using: 0 IPUs
HPU available: False, using: 0 HPUs
```

```
# NVIDIA GPUs
watch -n 1 nvidia-smi

# Should show Python process using GPU memory and compute
```

`crackz_cu118-VERSION-py3-none-any.whl`

`crackz_cu128-VERSION-py3-none-any.whl`

`crackz_cu118-VERSION-py3-none-any.whl`

```
nvcc --version  # If CUDA toolkit installed
nvidia-smi  # Shows driver version (different from CUDA toolkit)
```

1.

   ```
   python -c "import torch; print(torch.__version__)"
   # Should show: 2.5.0+cu118 or 2.5.0+cu128
   # If it shows just 2.5.0, you have CPU-only version!
   ```

2.

   ```
   pip uninstall torch torchvision
   pip install torch torchvision --index-url https://download.pytorch.org/whl/cu118
   ```

3.

   ```
   python -c "import torch; print(f'CUDA available: {torch.cuda.is_available()}')"
   ```

```
# GUI: Project → Settings → Training → Batch Size
# CLI:
crackz detect train --project my_project --batch-size 2
```

- `nvidia-smi`
-
-

```
CRACKZ_LOG_DIR=my_logs crackz detect run --project myproj
```

`$Env:CRACKZ_LOG_DIR='my_logs'`

- `pip install --require-hashes -r requirements.txt`
-

```
crackz migrate --project ./my_project         # applies changes + backup
crackz migrate --project ./my_project --dry-run  # preview only
```

`CRACKZ_AUTO_MIGRATE=1`

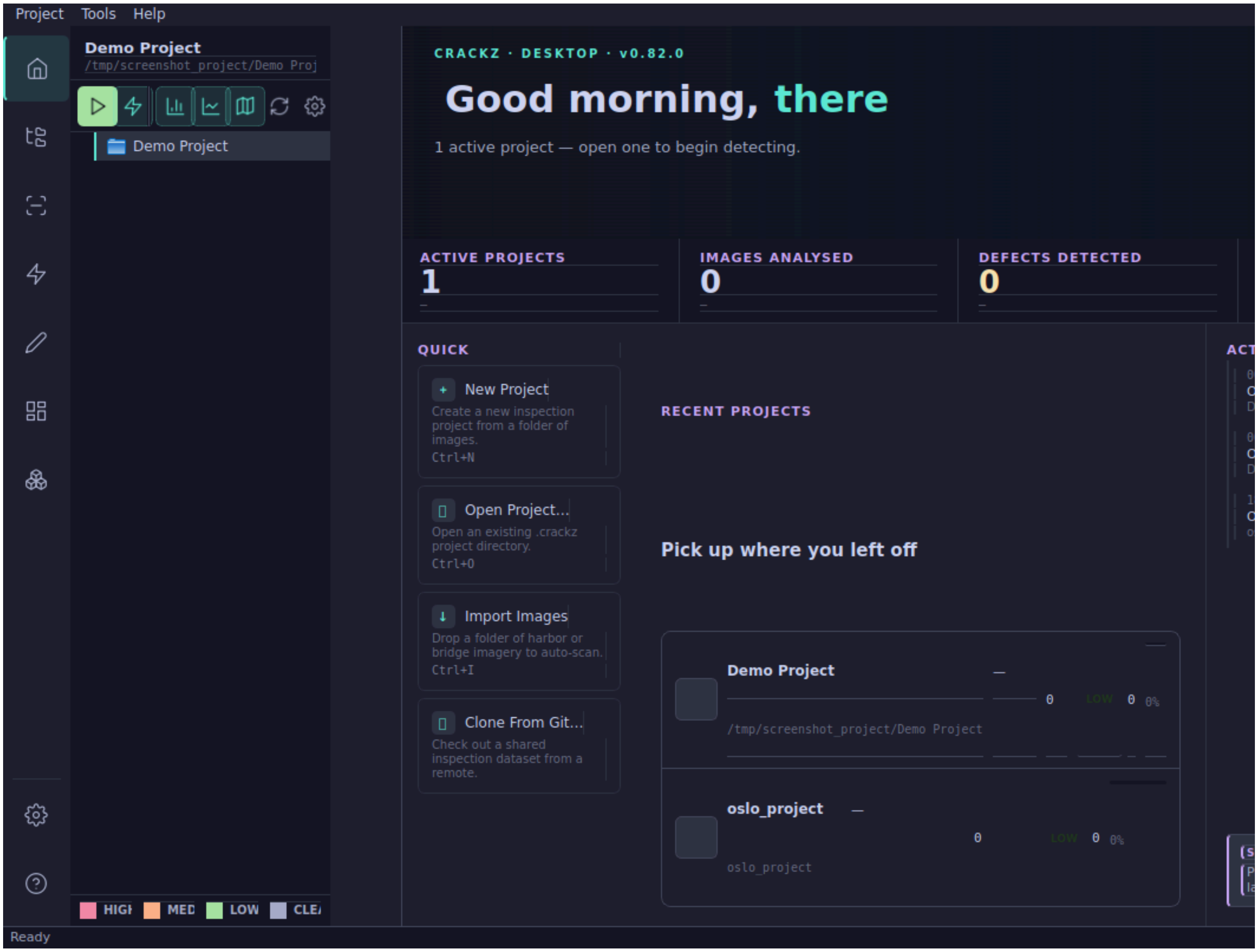
Quick Start

Getting Started with Crackz

```
# Install UV package manager
curl -LsSf https://astral.sh/uv/install.sh | sh  # macOS/Linux
# OR
irm https://astral.sh/uv/install.ps1 | iex       # Windows PowerShell

# Authenticate with GitHub
gh auth login

# Download and install Crackz
gh release download --repo anaxiatech/crackz --pattern '*.whl'
uv pip install crackz-*.whl

# Verify installation
crackz --version
```

```
# Pull the Docker image
docker pull ghcr.io/anaxiatech-ab/crackz:latest

# Verify it works
docker run --rm ghcr.io/anaxiatech-ab/crackz:latest --version
```

```
crackz-gui
```

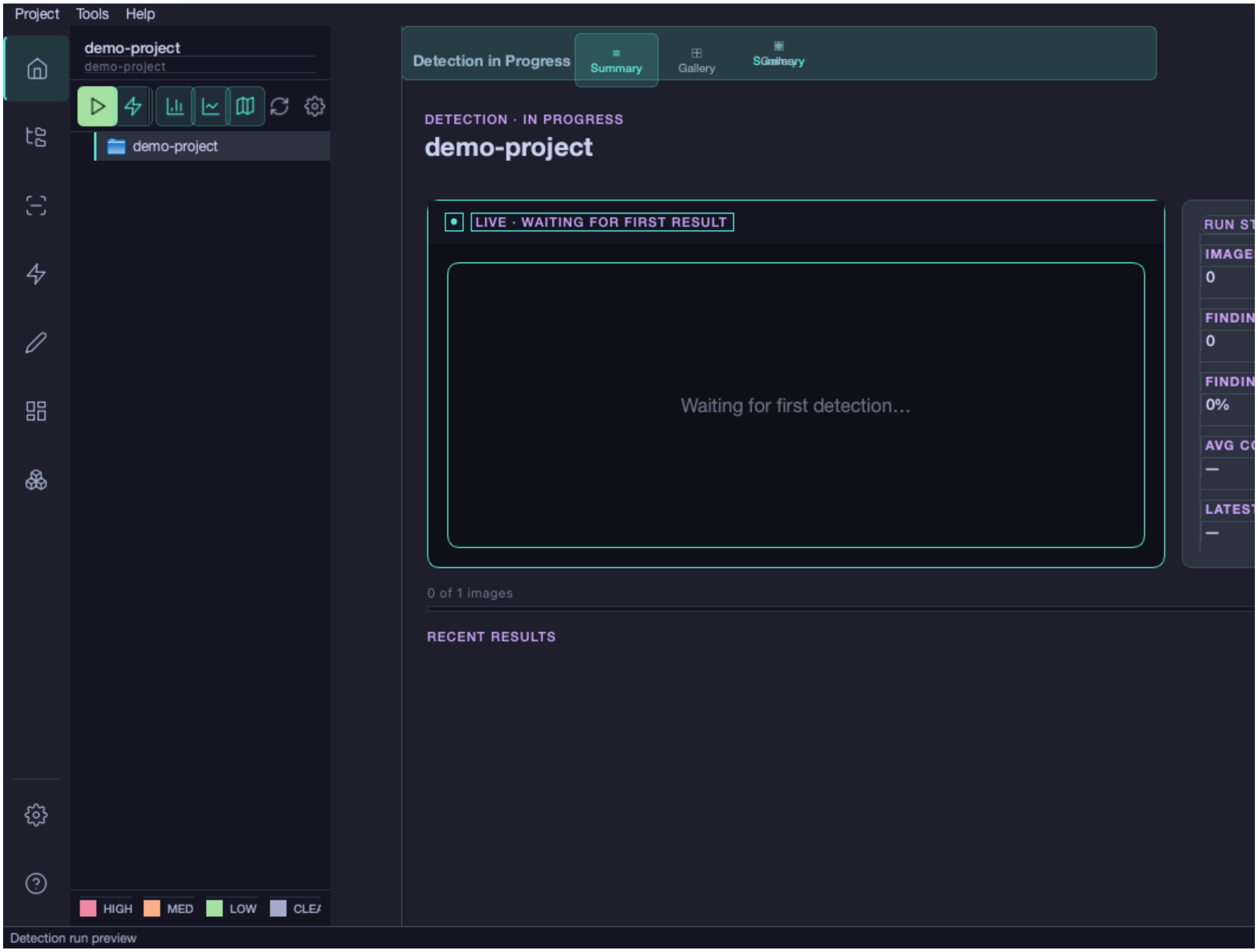


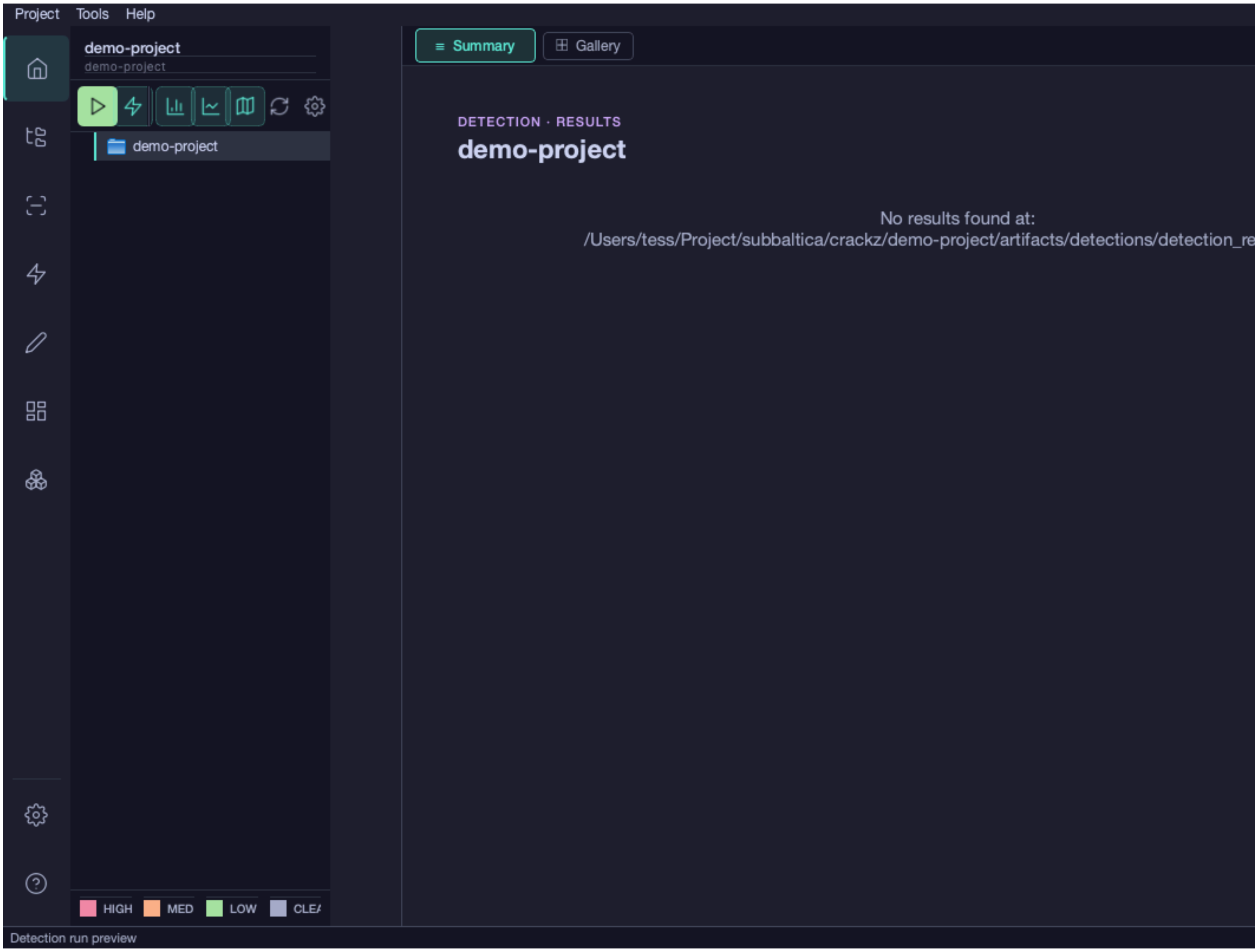
1.
2. `my-first-project`
3.
4.
5.
6.
7.

1.
2.
3.
4.


IMPORT IMAGES WITH AUTOMATIC SPLIT
Select a source directory with images and optional masks.
They will be split into train / val / test sets based on the ratios below.
SOURCE DIRECTORY
Select a directory containing images/ and optionally masks/ subdirectories.
Masks will be automatically matched to images by filename.
No directory selected
Select Source Directory...
SPLIT RATIOS (AUTO-ADJUSTING)
Train:
0.70
Validation:
0.15
Test:
0.15
Total: 1.00 ✓
RANDOM SEED (OPTIONAL)
Use random seed:
e.g. 42
IMAGE PROCESSING SETTINGS
Image size:
512
x
512
pixels
Mask label map:
Optional, e.g. 255=1,128=2
JPEG quality:
90
Import
Cancel

1.
2.
3.

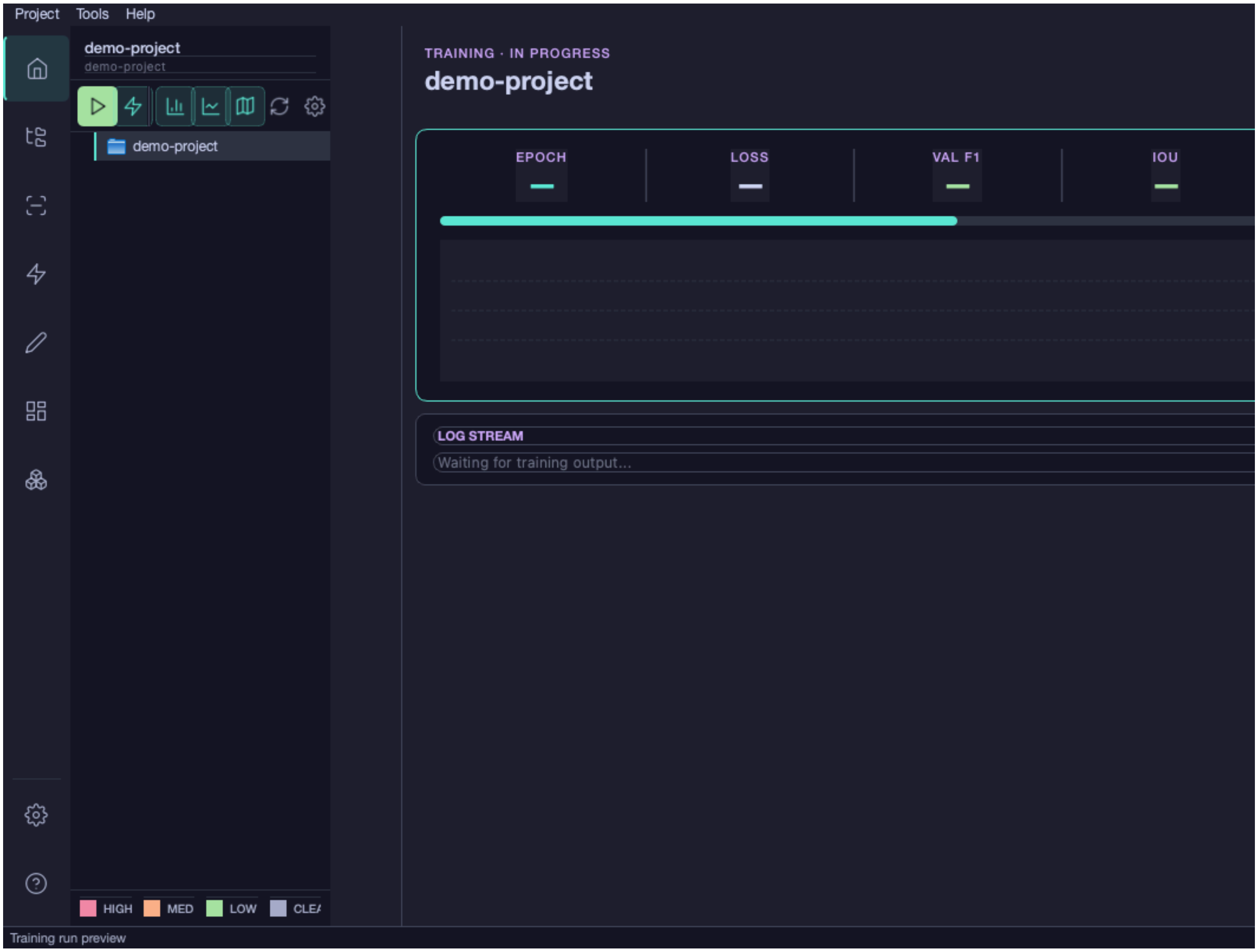

Project
Tools
Help
demo-project
demo-project
demo-project
HIGH
MED
LOW
CLEA
Detection in Progress
Summary
Gallery
DETECTION · IN PROGRESS
demo-project
LIVE · WAITING FOR FIRST RESULT
Waiting for first detection…
RUN ST
IMAGE
0
FINDIN
0
FINDIN
0%
AVG C
—
LATES
—
0 of 1 images
RECENT RESULTS
Detection run preview



`my-first-project/artifacts/detections/visualizations/` `my-first-project/artifacts/detections/detection_results.json`

1.
2. `raw`
3.
4.

1. `train` `val`
2.
3.
4.




```
my-first-project/
├── images/
│   ├── raw/        # Original images for detection
│   ├── train/      # Training images (if training a model)
│   ├── val/        # Validation images
│   └── test/       # Test images
├── masks/
│   ├── train/      # Ground truth masks for training
│   ├── val/        # Validation masks
│   └── test/       # Test masks
├── artifacts/
│   ├── models/     # Model checkpoints (best.ckpt, last.ckpt)
│   └── runtime/    # Detection results (images + JSON), training metrics
```

```
├── logs/            # Project-specific logs
└── my-first-project_project.yaml  # Configuration file (schema v6)
```

`my-first-project/my-first-project_project.yaml`

```yaml
ai:
  model:
    threshold: 0.5  # Increase to 0.7 for fewer false positives
                    # Decrease to 0.3 to catch more cracks
```

```bash
# Process several inspection campaigns
for project in site-A site-B site-C; do
    crackz detect run --project $project
done
```

```bash
# Run detection in container
docker run --rm \
  -v $(pwd)/my-first-project:/home/app \
  ghcr.io/anaxiatech-ab/crackz:latest \
  detect run --project /home/app
```


```bash
# Verify CUDA installation
nvidia-smi

# Check PyTorch CUDA support
python -c "import torch; print(torch.cuda.is_available())"

# Reinstall with CUDA support (see Installation Guide)
```

`images/raw/` `.jpg` `.jpeg` `.png` `.bmp` `.tif` `.tiff`

`project.yaml`

```
# Reduce batch size
crackz detect train --project my-project --batch-size 1

# Or reduce image size during import
crackz import-images --project my-project --src ./data --type train --size 256 256
```

`crackz init --name my-first-project`

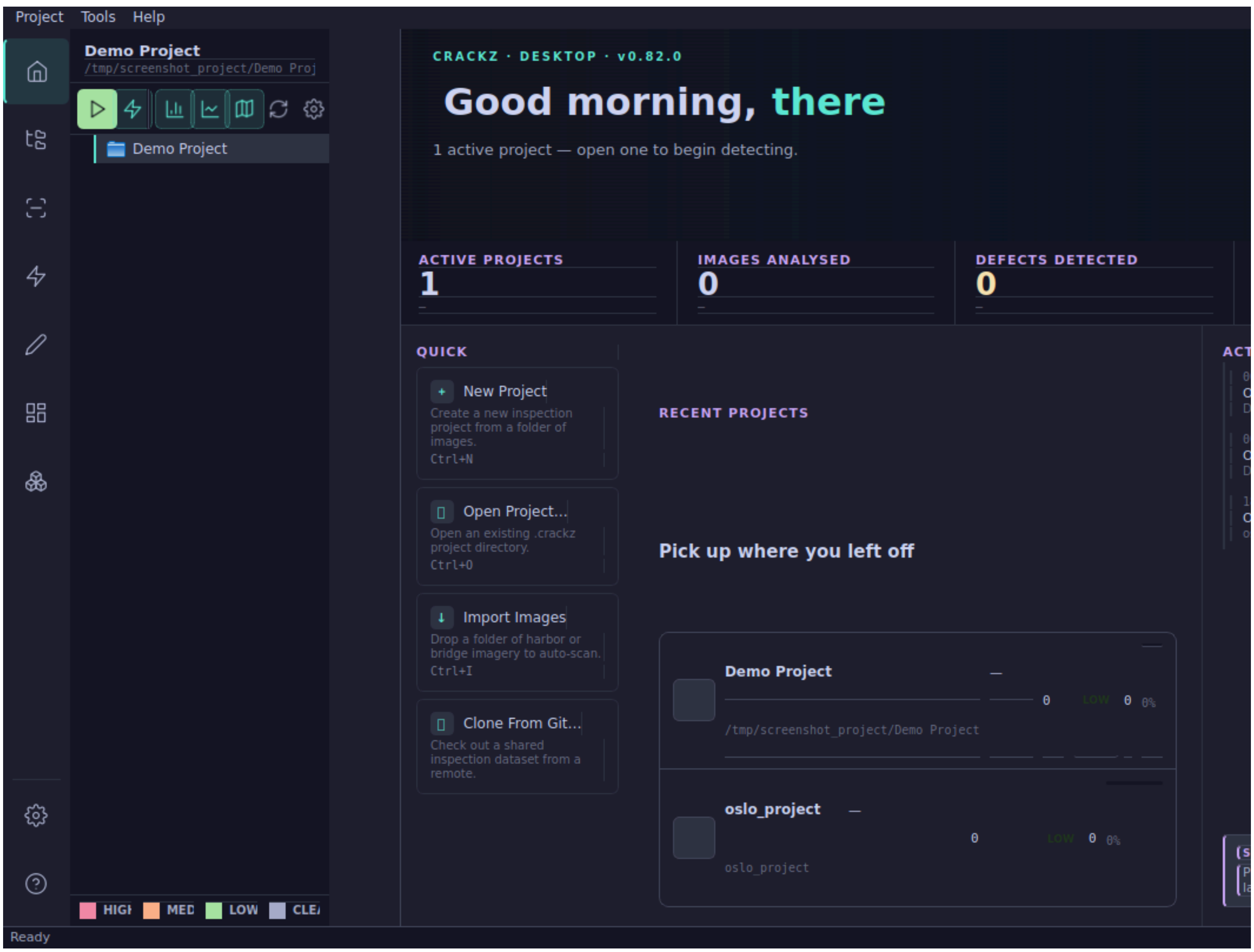
GUI Overview

GUI Guide

`crackz-gui`

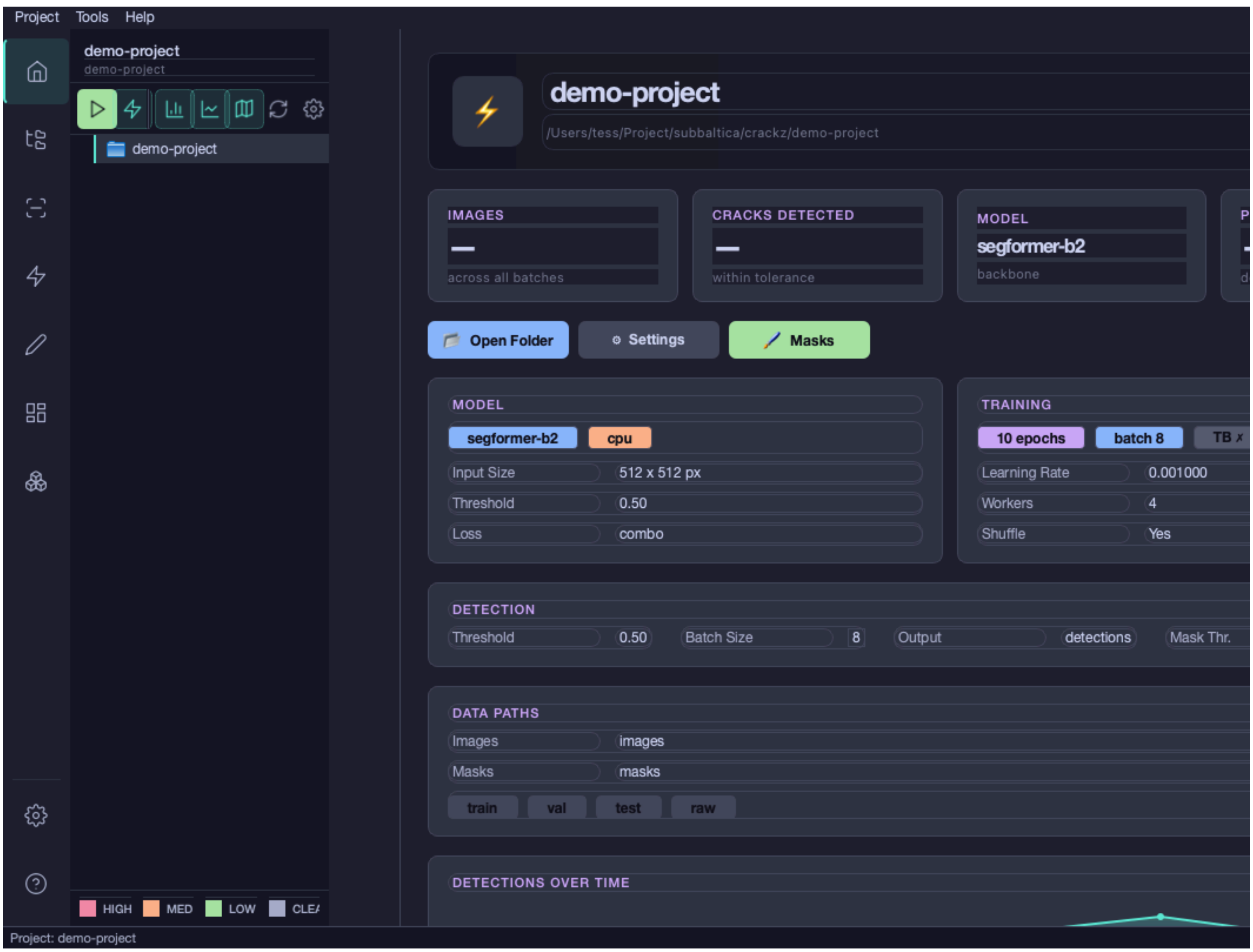
```
# After installing Crackz
crackz-gui
```

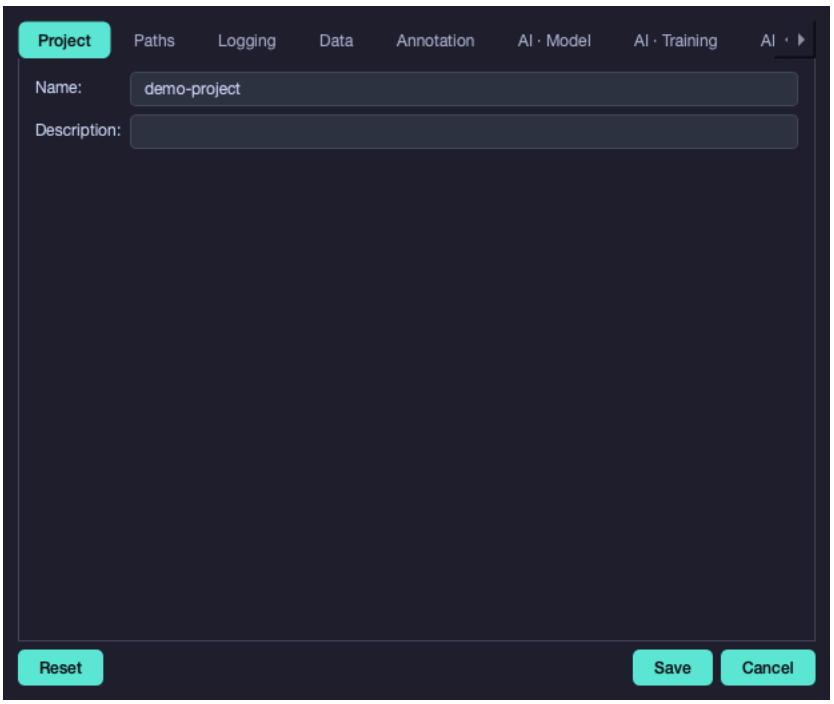
`crackz-gui.exe` `./crackz-gui`

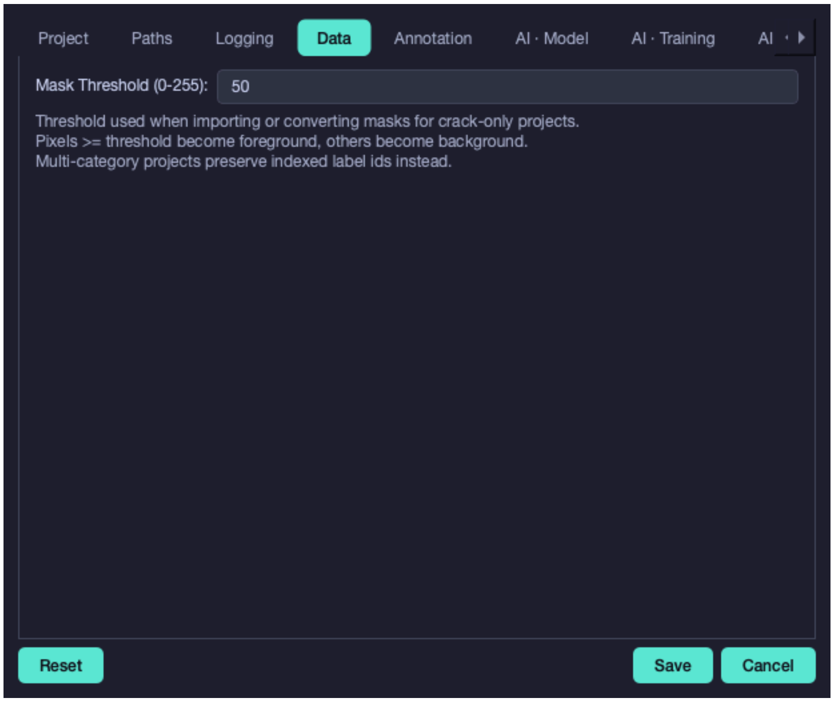


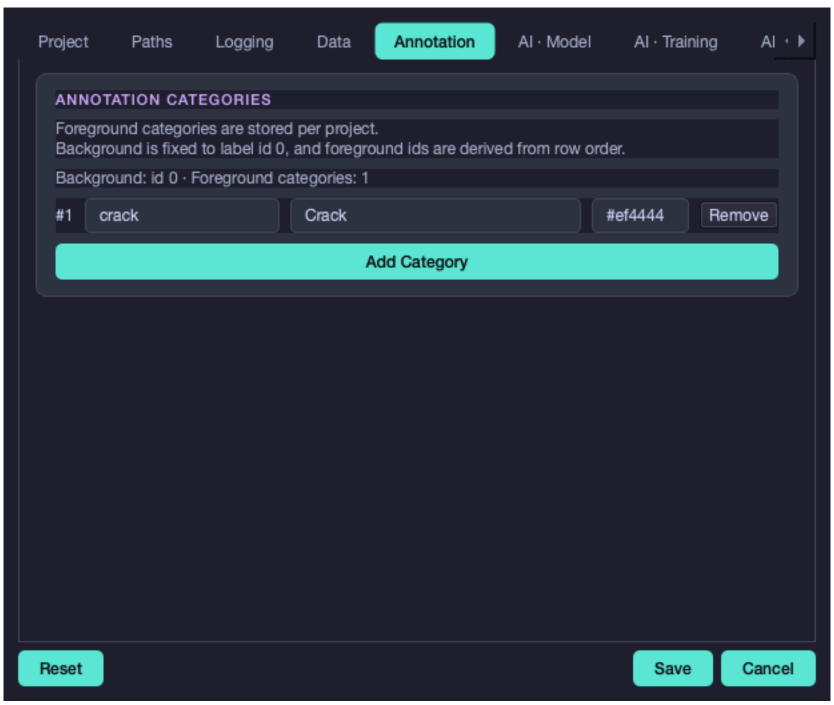
1.
2.
3.
4.
5.
6.
7.

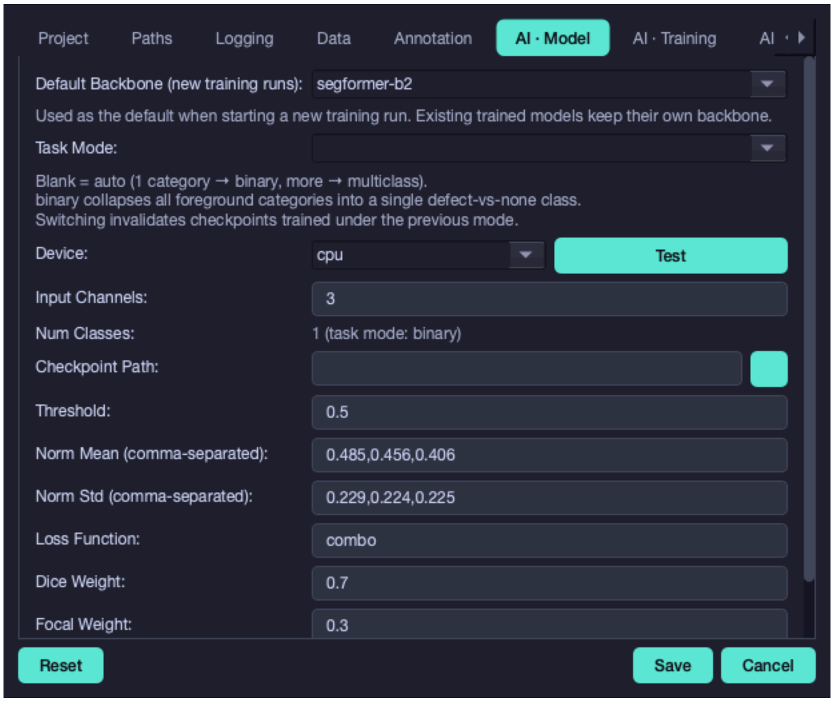
1.
2. `project.yaml`
3.

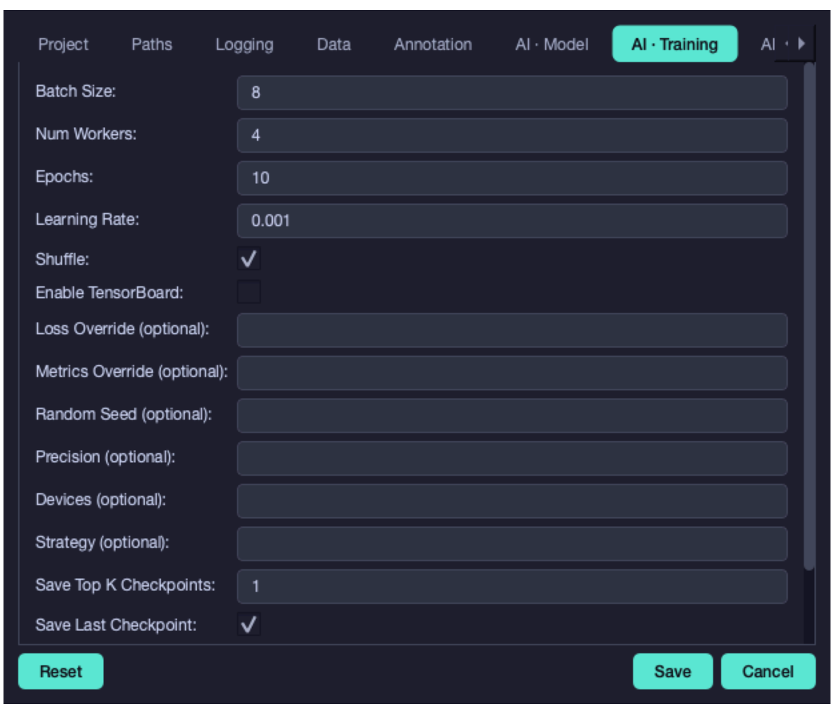


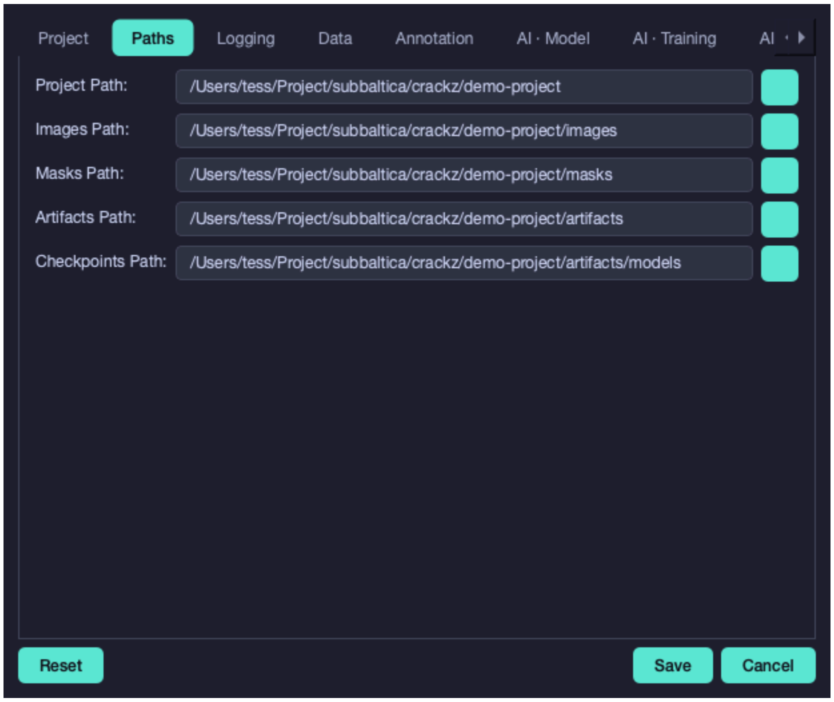

Project
Paths
Logging
Data
Annotation
AI · Model
AI · Training
AI
Name:
demo-project
Description:
Reset
Save
Cancel

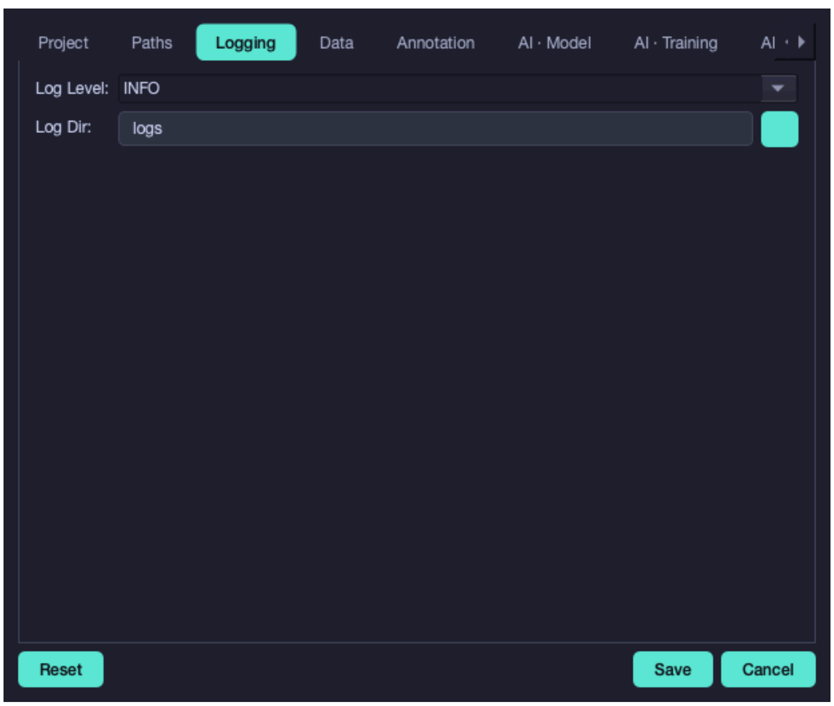

Project
Paths
Logging
Data
Annotation
AI · Model
AI · Training
AI
Mask Threshold (0-255):
50
Threshold used when importing or converting masks for crack-only projects.
Pixels >= threshold become foreground, others become background.
Multi-category projects preserve indexed label ids instead.
Reset
Save
Cancel


Project
Paths
Logging
Data
Annotation
AI · Model
AI · Training
AI
ANNOTATION CATEGORIES
Foreground categories are stored per project.
Background is fixed to label id 0, and foreground ids are derived from row order.
Background: id 0 · Foreground categories: 1
#1
crack
Crack
#ef4444
Remove
Add Category
Reset
Save
Cancel


Project
Paths
Logging
Data
Annotation
AI · Model
AI · Training
AI
Default Backbone (new training runs):
segformer-b2
Used as the default when starting a new training run. Existing trained models keep their own backbone.
Task Mode:
Blank = auto (1 category → binary, more → multiclass).
binary collapses all foreground categories into a single defect-vs-none class.
Switching invalidates checkpoints trained under the previous mode.
Device:
cpu
Test
Input Channels:
3
Num Classes:
1 (task mode: binary)
Checkpoint Path:
Threshold:
0.5
Norm Mean (comma-separated):
0.485,0.456,0.406
Norm Std (comma-separated):
0.229,0.224,0.225
Loss Function:
combo
Dice Weight:
0.7
Focal Weight:
0.3
Reset
Save
Cancel


Project
Paths
Logging
Data
Annotation
AI · Model
AI · Training
AI
Batch Size:
8
Num Workers:
4
Epochs:
10
Learning Rate:
0.001
Shuffle:
Enable TensorBoard:
Loss Override (optional):
Metrics Override (optional):
Random Seed (optional):
Precision (optional):
Devices (optional):
Strategy (optional):
Save Top K Checkpoints:
1
Save Last Checkpoint:
Reset
Save
Cancel





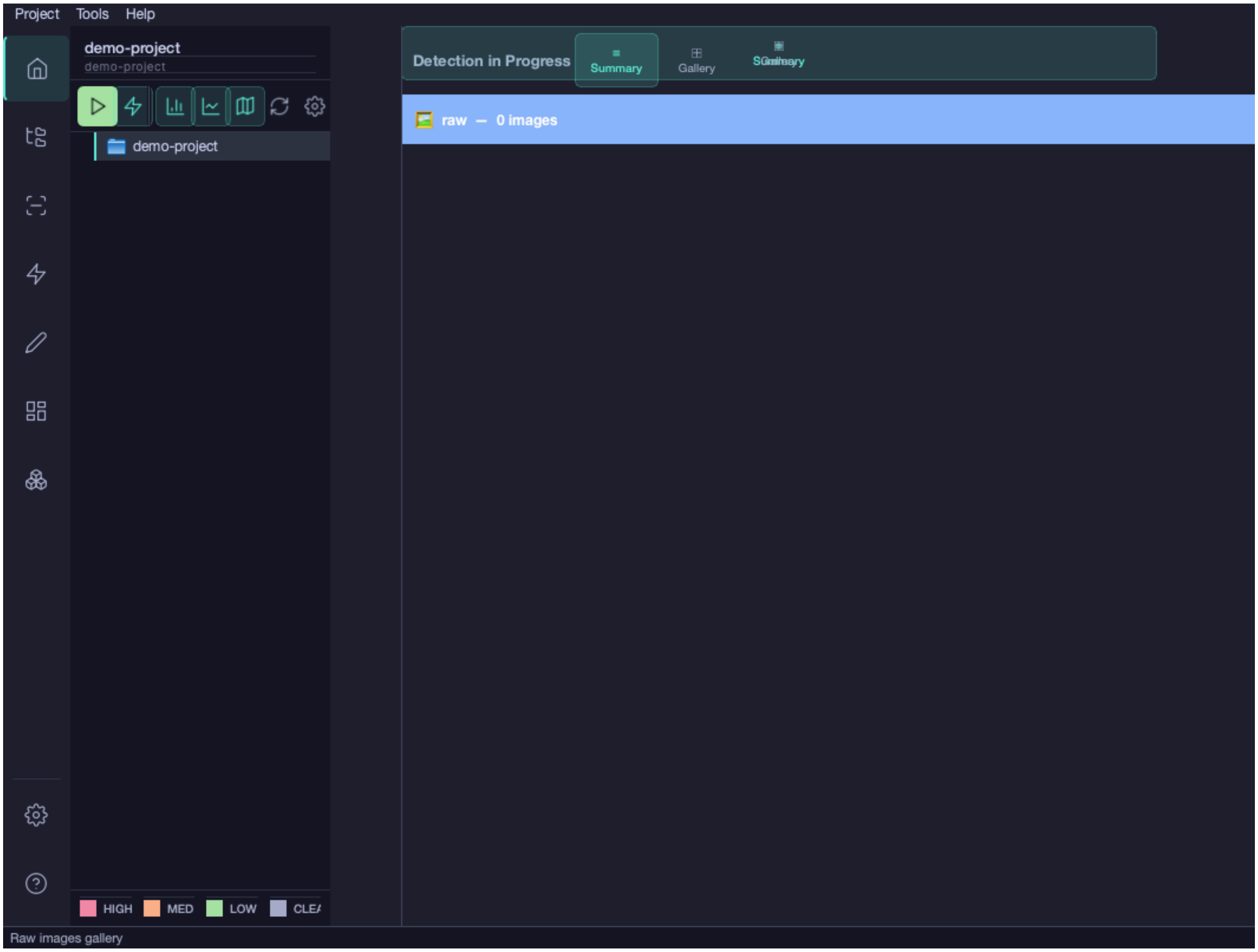
`crack` `spall`

`#RRGGBB`

- `binary`

- `multiclass` `1 + N`

1.
2. `masks/`
3.
4.
5.
6.

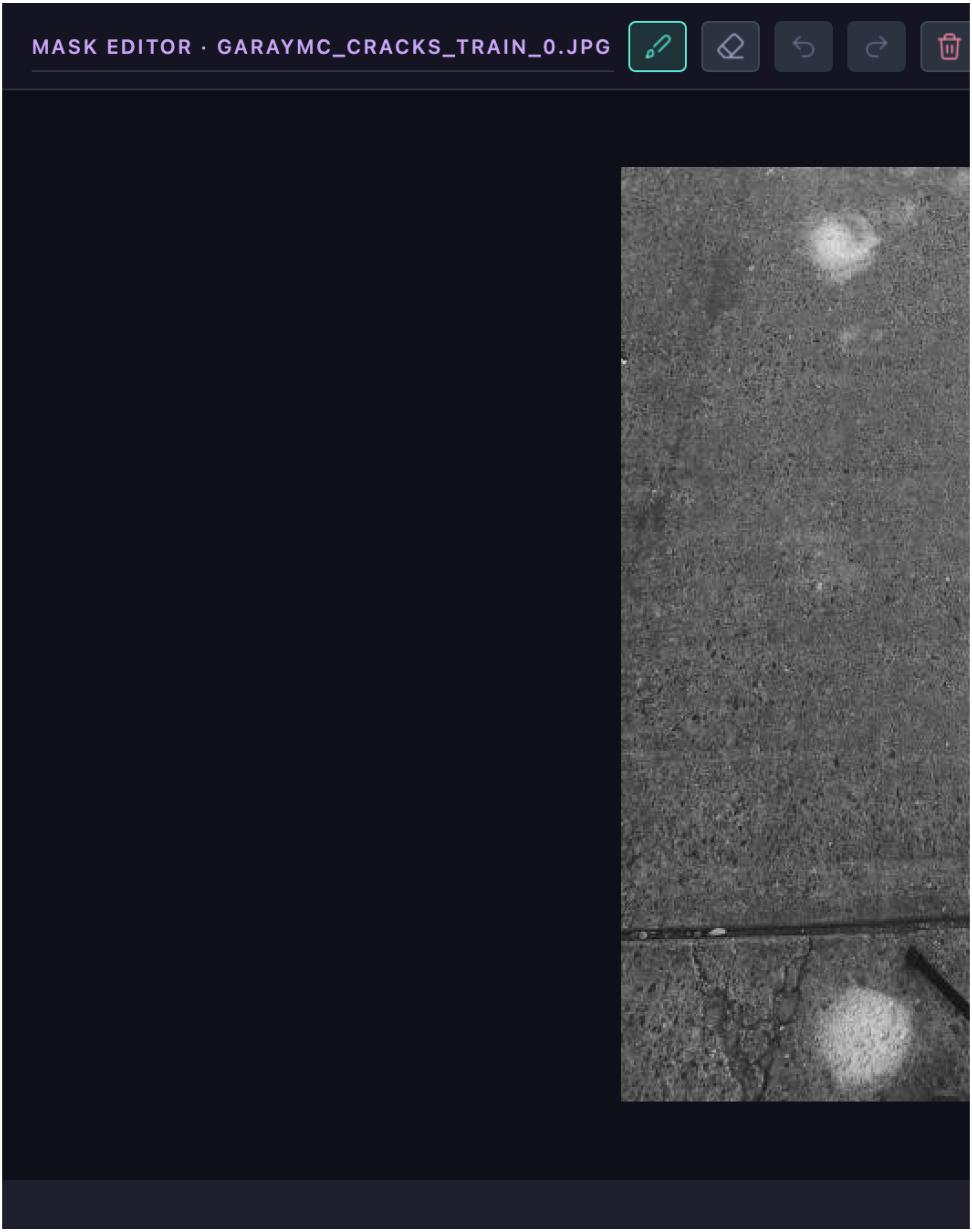


-
-

`images/`

`images/`

`images/`


MASK EDITOR · GARAYMC_CRACKS_TRAIN_0.JPG

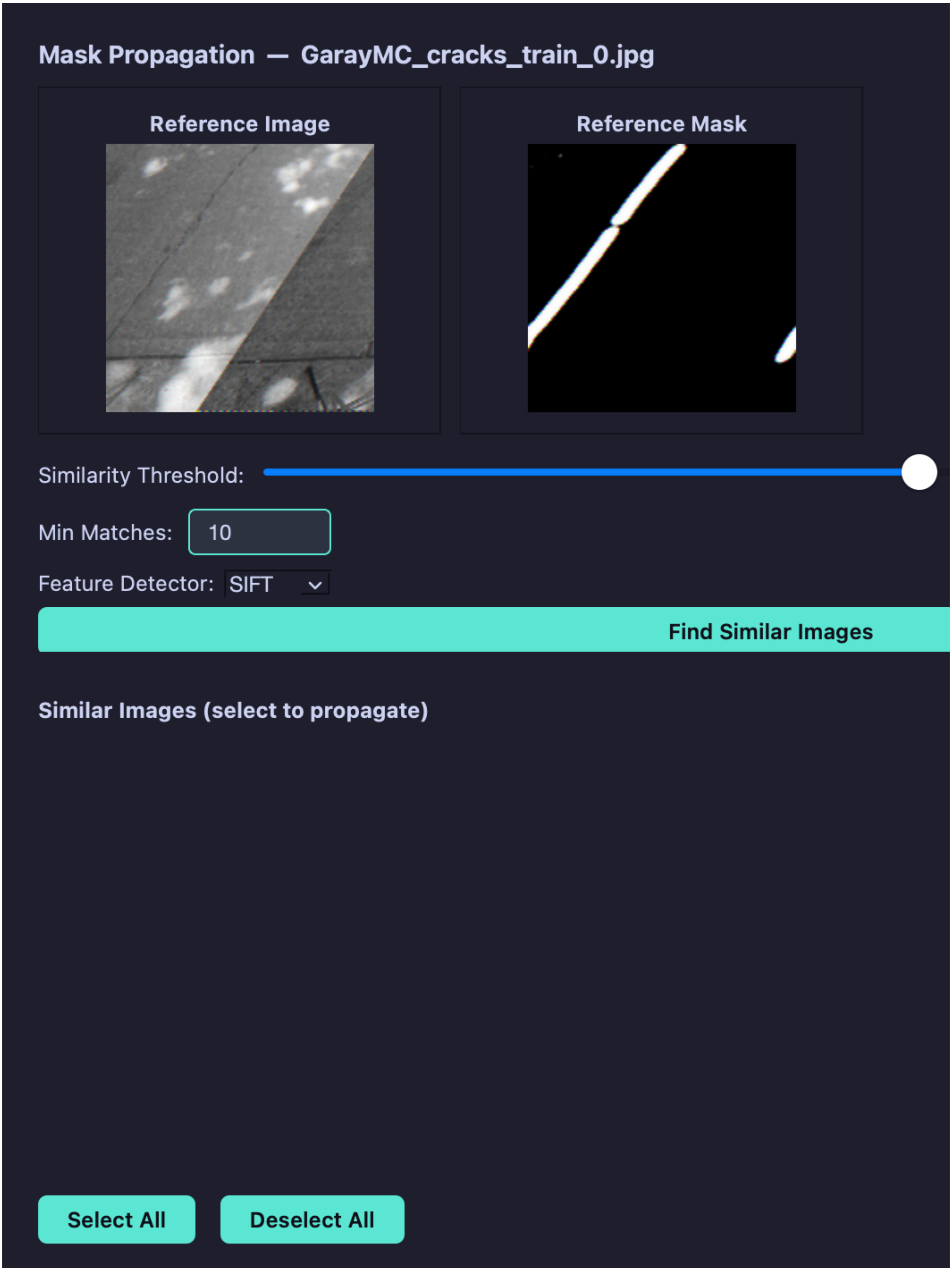
`masks/`

		`B`	
		`E`	
		`Ctrl+Z` `Cmd+Z`	
		`Ctrl+Y` `Cmd+Shift+Z`	
		`Ctrl+S` `Cmd+S`	
		`Esc`	

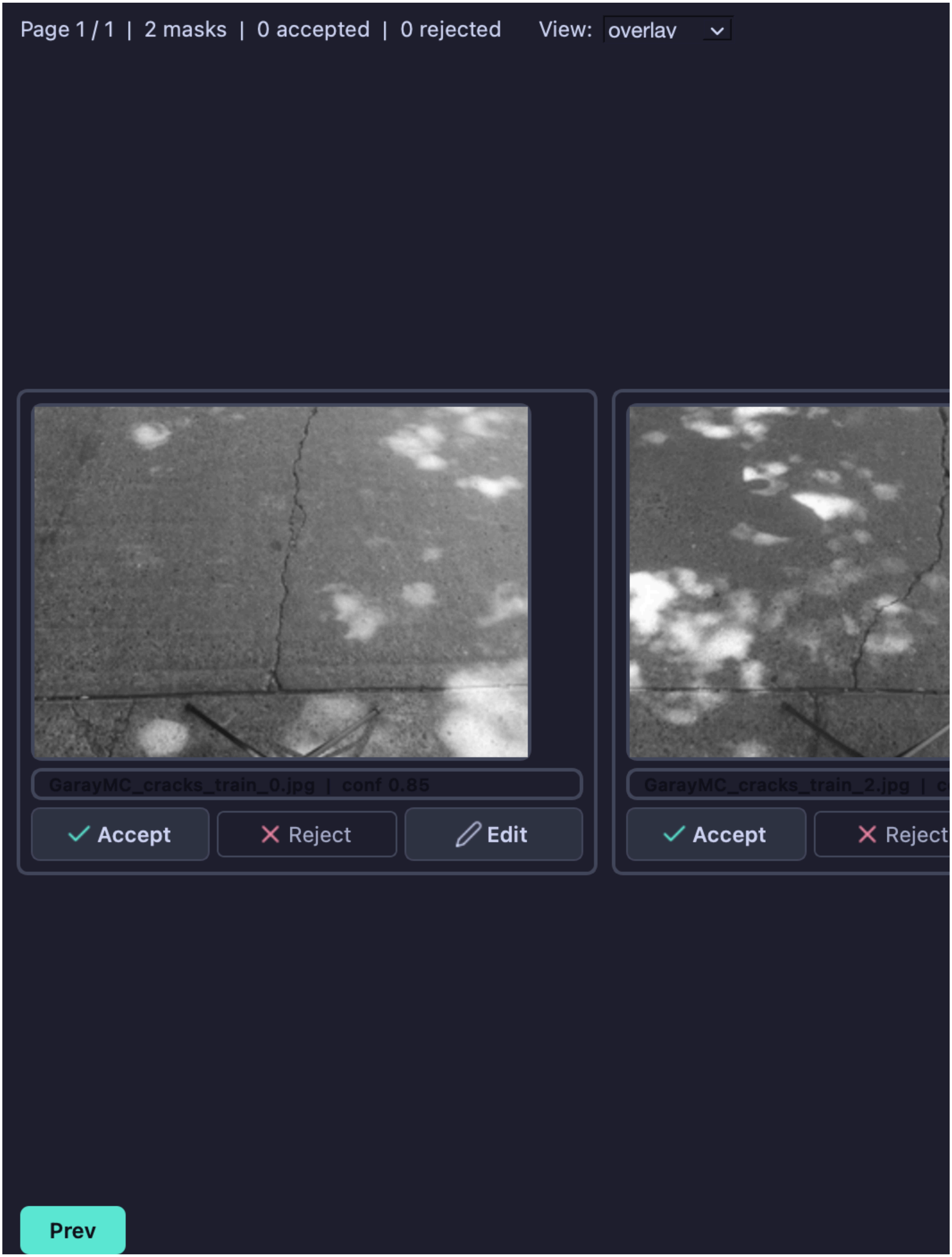
```
project/
├── images/
│   └── train/
│       └── crack_001.jpg        # Original image
└── masks/
    └── train/
        └── crack_001.png        # Indexed mask (auto-created)
```

`train/` `val/` `test/` `raw/`

1.
2.
3.
4.


Mask Propagation — GarayMC_cracks_train_0.jpg
Reference Image
Reference Mask
Similarity Threshold:
Min Matches:
10
Feature Detector:
SIFT
Find Similar Images
Similar Images (select to propagate)
Select All
Deselect All

1.
2.
3.
4.
5.



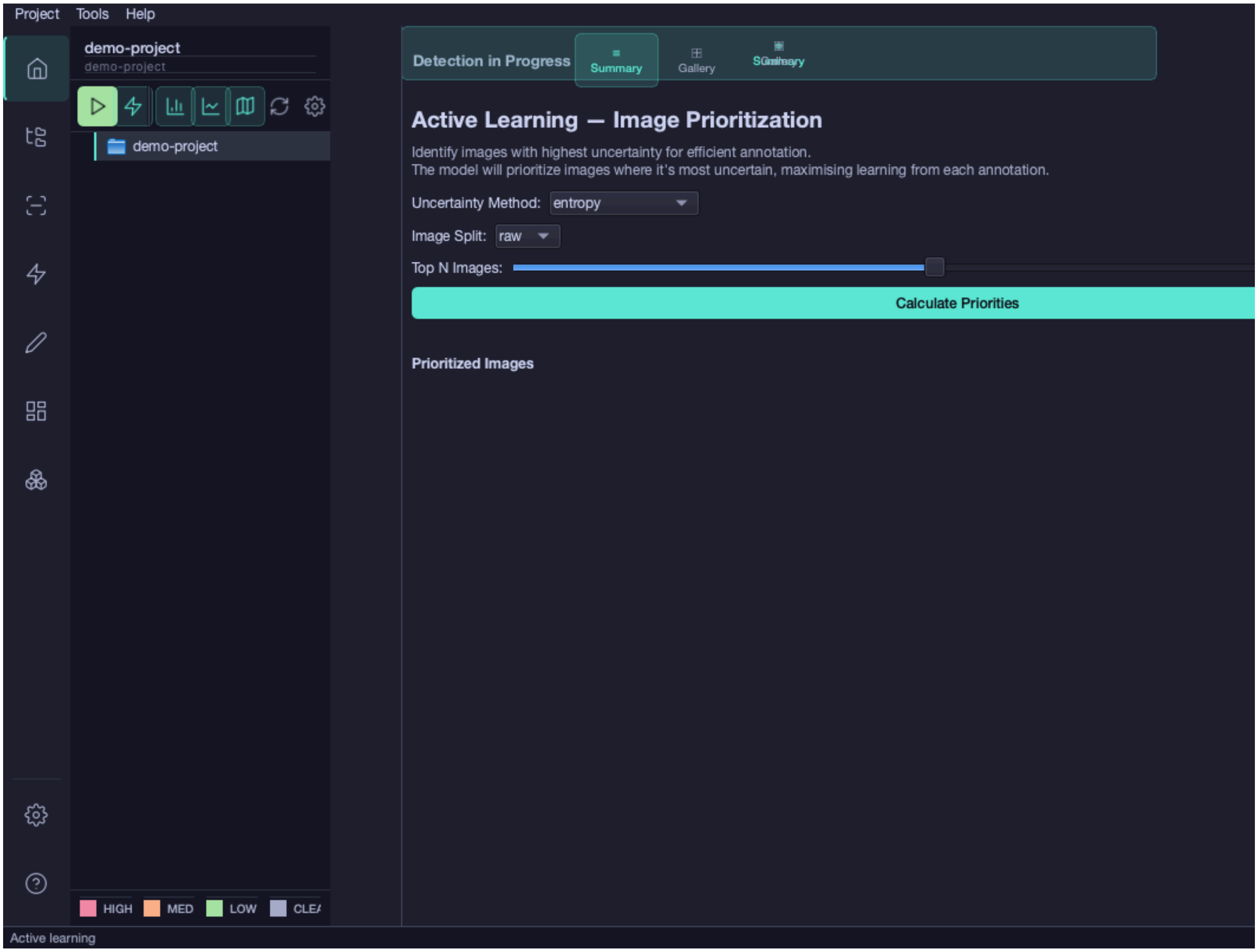
Page 1 / 1 | 2 masks | 0 accepted | 0 rejected
View: overlav
GarayMC_cracks_train_0.jpg | conf 0.85
Accept
Reject
Edit
GarayMC_cracks_train_2.jpg |
Accept
Reject
Prev

	`A`	
	`R`	
	`E`	

	`Right Arrow`	
	`Left Arrow`	

`masks/`

`A`
`R` `E`


Project Tools Help
demo-project
demo-project
demo-project
Detection in Progress
Summary
Gallery
Active Learning — Image Prioritization
Identify images with highest uncertainty for efficient annotation.
The model will prioritize images where it's most uncertain, maximising learning from each annotation.
Uncertainty Method: entropy
Image Split: raw
Top N Images:
Calculate Priorities
Prioritized Images
HIGH MED LOW
Active learning

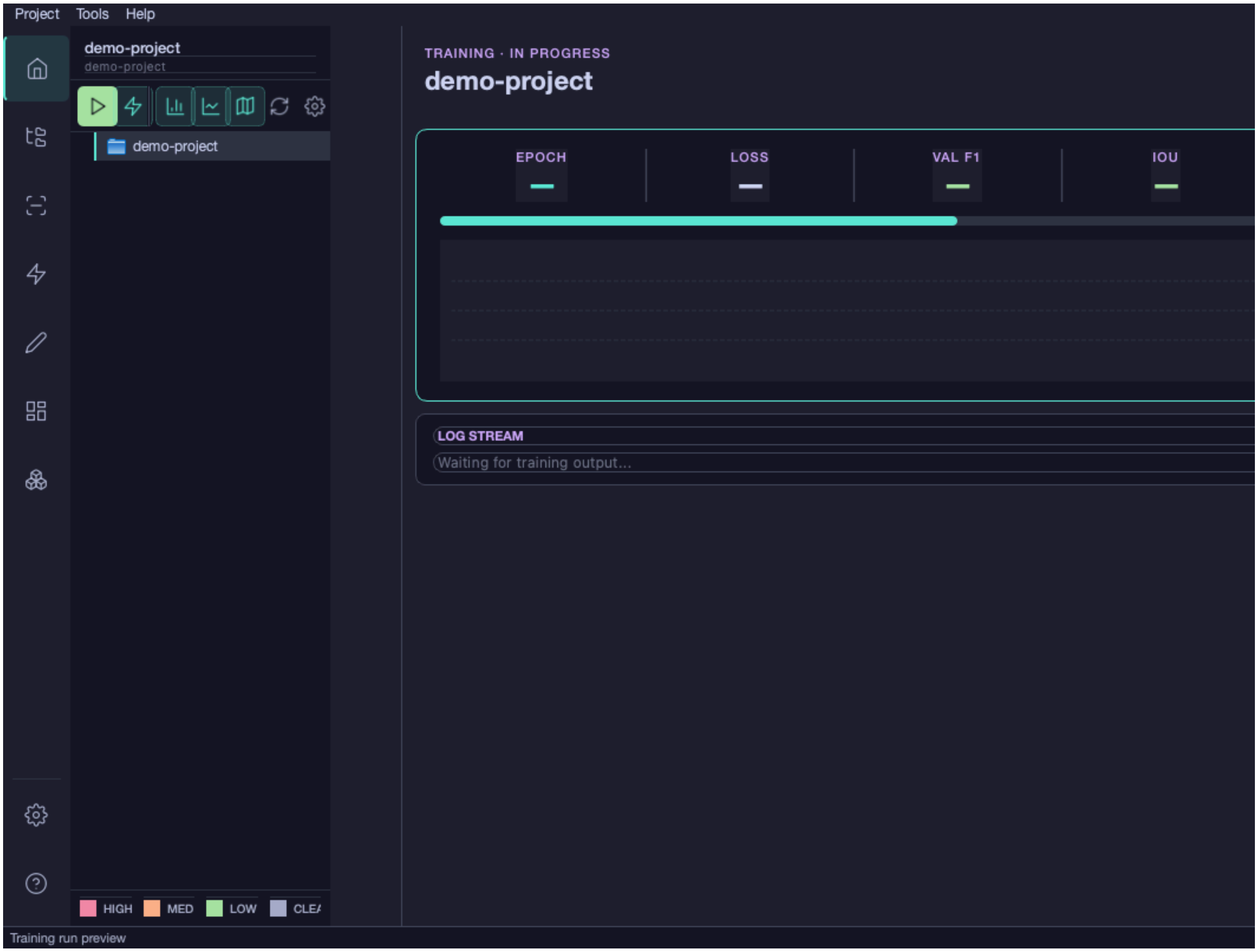
`raw/`

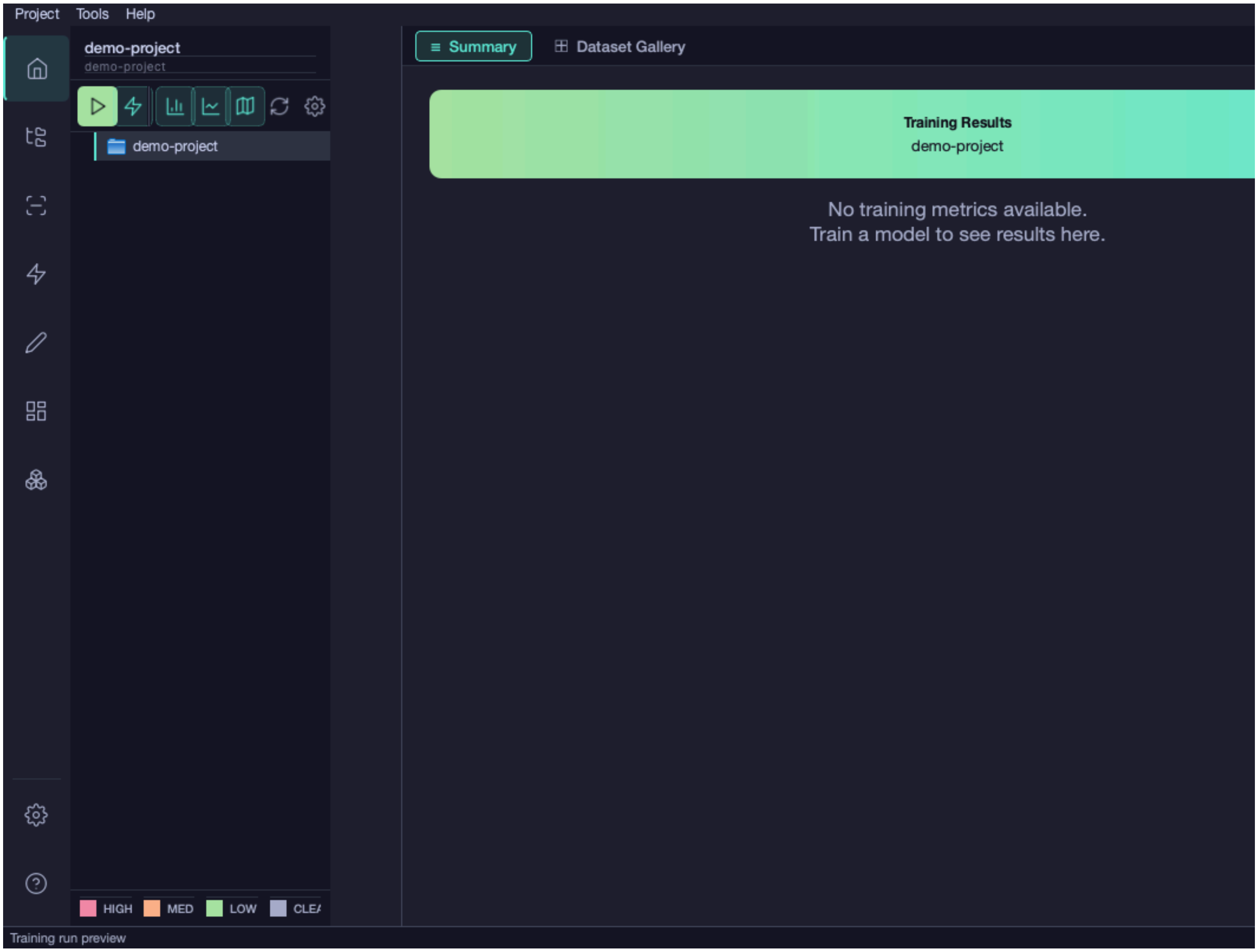
`raw/`

```
# Move annotated images from raw to train split
crackz move-images --project ./my-project --source raw --target train --pattern "crack_*.jpg"
```

`images/raw/`
`train` `val` `test`

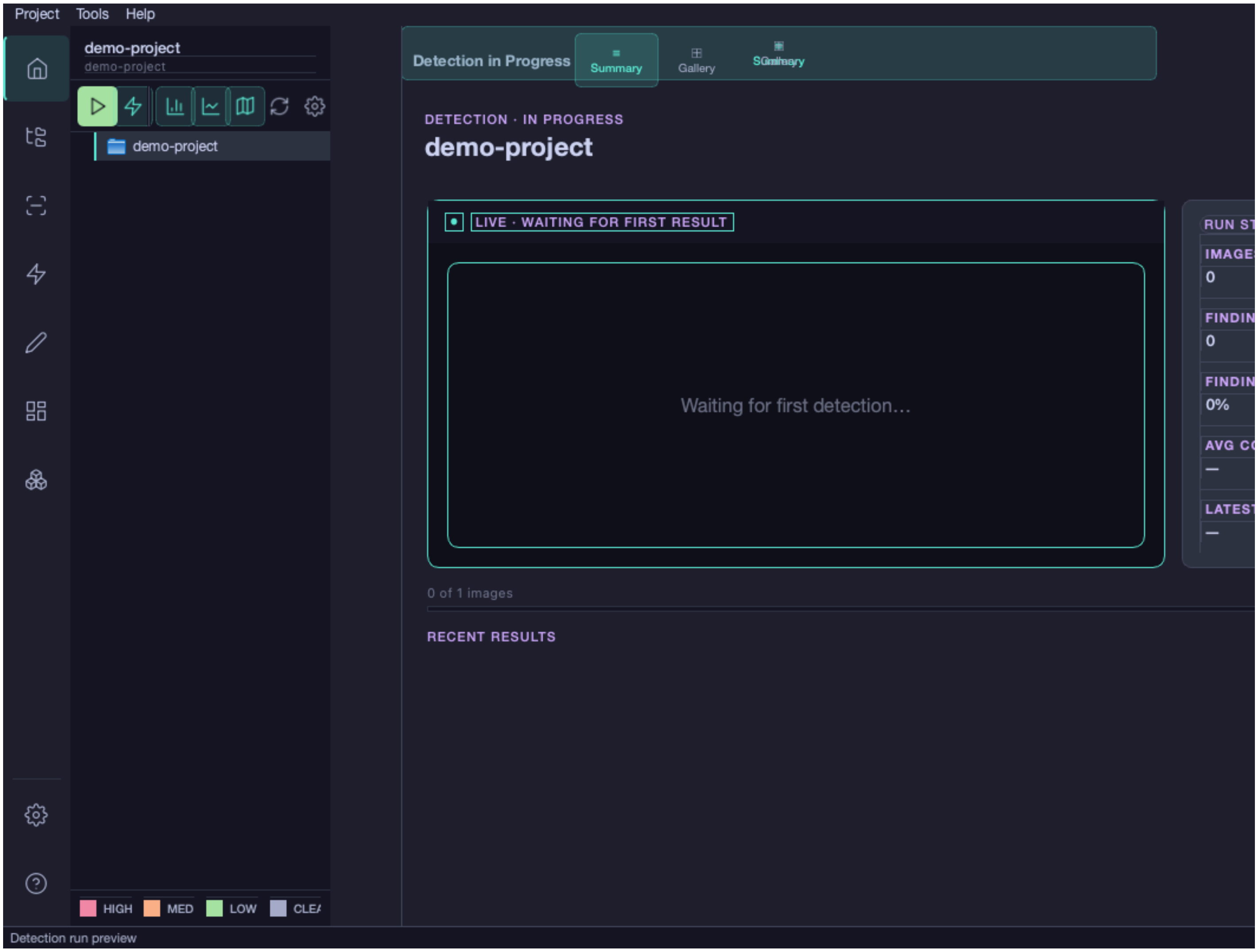
1.
2.
3.

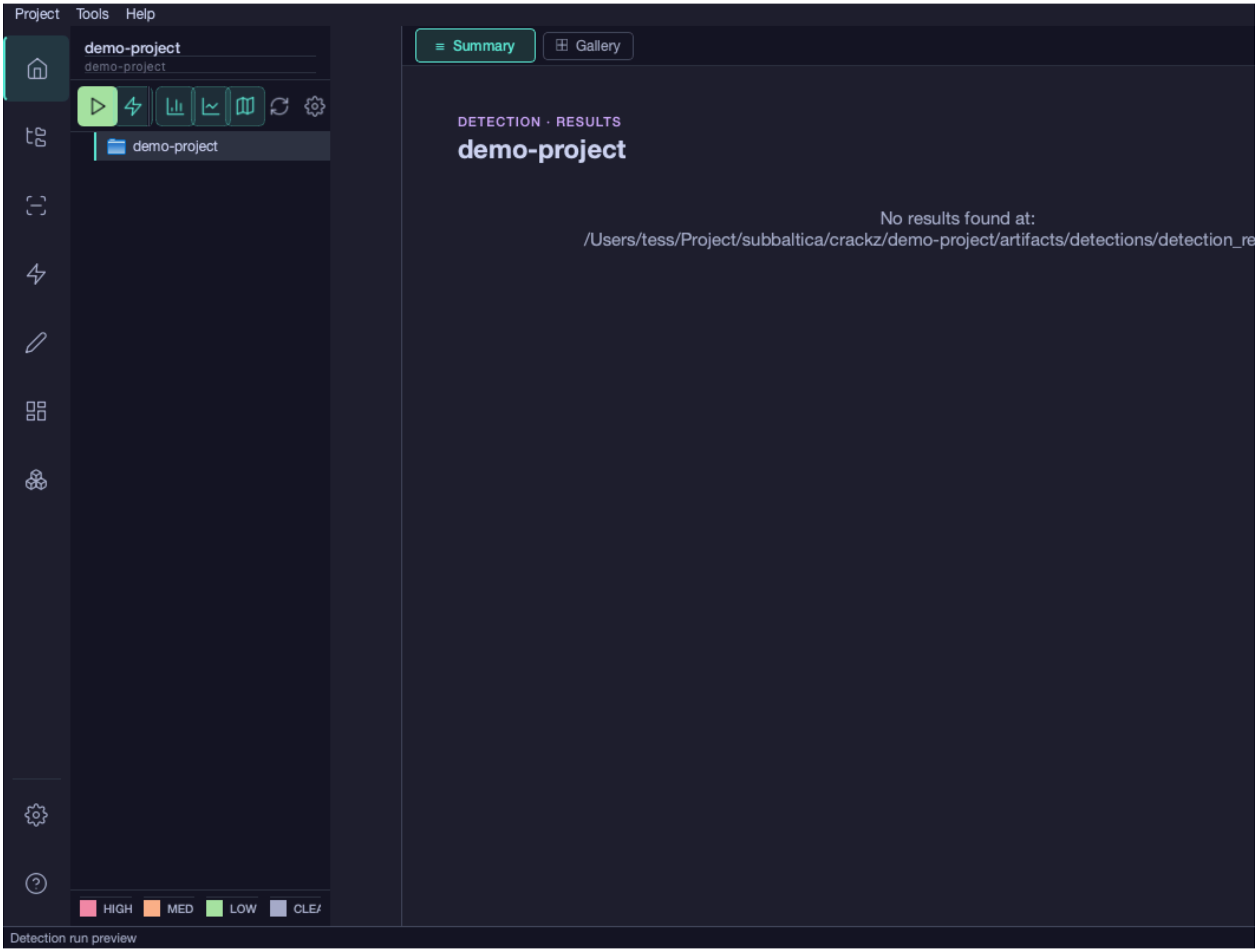

Project Tools Help
demo-project
demo-project
demo-project
TRAINING · IN PROGRESS
demo-project
EPOCH
LOSS
VAL F1
IOU
LOG STREAM
Waiting for training output...
HIGH
MED
LOW
CLEA
Training run preview

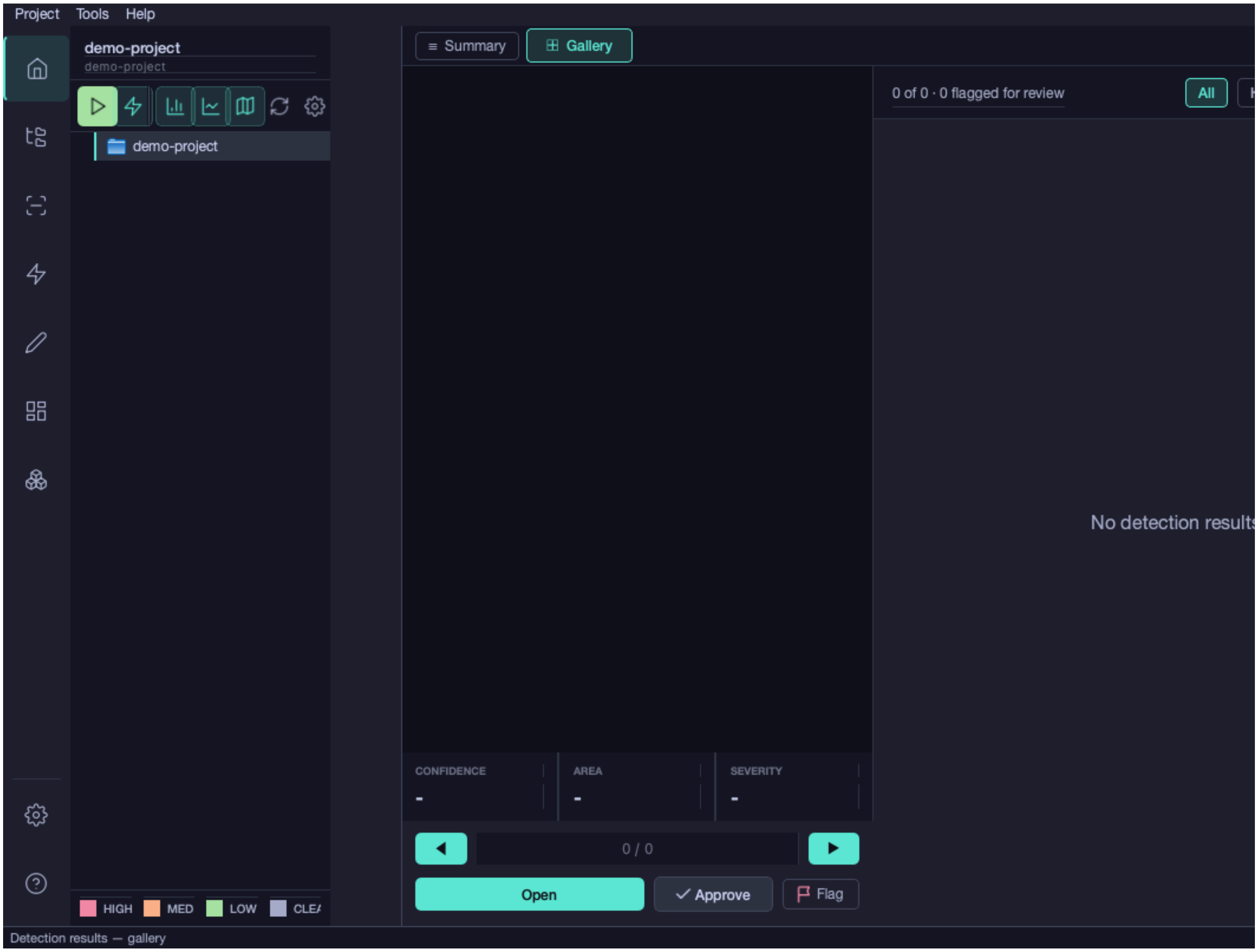

Project
Tools
Help
demo-project
demo-project
demo-project
Summary
Dataset Gallery
Training Results
demo-project
No training metrics available.
Train a model to see results here.
HIGH
MED
LOW
Training run preview

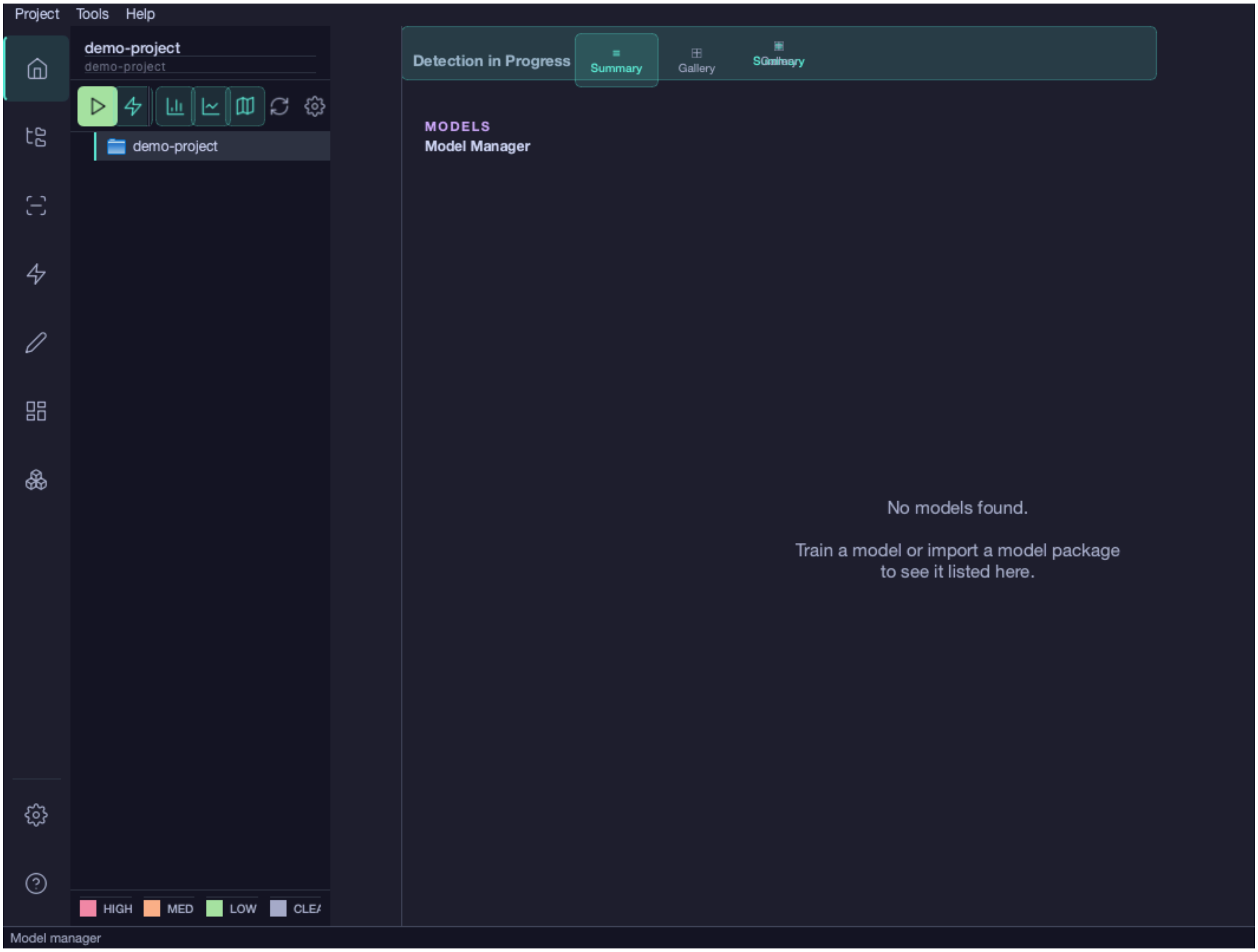
---

1. `raw`
2.
3.
4.
5.


Project Tools Help
demo-project
demo-project
demo-project
HIGH MED LOW CLEA
Detection in Progress
Summary
Gallery
DETECTION · IN PROGRESS
demo-project
LIVE · WAITING FOR FIRST RESULT
Waiting for first detection…
0 of 1 images
RECENT RESULTS
RUN ST
IMAGE
0
FINDIN
0
FINDIN
0%
AVG C
—
LATEST
—
Detection run preview

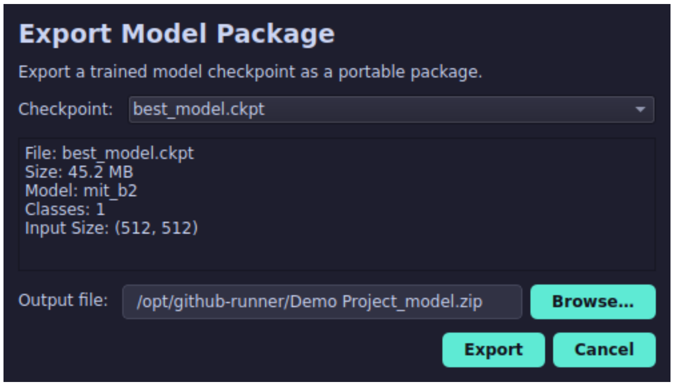

Project
Tools
Help
demo-project
demo-project
demo-project
Summary
Gallery
DETECTION · RESULTS
demo-project
No results found at:
/Users/tess/Project/subbaltica/crackz/demo-project/artifacts/detections/detection_re
HIGH
MED
LOW
CLEA
Detection run preview


Project Tools Help
demo-project
demo-project
demo-project
Summary
Gallery
0 of 0 · 0 flagged for review
All
No detection results
CONFIDENCE
-
AREA
-
SEVERITY
-
0 / 0
Open
Approve
Flag
HIGH
MED
LOW
Detection results — gallery

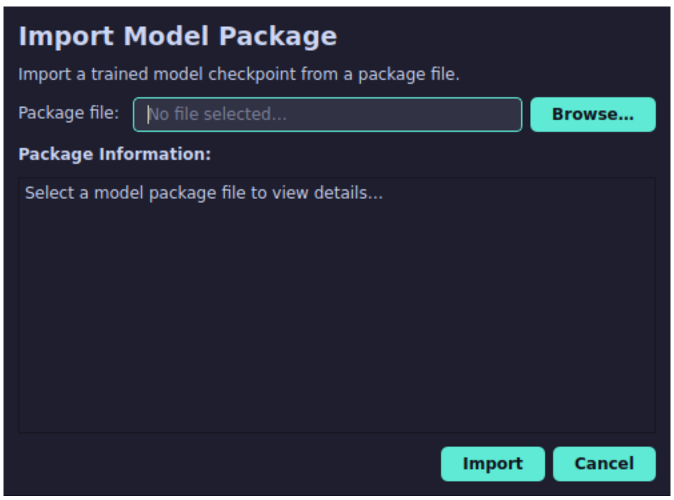


`Binary Segmentation` `Multiclass Segmentation`

1.
2.
3.
4.

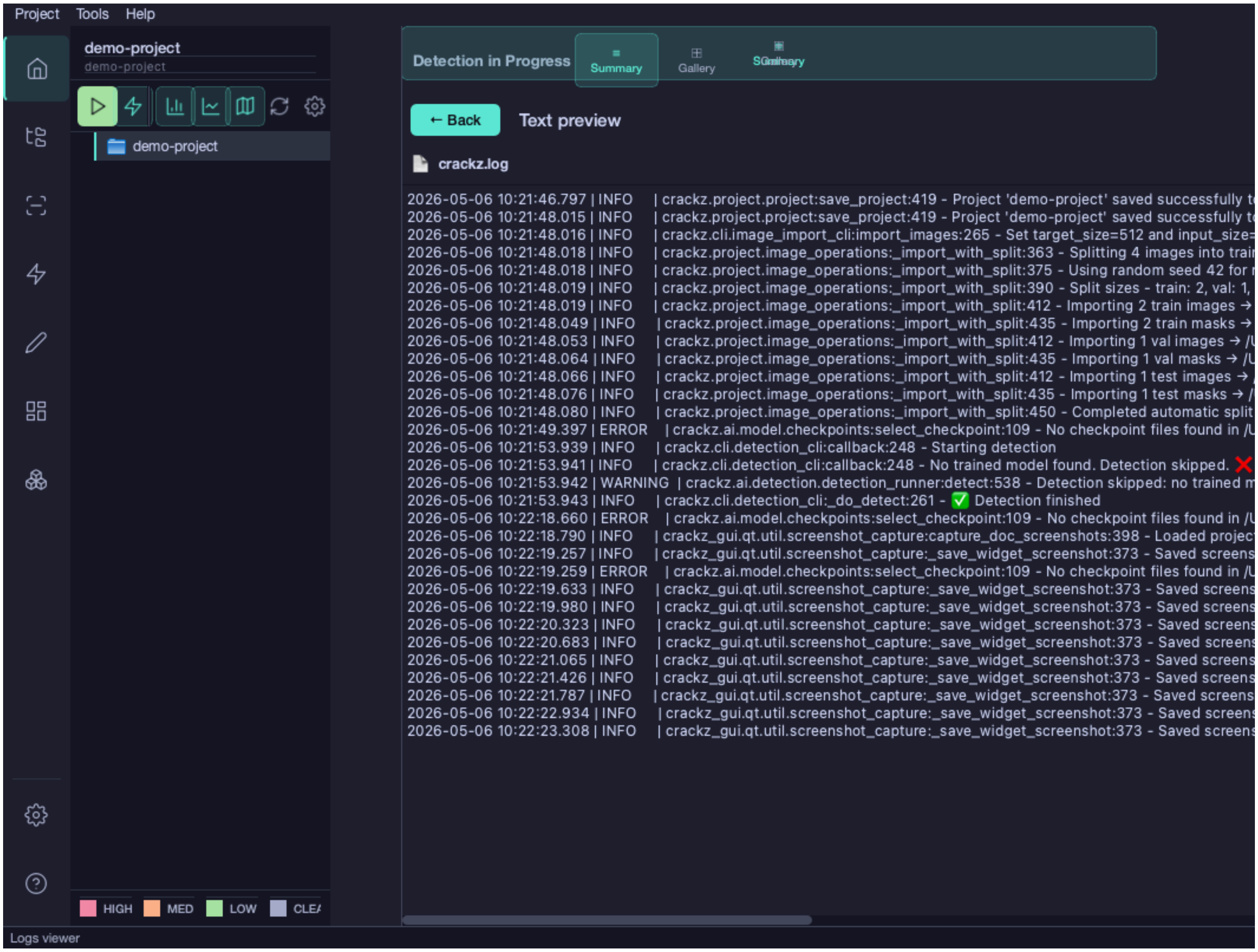


1.
2. `.zip`
3.
4.

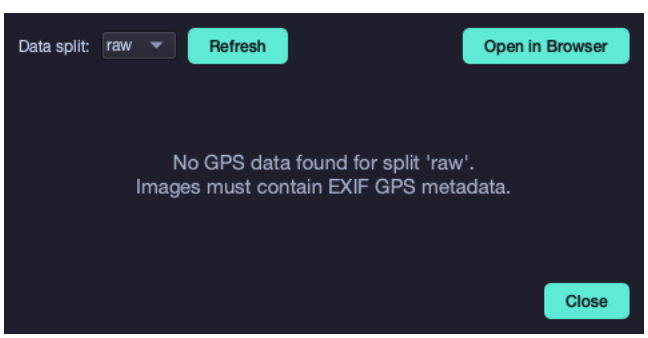


```
crackz clean --project ./my-project
```

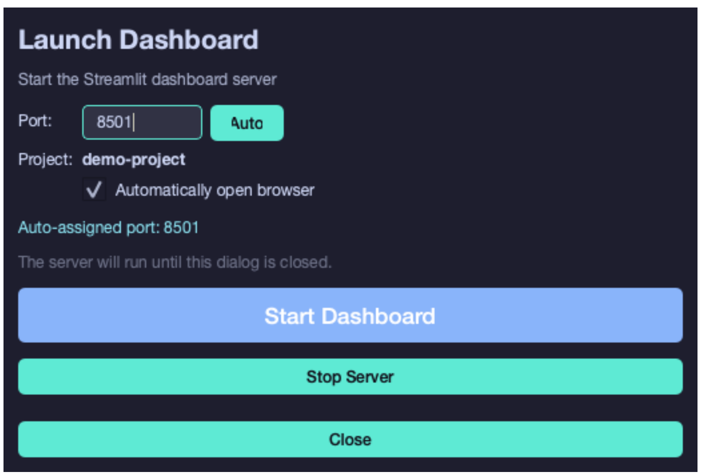
`project.yaml`


Project Tools Help
demo-project
demo-project
demo-project
Detection in Progress
Summary
Gallery
← Back
Text preview
crackz.log
2026-05-06 10:21:46.797 | INFO | crackz.project.project:save_project:419 - Project 'demo-project' saved successfully
2026-05-06 10:21:48.015 | INFO | crackz.project.project:save_project:419 - Project 'demo-project' saved successfully
2026-05-06 10:21:48.016 | INFO | crackz.cli.image_import_cli:import_images:265 - Set target_size=512 and
2026-05-06 10:21:48.018 | INFO | crackz.project.image_operations:_import_with_split:363 - Splitting 4 images into
2026-05-06 10:21:48.018 | INFO | crackz.project.image_operations:_import_with_split:375 - Using random seed 42 for
2026-05-06 10:21:48.019 | INFO | crackz.project.image_operations:_import_with_split:390 - Split sizes - train: 2, val: 1,
2026-05-06 10:21:48.019 | INFO | crackz.project.image_operations:_import_with_split:412 - Importing 2 train images →
2026-05-06 10:21:48.049 | INFO | crackz.project.image_operations:_import_with_split:435 - Importing 2 train masks →
2026-05-06 10:21:48.053 | INFO | crackz.project.image_operations:_import_with_split:412 - Importing 1 val images →
2026-05-06 10:21:48.064 | INFO | crackz.project.image_operations:_import_with_split:435 - Importing 1 val masks →
2026-05-06 10:21:48.066 | INFO | crackz.project.image_operations:_import_with_split:412 - Importing 1 test images →
2026-05-06 10:21:48.076 | INFO | crackz.project.image_operations:_import_with_split:435 - Importing 1 test masks →
2026-05-06 10:21:48.080 | INFO | crackz.project.image_operations:_import_with_split:450 - Completed automatic split
2026-05-06 10:21:49.397 | ERROR | crackz.ai.model.checkpoints:select_checkpoint:109 - No checkpoint files found in
2026-05-06 10:21:53.939 | INFO | crackz.cli.detection_cli:callback:248 - Starting detection
2026-05-06 10:21:53.941 | INFO | crackz.cli.detection_cli:callback:248 - No trained model found. Detection skipped. ❌
2026-05-06 10:21:53.942 | WARNING | crackz.ai.detection.detection_runner:detect:538 - Detection skipped: no trained
2026-05-06 10:21:53.943 | INFO | crackz.cli.detection_cli:_do_detect:261 - ✅ Detection finished
2026-05-06 10:22:18.660 | ERROR | crackz.ai.model.checkpoints:select_checkpoint:109 - No checkpoint files found in
2026-05-06 10:22:18.790 | INFO | crackz_gui.qt.util.screenshot_capture:capture_doc_screenshots:398 - Loaded
2026-05-06 10:22:19.257 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved
2026-05-06 10:22:19.259 | ERROR | crackz.ai.model.checkpoints:select_checkpoint:109 - No checkpoint files found in
2026-05-06 10:22:19.633 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved
2026-05-06 10:22:19.980 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved
2026-05-06 10:22:20.323 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved
2026-05-06 10:22:20.683 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved
2026-05-06 10:22:21.065 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved
2026-05-06 10:22:21.426 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved
2026-05-06 10:22:21.787 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved
2026-05-06 10:22:22.934 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved
2026-05-06 10:22:23.308 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved
HIGH
MED
LOW
Logs viewer

1.
2.
3.

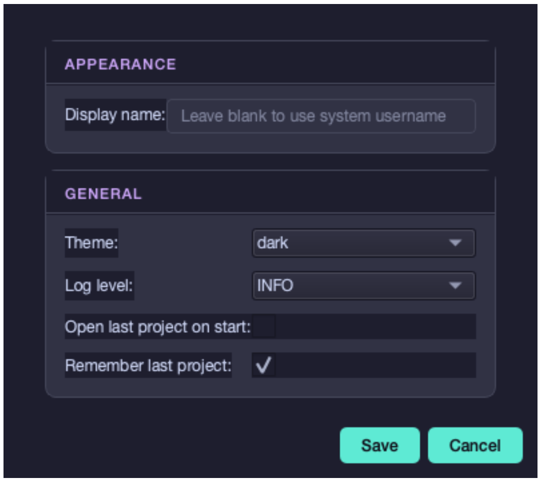

Data split:
raw
Refresh
Open in Browser
No GPS data found for split 'raw'.
Images must contain EXIF GPS metadata.
Close

1.
2.
3.
4.
5.
6.
7.



-
-
-
-
-

`Cmd/Ctrl+,`



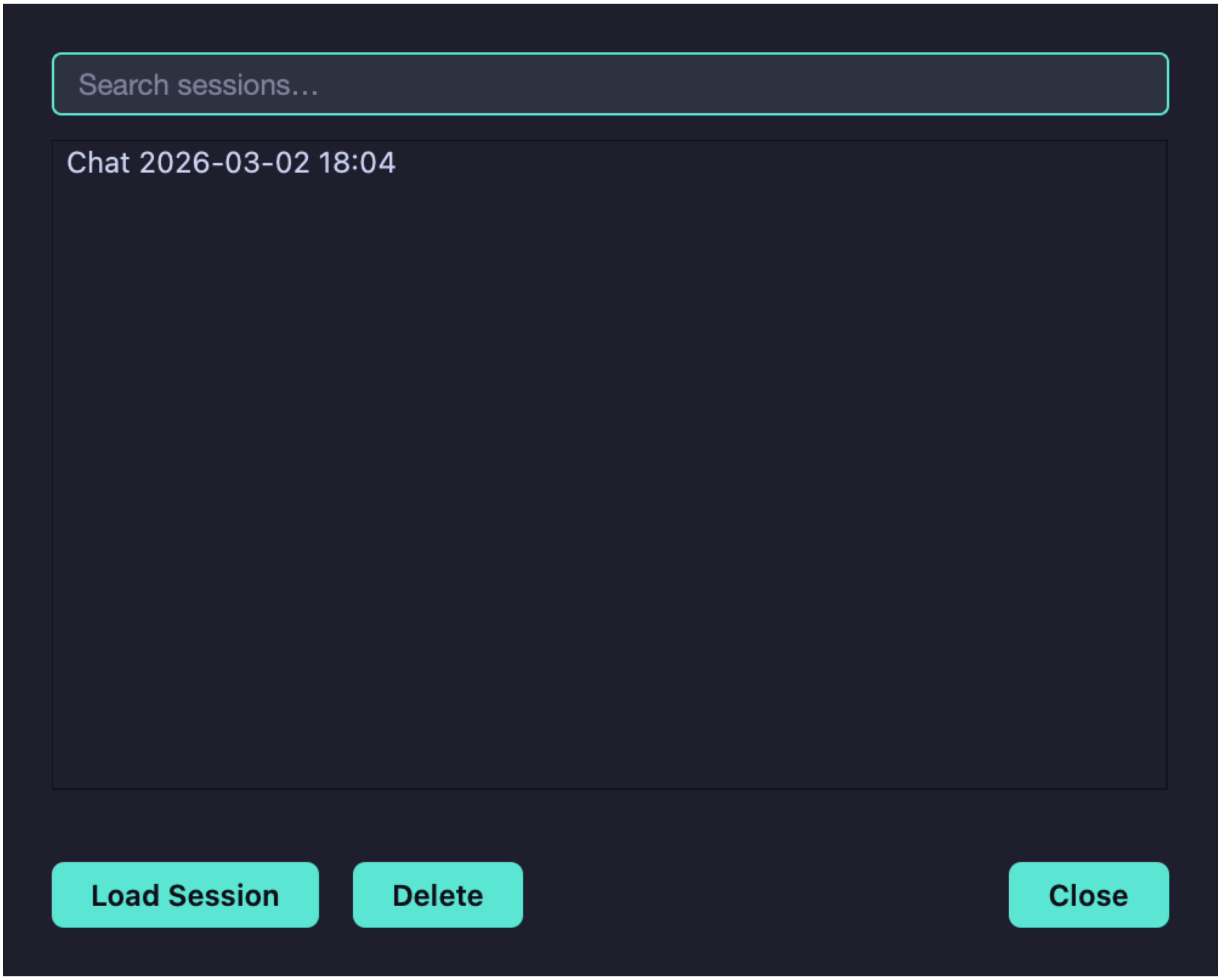
`dark` `light` `system`

`§` `~`

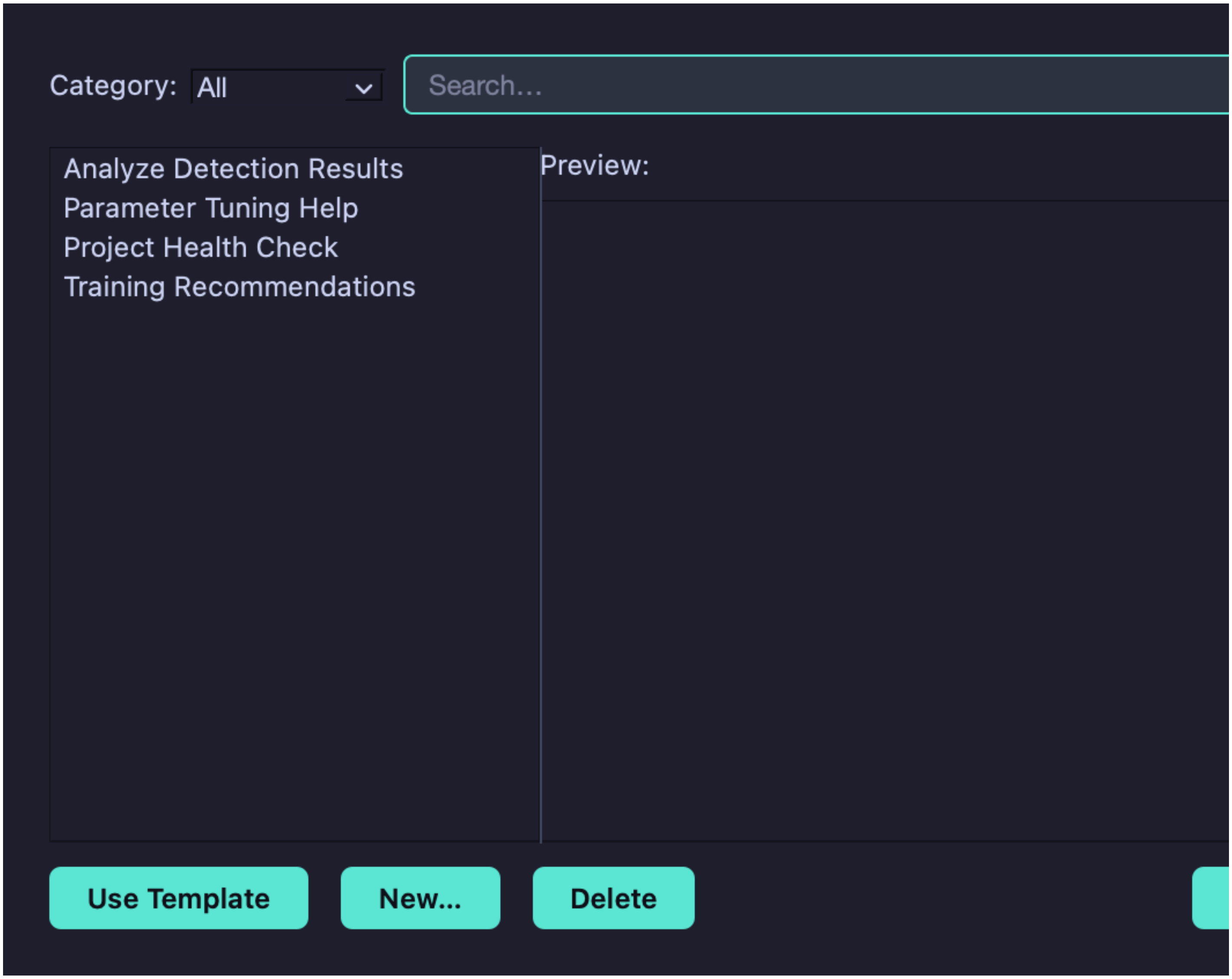
`§`

`~`

-


Search sessions…
Chat 2026-03-02 18:04
Load Session
Delete
Close

`§` `~` `Cmd/Ctrl+/`



`~/.crackz/chat_history.db`

```
# Check installation
python -c "from PySide6.QtWidgets import QApplication; print('PySide6 OK')"

# Launch with debug output
crackz-gui --debug
```

`CRACKZ_LOG_DIR/gui_app.log` `<project>/logs/gui_operations.log`

```
# Create and configure in GUI
crackz-gui

# Batch process via CLI
crackz detect run --project my-project --batch-size 16
```

```
# Review results in GUI
crackz-gui
```

AI Chat Assistant

AI Chat Assistant

```
# Install keyring support
uv add keyring

# Or with pip
pip install keyring
```

```
from crackz_gui.state.secure_storage import (
    store_api_key,
    retrieve_api_key,
    is_secure_storage_available,
)
```

```
# Check if keyring is available
if is_secure_storage_available():
    # Store API key securely
    store_api_key("your-api-key-here")

    # Retrieve API key
    api_key = retrieve_api_key()

    # Delete API key
    from crackz_gui.state.secure_storage import delete_api_key
    delete_api_key()
else:
    print("Keyring not available - falling back to plaintext storage")
```

2. `§`
3. `~`
4. `Ctrl+/` `Cmd+/`

- `sk-ant-...` `sk-or-...`

- `§`
- `~`
- `Ctrl+/` `Cmd+/`

```
You: My detection is finding too many false positives. How can I reduce them?

AI: Based on your project, I recommend:
1. Increase confidence_threshold from 0.25 to 0.4
2. Adjust iou_threshold to 0.5 for stricter overlap filtering
3. Consider retraining with more negative examples

Would you like me to explain how to adjust these in your project?
```

```
You: How many epochs should I train for?

AI: For your dataset with 150 training images:
- Start with 50 epochs as a baseline
- Monitor validation loss - it should decrease steadily
- If validation loss plateaus before 50 epochs, you can stop early
- If still improving at 50, try 75-100 epochs

Your current batch_size of 8 is good for this dataset size.
```

```
You: Analyze my project configuration

AI: Here's what I found in your oslo_project:
□ Good splits: 70/20/10 train/val/test distribution
⚠  Low image count (150 total) - consider data augmentation
□ Reasonable batch_size (8) for dataset
□ Suggestion: Enable data augmentation in training config
□ Suggestion: Consider collecting more diverse training data
```

```
You: How do I export detection results to CSV?

AI: To export detection results:
1. Run detection: `crackz detect run <project>`
2. Results are saved in project/artifacts/detections/
3. Use export command: `crackz export --format csv <project>`

Or in GUI:
1. Click "Run Detection" button
2. Results appear in Detection Results panel
3. Click "Export" → "CSV Format"
```

`claude-sonnet-4-5-20250929` `claude-haiku-4-5-20251001` `claude-opus-4-1-20250805`

`§` `~` `Ctrl+/`

`chat_window.py`

`ai_chat.py`

`chat_settings_dialog.py`

1. `ai_chat.py` `_get_context()`

```
def _get_context(self) -> str:
    if self.app_state is None or self.app_state.project is None:
```

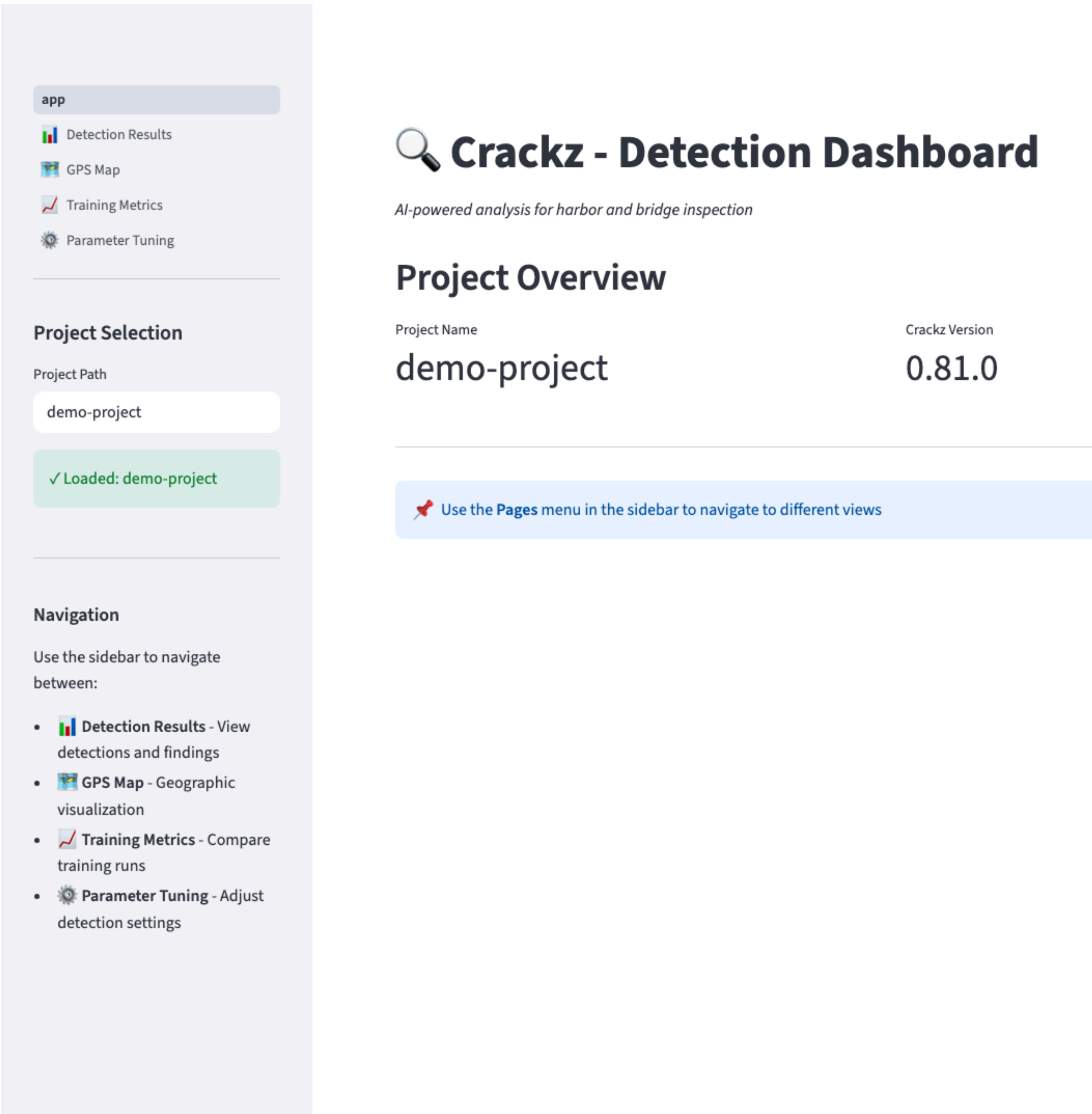
```
        return "No project loaded"

    project = self.app_state.project

    # Add your new context here
    return f"""
    Current project: {project.name}

    Your new context:
    - Additional info: {project.additional_info}
    """
```

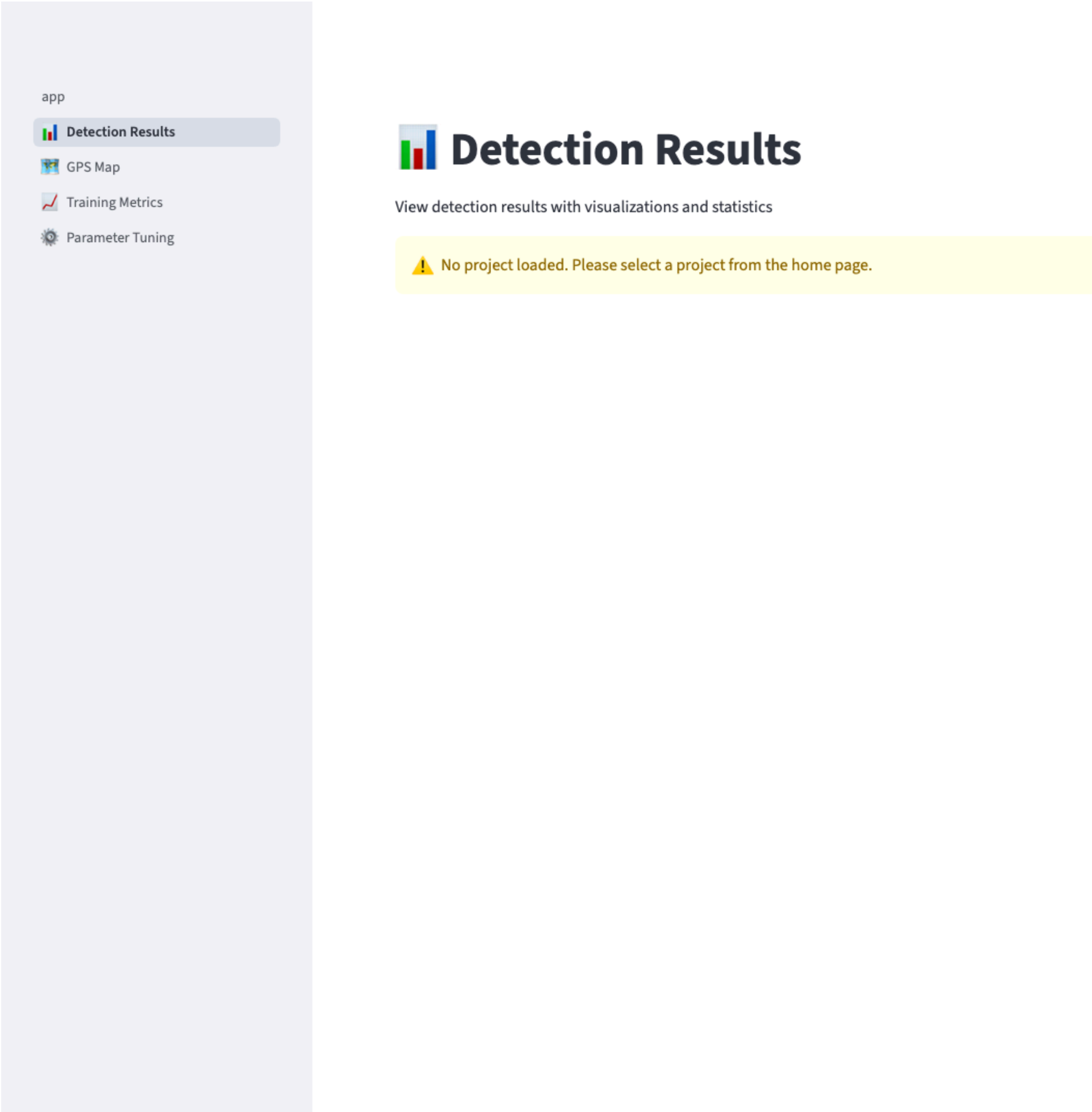
2.

```
uv run task tests
```

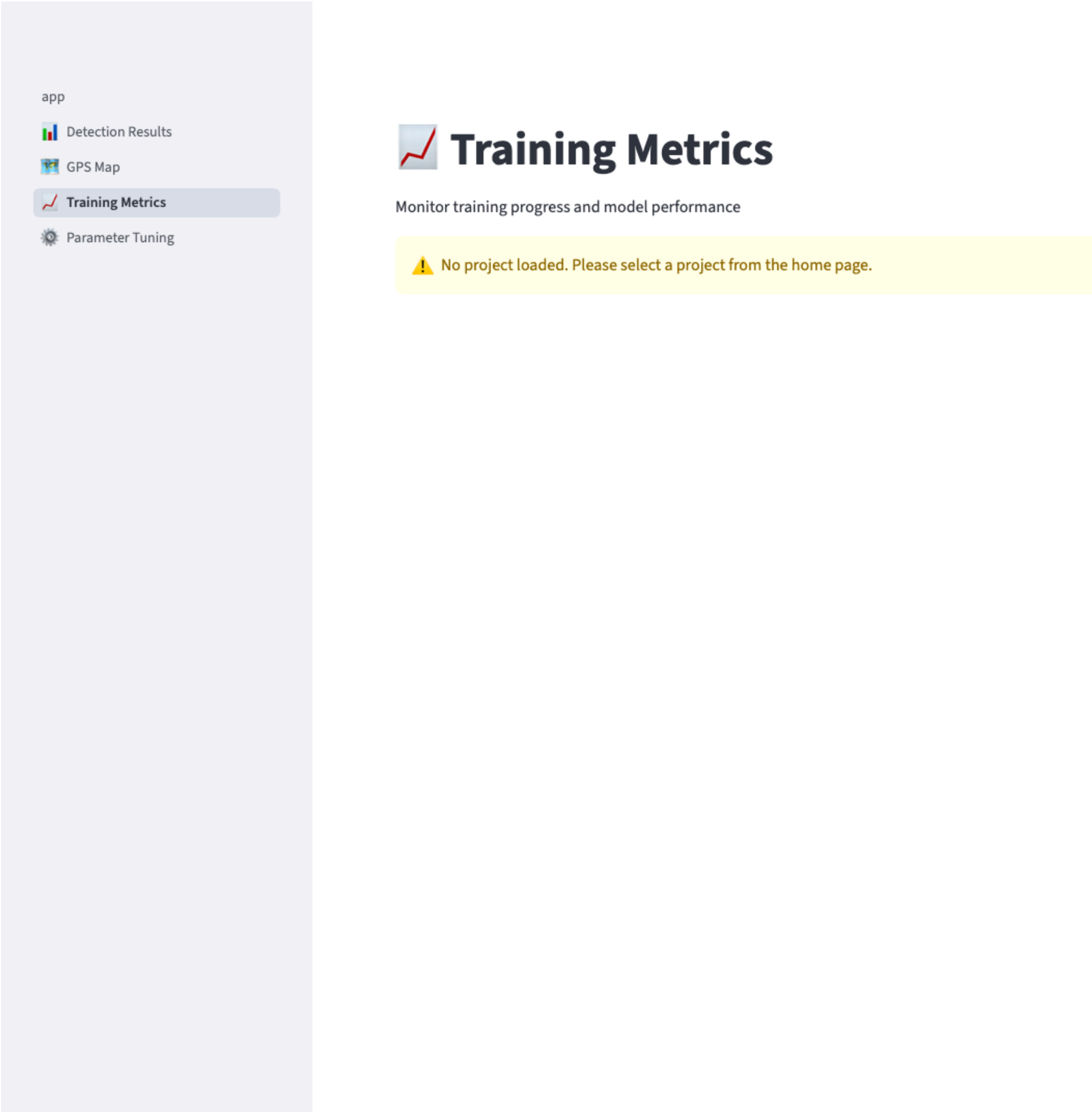
Dashboard

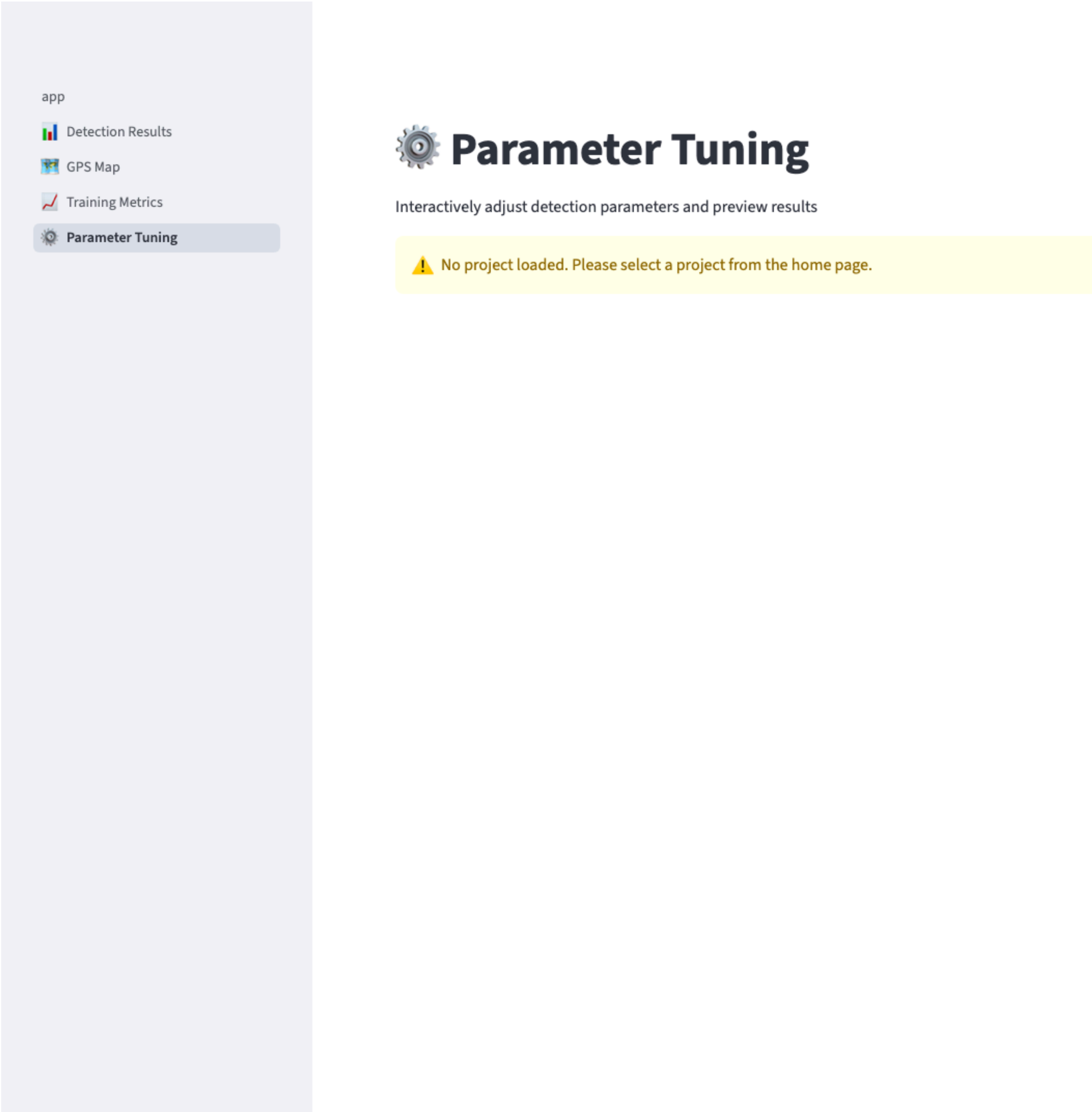
Streamlit Dashboard




app
Detection Results
GPS Map
Training Metrics
Parameter Tuning
Detection Results
View detection results with visualizations and statistics
No project loaded. Please select a project from the home page.


app
Detection Results
GPS Map
Training Metrics
Parameter Tuning
GPS Map Visualization
View detections and findings on an interactive map
No project loaded. Please select a project from the home page.


app
Detection Results
GPS Map
Training Metrics
Parameter Tuning
Training Metrics
Monitor training progress and model performance
No project loaded. Please select a project from the home page.


app
Detection Results
GPS Map
Training Metrics
Parameter Tuning
Parameter Tuning
Interactively adjust detection parameters and preview results
No project loaded. Please select a project from the home page.

```
# Launch with a specific project
crackz dashboard --project my-project

# Launch with custom port
crackz dashboard --project my-project --port 8502
```

1. `crackz-gui`
2.
3.
4.
5.
6.

```
streamlit run core/src/crackz/dashboard/app.py -- --project my-project
```

```
from crackz.dashboard import launch_dashboard

launch_dashboard(project_name="my-project", port=8501)
```

`crackz geo extract-gps --project <name>`

`artifacts/training/training_metrics.json`

`crackz train --project <name>`

```
core/src/crackz/dashboard/
├── __init__.py          # Launch function
├── app.py              # Main dashboard app
├── pages/              # Dashboard pages
│   ├── 1_□_Detection_Results.py
│   ├── 2_□_GPS_Map.py
│   ├── 3_□_Training_Metrics.py
│   └── 4_⚙_Parameter_Tuning.py
└── utils/              # Utility modules
    ├── data_loader.py  # Data loading functions
    ├── image_utils.py  # Image processing utilities
    ├── map_utils.py    # GPS mapping utilities
    └── viz_utils.py    # Visualization functions
```

	`artifacts/detections/visualizations/`
	`artifacts/detections/detection_results.json`
	`artifacts/training/training_metrics.json`
	`data/{test,val,train,raw}/`

```
crackz dashboard --project my-project --port 8080
```

`.streamlit/config.toml`

```
[server]
port = 8501
headless = true

[theme]
primaryColor = "#FF4B4B"
backgroundColor = "#FFFFFF"
secondaryBackgroundColor = "#F0F2F6"
```

```
textColor = "#262730"
font = "sans serif"
```

- `streamlit>=1.30.0`
- `streamlit-folium>=0.15.0`
- `folium>=0.15.0`
- `plotly>=5.18.0`
- `pandas>=2.0.0`

1.

   ```
   crackz detect --project my-project
   ```

2.
3.

1.

   ```
   crackz geo extract-gps --project my-project
   ```

2.
3.

1.
2.
3.

1.
2.
3.

`uv sync` `pip install streamlit` `--port 8502` `crackz detect --project <name>` `artifacts/detections/visualizations/`

`crackz geo extract-gps`

`crackz train --project <name>` `artifacts/training/training_metrics.json`

1.
2.
3.
4.

```
# 1. Initialize project
crackz init my-project

# 2. Import images
crackz import-images --source ./photos --project my-project

# 3. Train model
crackz train --project my-project

# 4. Run detection
crackz detect --project my-project

# 5. Launch dashboard to view results
crackz dashboard --project my-project
```

```
from crackz.dashboard import launch_dashboard

# Launch dashboard programmatically
launch_dashboard(
    project_name="my-project",  # Optional: project to load
    port=8501                   # Optional: server port
)
```

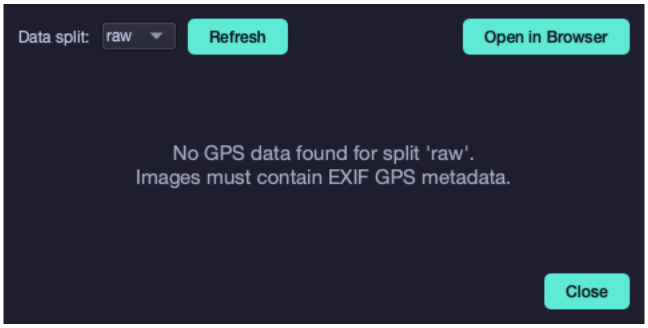
```
# Show help
crackz dashboard --help

# Launch with project
crackz dashboard --project my-project

# Launch with custom port
crackz dashboard --project my-project --port 8080

# Launch without initial project
crackz dashboard
```

```
# 1. Create and setup project
crackz init harbor-inspection
crackz import-images --source ./drone-photos --project harbor-inspection

# 2. Train model
crackz train --project harbor-inspection --epochs 50

# 3. View training progress
crackz dashboard --project harbor-inspection
# Navigate to "Training Metrics" page to review

# 4. Run detection on test set
crackz detect --project harbor-inspection --split test

# 5. Analyze results
crackz dashboard --project harbor-inspection
# Navigate to "Detection Results" page
# Navigate to "GPS Map" page to see locations

# 6. Tune parameters if needed
# Use "Parameter Tuning" page to optimize thresholds

# 7. Re-run with optimized parameters
crackz detect --project harbor-inspection --threshold 0.6

# 8. Generate final report
crackz report generate --project harbor-inspection --format pdf
```

3. `dashboard`

Map Viewer

Map Viewer Feature



- `gui/src/crackz_gui/components/map_viewer.py`
- `webbrowser.open()`

1.
2.
3.
4.
5.

```
from pathlib import Path
from crackz_gui.qt.views.map_viewer_view import MapViewerDialog
from crackz.project.project import load_project

# Load a project
project = load_project(Path("path/to/project"))

# Note: MapViewerDialog requires a parent GUI app instance and is typically
# used within the Crackz GUI application. For command-line access to GPS data, use:
from crackz.project.geoposition import get_project_geopositions

# Get GPS data for all images in the project
gps_data = get_project_geopositions(project)
for image_path, geo_pos in gps_data.items():
    print(f"{image_path}: {geo_pos.latitude}, {geo_pos.longitude}")
```

-
-
-

-
-
-

-
-
-

`tests/test_data/` `oslo_project/`

- `tests/test_data/images/`
- `oslo_project/images/`

`scripts/add_test_gps_data.py`

```
python scripts/add_test_gps_data.py
```

1.
2. `logs/crackz.log`
3.

-
- `crackz.project.geoposition.get_project_geopositions()`
- `python scripts/add_test_gps_data.py`

-
-
-

-
-
-

- `gui/src/crackz_gui/components/map_viewer.py`
- `gui/src/crackz_gui/components/browser/toolbar.py`
- `core/src/crackz/project/geoposition.py`
- `scripts/add_test_gps_data.py`

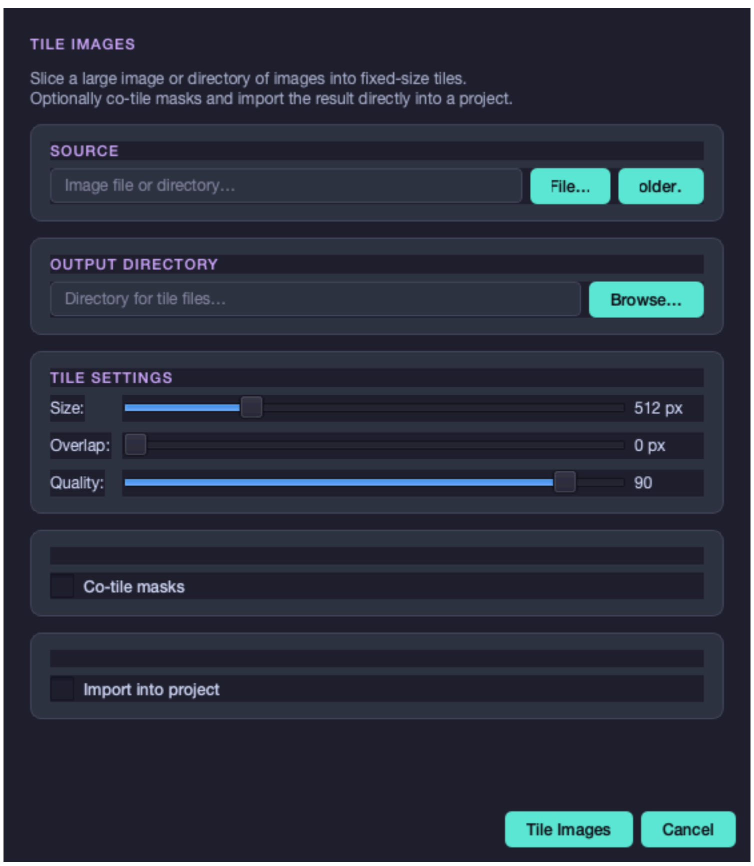
Image Tiling

Image Tiling

`crackz tile-image`

-
-
-
-
-

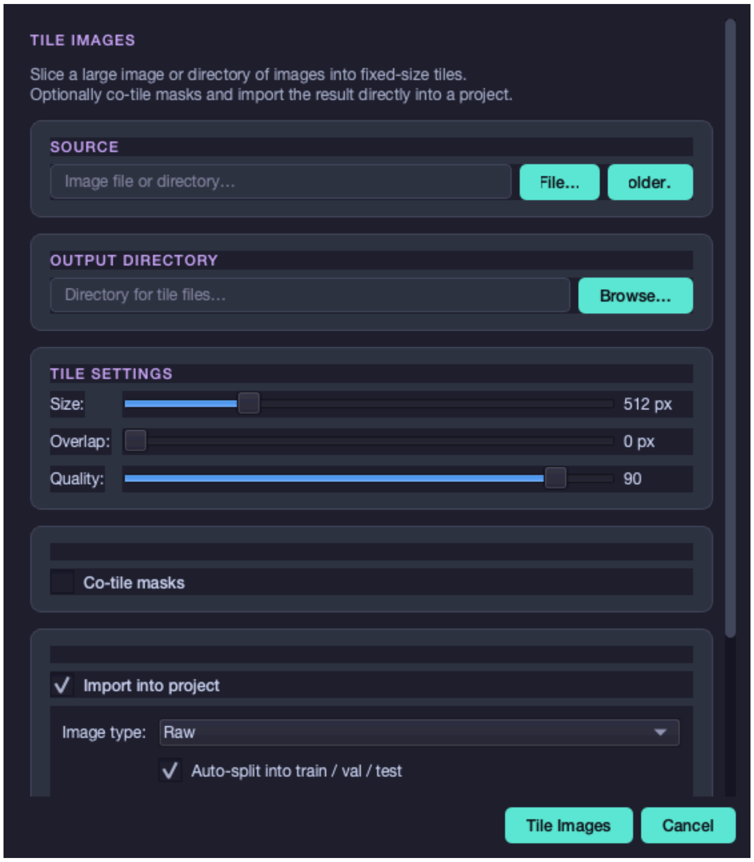
```
uv run python scripts/capture_qt_screenshots.py \
  --scenes tile-images-dialog,tile-images-dialog-masks,tile-images-dialog-import
```



- `2048 × 2048 px`
-

`≈ 16 tiles · 4 × 4 grid · 512 × 512 px each`

`size = 512` `overlap = 64`



```
crackz tile-image --src <source> --out <output> [OPTIONS]
```

```
crackz tile-image --src mosaic.jpg --out tiles/
```

```
crackz tile-image --src raw_images/ --out tiles/ --tile-size 256 --overlap 32
```

```
crackz tile-image \
 --src raw_images/ \
 --out tiles/images/ \
 --masks-src annotations/ \
 --masks-out tiles/masks/ \
 --tile-size 512
```

```
crackz tile-image \
 --src mosaic.jpg \
 --out tiles/ \
 --project ~/projects/harbor_survey \
 --type raw
```

```
crackz tile-image \
 --src raw_images/ \
 --out tiles/ \
 --masks-src annotations/ \
 --masks-out tiles/masks/ \
 --project ~/projects/harbor_survey \
 --split \
 --train-ratio 0.7 \
 --val-ratio 0.15 \
 --test-ratio 0.15 \
 --random-seed 42
```

`--src`	`-s`		
`--out`	`-o`		
`--masks-src`	`-ms`		
`--masks-out`	`-mo`		
`--tile-size`	`-z`	`512`	
`--overlap`		`0`	
`--quality`	`-q`	`90`	
`--project`	`-p`		
`--cfg`	`-c`		`--project`
`--type`	`-t`	`raw`	`raw` `train` `val` `test`
`--split`			
`--train-ratio`		`0.7`	`--split`
`--val-ratio`		`0.15`	`--split`
`--test-ratio`		`0.15`	`--split`
`--random-seed`			

`{stem}_{row:04d}_{col:04d}.{ext}`

`mosaic.jpg`

```
mosaic_0000_0000.jpg   mosaic_0000_0001.jpg
mosaic_0001_0000.jpg   mosaic_0001_0001.jpg
```

`.png`

```
mosaic_0000_0000.png   mosaic_0000_0001.png
mosaic_0001_0000.png   mosaic_0001_0001.png
```

```
step = max(tile_size - overlap, 1)
cols = max(ceil((width  - overlap) / step), 1)
```

```
rows = max(ceil((height - overlap) / step), 1)
n_tiles = cols × rows
```

```
step = 512 - 64 = 448
cols = ceil((4096 - 64) / 448) = ceil(9.0) = 9
rows = 9
n_tiles = 81
```

`.jpg` `.jpeg` `.png` `.tif` `.tiff` `.bmp` `.webp`	`.jpg`
`.png` `.jpg` `.jpeg` `.tif` `.tiff`	`.png`

`crackz.project.tiling`

```
from crackz.project.tiling import tile_image, tile_image_with_mask, tile_image_dir

# Tile a single image
tiles = tile_image(
    source=Path("mosaic.jpg"),
    dest_dir=Path("tiles/"),
    tile_size=(512, 512),
    overlap=0,
    quality=90,
)

# Tile image + mask together
img_tiles, msk_tiles = tile_image_with_mask(
    source=Path("mosaic.jpg"),
    mask=Path("mask.png"),
    dest_dir=Path("tiles/images/"),
    mask_dest_dir=Path("tiles/masks/"),
    tile_size=(512, 512),
    overlap=32,
    quality=90,
)

# Batch-tile a directory
img_tiles, msk_tiles = tile_image_dir(
    source_dir=Path("raw_images/"),
    dest_dir=Path("tiles/images/"),
    tile_size=(512, 512),
    overlap=0,
    quality=90,
    mask_source_dir=Path("annotations/"),  # optional
    mask_dest_dir=Path("tiles/masks/"),    # optional
)
```

Report Generation

Report Generation

```
crackz report generate --project demo-project
```

`demo-project_report.docx`

```
crackz report generate --project demo-project --output inspection_report.docx
```

```
crackz report generate --cfg project_config.yaml --output report.docx
```

```
ai:
  detection:
    min_confidence: 0.005      # 0.5% - Possible Crack threshold
    medium_confidence: 0.02    # 2% - Probable Crack threshold
    high_confidence: 0.05      # 5% - Crack Detected threshold
```

`detections/`

```
crackz report generate --project harbor_inspection_2024 \
  --output "Harbor_Inspection_Report_2024-Q4.docx"
```

```
# Monthly reports
crackz report generate --project monthly_monitoring \
  --output "reports/2024-11_monitoring.docx"
```

```
crackz report generate --project compliance_check \
  --output "compliance/inspection_$(date +%Y%m%d).docx"
```

```
crackz report generate --project qa_validation \
  --output "internal/qa_review.docx"
```

`crackz detect run`

```
crackz detect run --project demo-project
```

```
# Check for visualization images (new structure)
ls -la demo-project/artifacts/detections/visualizations/

# Or legacy location (older projects)
ls -la demo-project/artifacts/detections/*.png
```

```
crackz detect run --project demo-project
```

`artifacts/detections/visualizations/`

```
data:
  size: [512, 512]  # Use smaller images
```

`--output`

```
crackz report generate --project harbor_main \
  --output "Harbor_Main_Inspection_2024-11-05.docx"
```

```
for project in project1 project2 project3; do
  crackz report generate --project "$project" \
    --output "reports/${project}_$(date +%Y%m%d).docx"
done
```

```
crackz report generate --project demo-project  # Creates demo-project/demo-project_report.docx
```

```
# Add to project notes
project:
  notes: "Report generated 2024-11-05 with default thresholds"
```

CLI Reference

app

```
[OPTIONS] COMMAND [ARGS]...
```

```
 -v, --version         Show version and exit
 --debug               Enable debug logging
 --quiet               Only warnings and errors
 --log-dir DIRECTORY   Central log directory (precedence: env CRACKZ_LOG_DIR
                       > --log-dir > system temp). Created if missing.
                       Project logs are always under <project>/logs.
 --telemetry           Opt in to local telemetry: append structured events to
                       logs/events.jsonl. No image data, no PII. Also enabled
                       by CRACKZ_TELEMETRY_ENABLED=1. Off by default. See
                       ADR-021.
 --install-completion  Install completion for the current shell.
 --show-completion     Show completion for the current shell, to copy it or
                       customize the installation.
```

`dashboard`

```
crackz dashboard --project my-project
```

`http://localhost:8501`

```
crackz dashboard --project my-project --port 8080
```

```
crackz dashboard
```

1.
2.
3.
4.
5.
6.
7.
8.
9.
10.
11.
12.
13.
14.

15.
16.
17.
18.

```
crackz detect --project my-project
crackz dashboard --project my-project
# Navigate to "Detection Results" page
```

```
crackz train --project my-project --epochs 50
crackz dashboard --project my-project
# Navigate to "Training Metrics" page
```

```
crackz dashboard --project my-project
# Navigate to "GPS Map" page
```

```
crackz dashboard --project my-project
# Navigate to "Parameter Tuning" page
# Adjust thresholds and preview results
```

`clean`

```
crackz clean --project ./my_project
```

```
crackz clean -p ./my_project
```

- `*.log`
-
- `models/` `checkpoints/`
-
-

```
crackz clean --project ./my_project --dry-run
```

```
Dry run: The following would be cleaned:
  • /path/to/project/logs/crackz.log
  • /path/to/project/artifacts/training/tensorboard
  • /path/to/project/artifacts/models
  • /path/to/project/artifacts/detections

Re-run without --dry-run to apply changes.
```

```
crackz clean --project ./my_project --logs-only
```

`*.log`

`models/` `checkpoints/`

```
$ crackz clean --project ./my_project

Project cleaned! Removed 5 items (logs and artifacts).
```

`--logs-only`

```
$ crackz clean --project ./my_project --logs-only

Project cleaned! Removed 2 items (logs).
```

```
crackz clean -p ./my_project
```

```
crackz clean -p ./my_project --logs-only
```

```
crackz clean -p ./my_project --dry-run
```

`migrate`

`schema_version: 1` `ai.training.tensorboard` `false` `data.mask_threshold: 1` `crackz_version`

`checkpoints/` `models/` `training/tensorboard/` `runtime/tensorboard/` `training/lightning_logs/` `runtime/lightning_logs/` `detections/*.png` `detections/visualizations/`

```
crackz migrate --project ./my_project
```

```
crackz migrate --project ./my_project --migrate-files
```

```
crackz migrate --cfg ./my_project/my_project_project.yaml --migrate-files
```

```
crackz migrate -p ./my_project --migrate-files
crackz migrate -c ./my_project/my_project_project.yaml --migrate-files
```

```
# Schema changes only
crackz migrate --project ./my_project --dry-run

# Schema + file changes
crackz migrate --project ./my_project --migrate-files --dry-run
```

would

`.v{version}.{timestamp}.bak`

```
crackz migrate --project ./my_project --no-backup
```

```
[migrated] my_project schema 0 -> 1 (current=1) steps: 0->1: added tensorboard default + schema_version=1
```

```
[up-to-date] my_project schema 1 -> 1 (current=1) steps: none
```


`build.yml`	
`integration-tests.yml`	`pytest -m integration`
`release.yml`	
`docker-build.yml`	
`smoke-tests.yml`	
`benchmark-tests.yml`	`pytest -m benchmark`
`docs.yml`	

```
pytest -m integration -vv
```

```
pytest -m "not integration"
```

```
pytest -m benchmark --benchmark-only
```

`crackz/cli/*_cli.py`

`schema_version`

- `.bak`
- `CRACKZ_AUTO_MIGRATE=1`

```
crackz migrate --project <project-dir>
```

`--dry-run` `--no-backup`

`migration.md`

`model`

```
crackz model export --project ./my_project --out my_model.zip
```

```
crackz model export -p ./my_project -o my_model.zip
```

`<project_name>_model.zip`

```
crackz model export -p ./my_project -c final -o final_model.zip
```

`best` `last` `final`

```
crackz model import --project ./my_project --model my_model.zip
```

```
crackz model import -p ./my_project -m my_model.zip
```

```
crackz model import -p ./my_project -m my_model.zip --force
```

`--force`

```
# Export from training project
crackz model export -p ./training_project -o harbor_crack_v1.zip

# Import into deployment project
crackz model import -p ./deployment_project -m harbor_crack_v1.zip
```

```
# Export pre-trained model
crackz model export -p ./pretrained_project -o base_model.zip

# Import into new project for fine-tuning
crackz model import -p ./finetuning_project -m base_model.zip
```

```
# Export after successful training
crackz model export -p ./my_project -o backups/model_$(date +%Y%m%d).zip
```

`<checkpoint_name>.ckpt` `metadata.json`

```
{
  "project_name": "harbor_detection",
  "checkpoint_name": "best.ckpt",
  "model_config": {
    "encoder": "resnet34",
    "encoder_weights": "imagenet",
    "in_channels": 3,
    "num_classes": 1,
    "threshold": 0.5,
    "loss_name": "combo",
    "dice_weight": 0.7,
    "focal_weight": 0.3
  }
}
```

CLI Cookbook

CLI Cookbook

`crackz init --name demo-project`

```
demo-project/
  images/
    raw/ train/ val/ test/
  masks/
  artifacts/
  detections/
  checkpoints/
  project.yaml
```

- `crackz --help` `--help`
- `--project` `--cfg`
-
- `--debug` `--quiet`
-

```
crackz init --name demo-project
```

```
crackz init --name demo-project --force
```

```
# Preview what will be cleaned (safe)
crackz clean --project demo-project --dry-run

# Clean logs and artifacts
crackz clean --project demo-project

# Clean only logs, keep checkpoints and detections
crackz clean --project demo-project --logs-only
```

`*.log` `--logs-only`
`--logs-only` `--logs-only`

`--type`

```
crackz import-images \
  --project demo-project \
  --src ./samples \
  --type raw \
  --size 512 512 \
  --quality 90
```

```
crackz import-images --project demo-project --src ./samples --type train --size 512 512
crackz import-images --project demo-project --src ./samples --type val   --size 512 512
crackz import-images --project demo-project --src ./samples --type test  --size 512 512
```

```
crackz import-images \
 --project demo-project \
 --src ./samples \
 --split \
 --train-ratio 0.7 \
 --val-ratio 0.15 \
 --test-ratio 0.15 \
 --size 512 512
```

`--split`

`--split` `--type`

`--random-seed`

```
crackz import-images --project demo-project --src ./samples --split --random-seed 42
```

`--src/masks`

```
crackz import-images --project demo-project --src ./samples --masks_src ./samples/masks --type train
```

```
crackz import-images --project demo-project --src ./samples --type train --overwrite
```

`images/train` `val` `test` `raw/` `--type`
`train` `train/` `crackz detect train` `--split`

`scripts/import_masks.py`

```
# Basic usage - import from organized mask directories
python scripts/import_masks.py <project_dir> <masks_source_dir>

# Example: Import masks for skytteren_projekt from smasks directory
python scripts/import_masks.py ./skytteren_projekt ./smasks

# Custom size and quality
python scripts/import_masks.py ./my_project ./mask_sources --size 1024 1024 --quality 95
```

`masks_source_dir`

```
masks_source_dir/
 train/
   mask_001.jpg
   mask_002.png
   ...
 val/
   mask_101.jpg
   ...
 test/
   mask_201.png
   ...
```

`data.mask_threshold`
`_mask`

`--size WIDTH HEIGHT` `--quality QUALITY`

`data.mask_threshold`
`val/`

```
# Basic usage - edit an image (mask path auto-derived)
crackz masks edit images/train/crack_001.jpg

# With custom mask path
```

```
crackz masks edit photo.jpg --mask output/my_mask.png

# With project context (smart path derivation)
crackz masks edit images/raw/inspection.jpg --project my_harbor_project
```

`--mask`

1. `--project`

```
Image:  project/images/train/crack_001.jpg
Mask:   project/masks/train/crack_001.png  ← Auto-derived
```

2.

```
Image:  data/images/train/photo.jpg
Mask:   data/masks/train/photo.png  ← Auto-derived
```

3.

```
Image:  /tmp/photo.jpg
Mask:   /tmp/masks/photo.png  ← Auto-derived
```

`pyside6`

```
for img in images/train/*.jpg; do
    crackz masks edit "$img" --project my_project
done
```

```
ssh -X user@server
crackz masks edit /data/harbor/crack.jpg
```

```
# Convert all masks in a project (uses project threshold)
crackz masks convert --project my_harbor_project

# Convert with custom threshold
crackz masks convert --project my_harbor_project --threshold 128
```

-
- `train` `val` `test` `raw`
- `raw/`

1. `train` `val` `test` `image_01.jpg` `image_01.png`
2. `raw/`
3. `crackz detect train`
4. `crackz detect run --project <p>` `raw/`
5. `test`

`--type train`

`train` `val` `test`

1. `labelme` `labelme` `pip install labelme`
2.
3. `image_001.png` `images/train/image_001.jpg` `masks/train/image_001.png`
4. `masks/<split>/` `masks/train/` `masks/val/` `masks/test/`
5.
6.
7. `crackz import-images` `--masks_src` `masks/` `--split`

`cv2.threshold`

```
crackz detect run --project demo-project
```

`--project`

```
crackz detect run --cfg config/detect.yaml
```

`best.ckpt` `last.ckpt` `final.ckpt` `--model`

```
# Use specific checkpoint by name
crackz detect run --project demo-project --model best
crackz detect run --project demo-project --model last
crackz detect run --project demo-project --model final

# Use specific epoch checkpoint (e.g., epoch-0.9234.ckpt)
crackz detect run --project demo-project --model epoch-0.9234
```

--model best.ckpt last.ckpt
final.ckpt

--model best
--model last **--model epoch-X**

```
crackz detect train --project demo-project
```

train/val/test

```
crackz detect train --project demo-project --epochs 1 --batch-size 1
```

```
# Auto-selects best checkpoint, outputs to <project-name>_model.zip
crackz model export --project demo-project

# Specify output path
crackz model export --project demo-project --out models/harbor_v1.zip

# Export specific checkpoint (best, last, or final)
crackz model export --project demo-project --checkpoint best --out best_model.zip
```

```
crackz model export -p demo-project -o my_model.zip
crackz model export -p demo-project -c final -o final_model.zip
```

.ckpt

best.ckpt last.ckpt final.ckpt
<project_name>_model.zip --out

```
# Import model package
crackz model import --project demo-project --model harbor_v1.zip

# Overwrite existing checkpoint with same name
crackz model import --project demo-project --model harbor_v1.zip --force
```

```
crackz model import -p demo-project -m my_model.zip
crackz model import -p demo-project -m my_model.zip --force
```

checkpoints/

--force

```
# Team member A: Export trained model
crackz model export -p training-project -o shared/harbor_crack_v1.zip

# Team member B: Import into their project
crackz model import -p deployment-project -m shared/harbor_crack_v1.zip
```

```
# Export base model trained on general cracks
crackz model export -p general-cracks -o pretrained/base_model.zip

# Import into specialized project for fine-tuning
crackz model import -p harbor-specific -m pretrained/base_model.zip

# Fine-tune on harbor-specific data
crackz detect train -p harbor-specific --epochs 10
```

```
# Backup current best model before trying new hyperparameters
crackz model export -p demo-project -c best -o backups/before_experiment.zip

# Run experiment
crackz detect train -p demo-project --epochs 50

# If results are worse, can import the backup
crackz model import -p demo-project -m backups/before_experiment.zip --force
```

```
# Export with timestamp
crackz model export -p demo-project -o models/model_2025-10-11.zip

# Or use git-style versioning
crackz model export -p demo-project -o models/v1.0.0.zip
```

```
# Export from training environment
crackz model export -p training-project -o production/model.zip

# Transfer to production server and import
crackz model import -p production-project -m production/model.zip

# Run detection in production
crackz detect run -p production-project
```

`<checkpoint_name>.ckpt`
`metadata.json`

```
# Linux/Mac
unzip -l my_model.zip

# Windows PowerShell
Expand-Archive -Path my_model.zip -DestinationPath temp -Force
cat temp\metadata.json
```

`metadata.json`

```
{
  "project_name": "harbor_detection",
  "checkpoint_name": "best.ckpt",
  "model_config": {
    "encoder": "resnet34",
    "encoder_weights": "imagenet",
    "in_channels": 3,
    "num_classes": 1,
    "threshold": 0.5,
    "loss_name": "combo",
    "dice_weight": 0.7,
    "focal_weight": 0.3
  }
}
```

```
# Generate map for raw split (default)
crackz geo create-map --project demo-project --output harbor_map.html

# Map for specific split (train/val/test)
crackz geo create-map --project demo-project --output train_map.html --split train
```

```
crackz geo create-map -p demo-project -o map.html -s raw
```

```
# Export all images with GPS data
crackz geo export-geojson --project demo-project --output detections.geojson

# Export specific split
crackz geo export-geojson --project demo-project --output train.geojson --split train
```

```
crackz geo export-geojson -p demo-project -o data.geojson -s raw
```

```
{
  "type": "FeatureCollection",
  "features": [
    {
      "type": "Feature",
      "geometry": {
        "type": "Point",
        "coordinates": [longitude, latitude]
      },
      "properties": {
        "image_name": "IMG_0001.jpg",
        "image_path": "/path/to/image.jpg",
        "has_detection": true,
        "latitude": 59.123456,
        "longitude": 10.123456
      }
    }
  ],
  "metadata": {
    "project": "demo-project",
    "split": "raw",
    "total_images": 42,
    "images_with_detections": 15
  }
}
```

```
# Linux/Mac
exiftool image.jpg | grep GPS

# Python (Pillow)
from PIL import Image
img = Image.open("image.jpg")
exif = img.getexif()
gps_ifd = exif.get_ifd(0x8825)  # 0x8825 is GPS IFD tag
print(gps_ifd)
```

```
# 1. Import drone photos (GPS data preserved in EXIF)
crackz import-images --project harbor --src ./drone_photos --type raw --size 512 512

# 2. Run detection
crackz detect run --project harbor

# 3. Generate map to visualize results
```

```
crackz geo create-map --project harbor --output harbor_results.html

# 4. Export to GeoJSON for GIS analysis
crackz geo export-geojson --project harbor --output harbor.geojson

# 5. Open map in browser to review
# (or import GeoJSON into QGIS/ArcGIS)
```

```
crackz --debug detect run --project demo-project
crackz --quiet detect run --project demo-project
crackz --log-dir run_logs detect run --project demo-project
```

```
CRACKZ_LOG_DIR=logs_debug crackz detect run --project demo-project
```

`--debug`

```
# Start training
crackz detect train --project demo-project

# Press Ctrl+C to cancel at any time
# The operation will be cancelled gracefully
# Partial results (e.g., checkpoints) are preserved
```

```
crackz detect run --cfg configs/detect.yaml
crackz detect train --cfg configs/train.yaml
```

```
for P in demo-project other-project; do
  crackz detect run --project "$P" || echo "Failed: $P" >&2
done
```

`/home/app`

```
# Init project (host ./project mapped to /home/app)
mkdir -p project
docker run --rm -v %cd%\project:/home/app crackz-cli init --name demo-project

# Import images (mount samples to /data)
docker run --rm -v %cd%\project:/home/app -v %cd%\samples:/data \
  crackz-cli import-images --project /home/app/demo-project --src /data --type raw --size 512 512

# Detection
docker run --rm -v %cd%\project:/home/app crackz-cli detect run --project /home/app/demo-project
```

```
docker run --gpus all --rm -v %cd%\project:/home/app crackz-cli:gpu detect run --project /home/app/demo-project
```


`--project` `--cfg`

`--src`

`images/train` `raw`

`torch.cuda.is_available()`

```
# Generate report with default name (project_name_report.docx)
crackz report generate --project demo-project

# Specify custom output path
crackz report generate --project demo-project --output inspection_report.docx

# Use config file instead of project path
crackz report generate --cfg project.yaml --output report.docx
```

`crackz detect run`

```
# Find top 50 most uncertain images
uv run crackz active-learning run \
  --project demo-project \
  --top-n 50 \
  --method entropy \
  --out priorities.json
```

`entropy` `max_probability`
`variance` `mean_confidence`

```
uv run crackz active-learning info
```

```
# Prioritize images in 'raw' split (default)
uv run crackz active-learning run --project demo-project --split raw

# Prioritize images in 'train' split
uv run crackz active-learning run --project demo-project --split train
```

```
{
  "method": "entropy",
  "total_images": 500,
  "prioritized_count": 50,
  "images": [
    {
      "path": "image_042.jpg",
      "uncertainty_score": 0.8234,
      "rank": 1
    }
```

```
  ]
}
```

```
# Find similar images and propagate mask
uv run crackz propagate-masks run \
  --project demo-project \
  --reference wall_01.jpg \
  --mask wall_01_mask.png \
  --split raw \
  --threshold 0.8 \
  --out ./propagated_masks
```

`--threshold` `--min-matches` `--sift` `--orb`

```
uv run crackz propagate-masks find-similar \
  --project demo-project \
  --reference wall_01.jpg \
  --split raw \
  --out similarities.json
```

```
# More permissive matching (more propagations, less accurate)
uv run crackz propagate-masks run \
  --reference ref.jpg \
  --mask ref.png \
  --project demo-project \
  --threshold 0.6 \
  --min-matches 5

# Stricter matching (fewer propagations, more reliable)
uv run crackz propagate-masks run \
  --reference ref.jpg \
  --mask ref.png \
  --project demo-project \
  --threshold 0.9 \
  --min-matches 20 \
  --sift
```

```
uv run crackz propagate-masks run \
  --reference ref.jpg \
  --mask ref.png \
  --project demo-project \
  --orb
```

- `api.md` `detect()` `train()`

  `crackz export-config`

Shell Completion

Shell Completion

`crackz`

-
-
-
-

```
crackz --show-completion bash
crackz --show-completion zsh
crackz --show-completion fish
crackz --show-completion powershell
```

`source` `Invoke-Expression`

```
crackz --install-completion SHELL
```

`SHELL` `bash` `zsh` `fish` `powershell`

```
crackz det<TAB>
```

```
crackz detect
```

```
crackz --install-completion bash
# New shell OR
source ~/.bash_completion
```

`bash-completion` `--show-completion`

```
crackz --install-completion zsh
# Then restart shell or:
autoload -U compinit && compinit
```

`$fpath` `fpath`

```
crackz --install-completion fish
# Typically installs into ~/.config/fish/completions/
# Reload:
exec fish
```

```
pwsh -File scripts/setup_completion.ps1
# Or force re-install:
pwsh -File scripts/setup_completion.ps1 -Force
```

`crackz` `crackz --install-completion powershell`

```
crackz --show-completion powershell | Out-String | Invoke-Expression
```

`. $PROFILE`

```
crackz --install-completion powershell
# If the profile wasn't updated, add manually:
Add-Content $PROFILE "crackz --show-completion powershell | Out-String | Invoke-Expression"
. $PROFILE
```

	`crackz --show-completion bash > /tmp/crackz.sh && source /tmp/crackz.sh`
	`source <(crackz --show-completion zsh)`
	`crackz --show-completion fish | source`
	`crackz --show-completion powershell | Out-String | Invoke-Expression`

		`. $PROFILE`
		`uv run crackz --install-completion ...` `uv run crackz --show-completion ...`
	`--` `Force`	`-Force`
		`autoload -U compinit && compinit`
`_completion_loader`		`--show-completion`

`~/.bashrc` `~/.zshrc` `$PROFILE` `~/.config/fish/completions/`

```
crackz --show-completion bash > .crackz_comp.sh
source .crackz_comp.sh
# run tests that exercise completion
```

Project Configuration

Configuration Reference

`crackz init`

```
CLI flag > ENV var > Project YAML > Code default
```

`crackz init --name demo-project` `demo-project/_project.yaml`
`<name>_project.yaml`

```yaml
schema_version: 6
project:
  name: demo-project
  description: Demo project
  created: 2025-01-01T12:00:00+00:00
annotation:
  background:
    id: 0
    name: Background
    color: "#000000"
  categories:
    - id: 1
      key: crack
      name: Crack
      color: "#ef4444"
paths:
  project_path: demo-project
  image_path: demo-project/images
  image_mask_path: demo-project/masks
  artifacts_path: demo-project/artifacts
logging:
  level: INFO
  log_dir: logs
ai:
  model:
    backbone: segformer-b2
    active_checkpoint: best
    device: cpu
    in_channels: 3
    input_size: [1024, 1024]
    threshold: 0.5
    loss_name: combo
    dice_weight: 0.7
    focal_weight: 0.3
  training:
    batch_size: 8
    num_workers: 4
    epochs: 10
    learning_rate: 0.001
    shuffle: true
    augmentation: true
  detection:
    threshold: 0.5
    batch_size: 8
    output_dir_name: detections
data:
  mask_threshold: 50
  image_quality: 90
  target_size: 1024
```

`crackz migrate`

`--project <path>`

`ProjectConfig`

```python
from pathlib import Path
from crackz.config.project_config import ProjectConfig
```

```
pc = ProjectConfig(
    name="demo-project",
    project_path=Path("demo-project"),
    image_path=Path("demo-project/images"),
    image_mask_path=Path("demo-project/masks"),
    artifacts_path=Path("demo-project/artifacts"),
)
pc.save(Path("project_config.yaml"))

# Later
loaded = ProjectConfig.load(Path("project_config.yaml"))
project = loaded.to_project()
```

`Project`

```
from crackz.config.project_config import ProjectConfig
pc = ProjectConfig.from_project(project)
pc.save(Path("backup_config.yaml"))
```

	`ModelConfig`	
	`TrainingConfig`	
	`DetectionConfig`	

`binary` `multiclass`

`num_classes` `num_classes` `1 + num_categories`

`backbone` `encoder` `encoder_weights` `backbone` `segmentation-models-pytorch` `segformer-b0` `segformer-b5`

`segformer-b0`				
`segformer-b1`				
`segformer-b2`				
`segformer-b3`				
`segformer-b4`				
`segformer-b5`				

`segmentation-models-pytorch`

`efficientnet-b0`				
`efficientnet-b3`				
`efficientnet-b7`				
`resnet18`				
`resnet34`				
`resnet50`				
`mobilenet_v2`				

`imagenet` `ssl`
`swsl` `None` `null`

Binary Segmentation — SegFormer (Default)

```
ai:
  model:
    backbone: segformer-b2   # Default SegFormer model
    input_size: [1024, 1024]
    threshold: 0.5
annotation:
  categories:
    - id: 1
      key: crack
      name: Crack
      color: "#ef4444"
```

Multi-Class Segmentation — Harbor Defects

```
annotation:
  categories:
    - id: 1
      key: crack
      name: Crack
      color: "#ef4444"
    - id: 2
      key: spall
      name: Spall
      color: "#f59e0b"
    - id: 3
      key: corrosion
      name: Corrosion
      color: "#3b82f6"
ai:
  model:
    backbone: segformer-b2
    # num_classes is derived automatically: 1 + 3 categories = 4 output channels
```

Binary Segmentation — U-Net with EfficientNet (legacy or fallback)

```
ai:
  model:
    backbone: efficientnet-b3   # U-Net decoder, EfficientNet-B3 encoder
    input_size: [512, 512]
    threshold: 0.5
```

Grayscale Input

```
ai:
  model:
    backbone: segformer-b2
    in_channels: 1          # Grayscale images
    normalization:
      mean: [0.5]
      std: [0.5]
```

Force binary task mode with multiple annotation categories

```
ai:
  model:
    backbone: segformer-b2
    task_mode: binary       # All categories collapse into one defect-vs-background class
```


`TrainingConfig`

`int` `int | str`

`tensorboard`
`artifacts/runtime/tensorboard/` `False`
`getattr(project.ai.training, "tensorboard", False)`

```
# Safe access pattern inside runtime code
use_tb = getattr(project.ai.training, "tensorboard", False)
if use_tb:
    ... # create logger
```

`getattr(..., default)`

`tests/unit/crackz/ai/test_detection.py::test_tensorboard_logger_disabled_by_default`
`False` `model_fields`

`TrainingConfig`

`min_confidence` `medium_confidence` `high_confidence`

-
-
-


```yaml
ai:
 detection:
   min_confidence: 0.005     # Save detections with ≥0.5% crack area
   medium_confidence: 0.02   # "Probable Crack" at ≥2% crack area
   high_confidence: 0.05     # "Crack Detected" at ≥5% crack area
   min_crack_area_px: 100    # Suppress detections smaller than 100 pixels
```

`data`

	`DataConfig`	

`mask_threshold`

-
-
-


```yaml
data:
 mask_threshold: 50  # Default - balanced filtering
```

`crackz import-images --masks_src <path>` `project.import_images()` `scripts/import_masks.py`

`crackz migrate` `mask_threshold: 50`

`annotation`

```yaml
annotation:
  background:
    id: 0
    name: Background
    color: "#000000"
  categories:
    - id: 1
      key: crack
      name: Crack
      color: "#ef4444"
    - id: 2
      key: spall
      name: Spall
      color: "#f59e0b"
    - id: 3
      key: corrosion
      name: Corrosion
      color: "#3b82f6"
```


		`1..N`
		`crack` `spall`
		`Crack` `Spall`
		`#RRGGBB`

-
- `1, 2, 3, …N`
-
- `1`
- `1 + N`
- **`task_mode: binary`** `ai.model`

`crack-only`		
`harbor-defects`		
`survey-general`		

```
# CLI: initialize a project with a harbor-defects annotation template
crackz init --name harbor-project --annotation-template harbor-defects
```

```
crackz export-config --project demo-project --out tuned_config.yaml
```

```
from crackz.config.project_config import ProjectConfig
pcfg = ProjectConfig.from_project(project)
pcfg.save(Path("tuned_config.yaml"))
```

```
CRACKZ_LOG_DIR=run_logs crackz detect run --project demo-project
```

```
$env:CRACKZ_LOG_DIR='run_logs'; crackz detect run --project demo-project
```

`ai.model.device`
`resolve_device_and_accelerator` `ai.model.device:`
`cuda` `python -c "import torch; print(torch.cuda.is_available())"`

`tensorboard: true`

```
ai:
  training:
    tensorboard: true
    epochs: 10
    batch_size: 8
```

```
from crackz import Project

project = Project.load(Path("demo-project"))
project.ai.training.tensorboard = True
project.save()
```

```
<artifacts_path>/runtime/tensorboard/
```

```
tensorboard --logdir demo-project/artifacts/runtime/tensorboard
```

```
docker run --rm -v $(pwd)/demo-project:/workspace -p 6006:6006 crackz-cli \
  tensorboard --logdir /workspace/artifacts/runtime/tensorboard
```

	`crackz detect run --project p --cfg cfg.yaml`
	`training.epochs` `batch_size`
	`detection.batch_size`

`<artifacts_path>/models/` `best.ckpt` `last.ckpt`
`epoch={N}-val_iou={score}.ckpt` `epoch=9-val_iou=0.8234.ckpt`

`<artifacts_path>/checkpoints/`

```
ai:
  training:
    save_top_k: 1          # Keep N best checkpoints (by val_iou), default: 1
    save_last: true        # Always save last.ckpt, default: true
    auto_prune_checkpoints: false  # Auto-prune after training, default: false
    checkpoint_keep: 3     # Checkpoints to keep when auto-pruning, default: 3
```

1. `save_top_k: 1, save_last: true`
2. `save_top_k: 3, auto_prune_checkpoints: true, checkpoint_keep: 3`
3. `save_top_k: 1, save_last: false, auto_prune_checkpoints: true, checkpoint_keep: 1`

```
crackz detect prune-checkpoints --project demo-project --keep 3
```

```
docker run --gpus all --rm -v %cd%\project:/workspace crackz-cli:gpu \
  detect prune-checkpoints --project /workspace/demo-project --keep 2
```

```
from crackz.ai.checkpoints import prune_checkpoints
removed = prune_checkpoints(project, keep=2)
```

`best.ckpt` `003-0.8234.ckpt` `last.ckpt`

```
<artifacts_path>/detections/detection_results.json
```

```
from crackz.ai.results import load_detection_results
from crackz.ai.eval import basic_summary
results = load_detection_results(project)
print(basic_summary(results))
```

`crack_severity`

`seed`

```yaml
# project_config.yaml (unified example)
name: demo-project
description: Example project
project_path: demo-project
image_path: demo-project/images
image_mask_path: demo-project/masks
artifacts_path: demo-project/artifacts
logging:
  level: INFO
  log_dir: logs
model:
  encoder: resnet34
  device: cpu
  input_size: [512, 512]
training:
  epochs: 5
  batch_size: 4
  learning_rate: 0.0005
  tensorboard: false  # Set to true to enable TensorBoard logging
detection:
  threshold: 0.55
  batch_size: 4
```

`schema_version`

```
crackz migrate --project ./my_project
```

`schema_version` `schema_version: 1` `tensorboard: false` `.bak`

```
crackz migrate --cfg ./my_project/my_project_project.yaml
```

```
crackz migrate --project ./my_project --dry-run
```

```
crackz migrate --project ./my_project --no-backup
```

`up-to-date` `noop`

```
for d in projects/*; do
  [ -d "$d" ] && crackz migrate --project "$d" --no-backup;
done
```

`ProjectConfig`

Training Configuration

TrainingConfig Contract

`TrainingConfig` `crackz.project.config` `ai.training`

`batch_size`			
`num_workers`			
`epochs`			
`learning_rate`			
`shuffle`			
`loss`			
`metrics`			
`seed`			
`precision`			
`devices`			
`strategy`			
`tensorboard`			`artifacts/training/tensorboard`

`tests/unit/crackz/project/test_training_config_schema.py`
`TrainingConfig`

1.
2.
3.
4. `CURRENT_SCHEMA_VERSION`

1.
2.
3.
4.

`--epochs` `--batch-size`

`tensorboard=True` `<project>/artifacts/training/tensorboard`

`seed`

Data Splits & Import

Data Splits and Import Guide

`images/raw/`

`raw/`

```
crackz import-images --project my-project --src ./photos --type raw --size 512 512
```

`images/train/` `masks/train/`

`img001.jpg` `img001.png`

`images/val/` `masks/val/`

`images/test/` `masks/test/`

`--type raw`

```
crackz import-images \
  --project my-project \
  --src ./labeled-data/images \
  --src-masks ./labeled-data/masks \
```

```
  --split-ratios 0.7 0.15 0.15 \
  --size 512 512
```

```
crackz import-images \
  --project my-project \
  --src ./labeled-data/images \
  --src-masks ./labeled-data/masks \
  --split-ratios 0.7 0.15 0.15 \
  --seed 42 \
  --size 512 512
```

```
# Small dataset: larger validation set
crackz import-images \
  --project my-project \
  --src ./small-dataset/images \
  --src-masks ./small-dataset/masks \
  --split-ratios 0.6 0.25 0.15 \
  --size 512 512

# Large dataset: more for training
crackz import-images \
  --project my-project \
  --src ./large-dataset/images \
  --src-masks ./large-dataset/masks \
  --split-ratios 0.8 0.1 0.1 \
  --size 512 512
```

1.
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24.
25.
26.

-
-
-
-

```
crackz init --name harbor-inspection
cd harbor-inspection
```

```
crackz import-images \
  --project . \
  --src ~/datasets/harbor-cracks/images \
  --src-masks ~/datasets/harbor-cracks/masks \
  --split-ratios 0.7 0.15 0.15 \
  --seed 42 \
  --size 512 512
```

`images/train/` `images/val/` `images/test/`
`masks/`

```
# Count images
ls images/train/ | wc -l   # Should show 140
ls images/val/ | wc -l     # Should show 30
ls images/test/ | wc -l    # Should show 30

# Count masks (should match images)
ls masks/train/ | wc -l    # Should show 140
ls masks/val/ | wc -l      # Should show 30
ls masks/test/ | wc -l     # Should show 30
```

```
crackz detect train --epochs 50 --batch-size 8
```

`images/train/` `masks/train/`
`images/val/` `masks/val/`

`artifacts/metrics.json` `artifacts/loss_curve.png` `checkpoints/best.ckpt`

```
crackz detect run --split test
```

`images/test/`
`masks/test/`

`detections/`

```
# Import new unlabeled images
crackz import-images \
  --project . \
  --src ~/new-harbor-photos \
  --type raw \
  --size 512 512

# Run detection
crackz detect run --split raw
```

`detections/`

1.

```
--seed 42  # Same split every time
```

2.

```
cp -r original-data backup/
```

3.

```
# Check if splits are representative
ls images/train/ | wc -l
ls images/val/ | wc -l
```

4.

```
# For 50 images: 0.6, 0.25, 0.15 ensures val gets ~13 images
--split-ratios 0.6 0.25 0.15
```

5.

```
# In project documentation
split_info:
  seed: 42
  ratios: [0.7, 0.15, 0.15]
  date: 2025-10-19
```

1.
2.
3.
4.
5.
6.

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-
-
-
-

```
# Wrong: 0.7 + 0.2 + 0.2 = 1.1 ×
# Right: 0.7 + 0.15 + 0.15 = 1.0 
--split-ratios 0.7 0.15 0.15
```

`images/val/`

```
# If you have 10 images and use 0.7/0.15/0.15:
# train=7, val=1, test=2 (works)

# If you have 5 images and use 0.8/0.1/0.1:
# train=4, val=0, test=1 (val is empty!)

# Solution: Use larger val ratio or more images
--split-ratios 0.6 0.25 0.15  # Ensures val gets at least 1
```

```
# Image: images/crack_001.jpg
# Mask must be: masks/crack_001.png (same stem)

# Check your filenames:
ls images/ > image_list.txt
ls masks/ > mask_list.txt
# Compare the stems (filename without extension)
```

```
# Add --seed for reproducible splits
--seed 42
```

```
# Increase validation ratio
--split-ratios 0.65 0.25 0.1  # Gives more to validation
```

```
# Import with smaller size
--size 256 256  # Instead of 512 512

# Or reduce batch size during training
crackz detect train --batch-size 2  # Instead of 8
```

```
# Separate images by class first
mkdir temp/crack temp/no-crack

# Move files to class folders
# ... organize manually or with script ...

# Import each class separately with same ratio
crackz import-images --src temp/crack --split-ratios 0.7 0.15 0.15 --seed 42
crackz import-images --src temp/no-crack --split-ratios 0.7 0.15 0.15 --seed 42
```

```
# Import full dataset to train
crackz import-images --src data --type train --size 512 512

# Manually create folds by moving files between train/val
# Train 5 models with different train/val splits
# Average results for robust performance estimate
```

```
# Import new batch with same seed and ratios
crackz import-images \
  --src new-data \
  --split-ratios 0.7 0.15 0.15 \
  --seed 42 \
  --size 512 512

# WARNING: This adds to existing splits
# Consider fresh re-import of all data for clean splits
```

	`images/raw/`				
	`images/train/`				
	`images/val/`				
	`images/test/`				

```
# Import for training (with split)
crackz import-images --src data/images --src-masks data/masks \
  --split-ratios 0.7 0.15 0.15 --seed 42 --size 512 512

# Import for detection (no split)
crackz import-images --src photos --type raw --size 512 512

# Train on train split, validate on val split
crackz detect train --epochs 50 --batch-size 8

# Test on test split
crackz detect run --split test

# Detect on raw split
crackz detect run --split raw
```

Docker

Docker Usage

```
# Pull from registry
docker pull ghcr.io/anaxiatech-ab/crackz:latest
docker pull ghcr.io/anaxiatech-ab/crackz:slim
docker pull ghcr.io/anaxiatech-ab/crackz:gpu

# Run from registry
docker run --rm ghcr.io/anaxiatech-ab/crackz:latest --help
```

`latest`	`Dockerfile`		
`slim`	`Dockerfile.slim`		
`gpu`	`Dockerfile.gpu`		

```
# Full
docker build -t crackz-cli .
# Slim
docker build -f Dockerfile.slim -t crackz-cli:slim .
# GPU (CUDA 12.8)
docker build -f Dockerfile.gpu -t crackz-cli:gpu .
```

```
docker build --build-arg PYTHON_VERSION=3.13 -t crackz-cli .
```

1. `ghcr.io/astral-sh/uv:bookworm-slim` `uv build --wheel`
2.
3. `python:3.13-slim`
4. `nvidia/cuda:12.8.0-runtime-ubuntu22.04`
5.
6. `--build-arg TORCH_INDEX_URL=...`

- `uv build`
-
-
-

- `nvidia/cuda:12.8.0-runtime-ubuntu22.04`
-
- `git` `libglib2.0-0` `libjpeg-turbo8` `libgomp1` `ca-certificates`
-
-

```
# Default: CUDA 12.8 (recommended, matches PyTorch cu128)
docker build -f Dockerfile.gpu -t crackz-cli:gpu .

# Override for CUDA 12.1
docker build -f Dockerfile.gpu \
  --build-arg TORCH_INDEX_URL=https://download.pytorch.org/whl/cu121 \
  -t crackz-cli:gpu-cu121 .

# Override for CUDA 12.4 (if needed for compatibility)
docker build -f Dockerfile.gpu \
  --build-arg TORCH_INDEX_URL=https://download.pytorch.org/whl/cu124 \
  -t crackz-cli:gpu-cu124 .
```

`TORCH_INDEX_URL`

```
#18 [runtime 7/9] RUN /app/.venv/bin/python3 -c 'import torch; print("Torch:", torch.__version__, "CUDA available:", torch.cuda.is_available())'
#18 0.699 Torch: 2.8.0+cpu CUDA available: False
```

`CUDA available: False`

`linux/amd64`

`/home/app`

```
# Help
docker run --rm crackz-cli --help

# Init project (creates /home/app/demo-project inside container)
mkdir -p project
docker run --rm -v "$(pwd)/project:/home/app" crackz-cli init --name demo-project

# Import images (mount samples to /data)
mkdir -p samples
# (populate samples/ with images and optional masks/ first)
docker run --rm \
  -v "$(pwd)/project:/home/app" \
  -v "$(pwd)/samples:/data" \
  crackz-cli import-images --project /home/app/demo-project --src /data --type raw --size 512 512

# Detection
docker run --rm -v "$(pwd)/project:/home/app" crackz-cli detect run --project /home/app/demo-project
```

```
docker run --gpus all --rm -v "$(pwd)/project:/home/app" crackz-cli:gpu detect run --project /home/app/demo-project
```

`/home/app`		`project/`
`/data`		`samples/`

`/app` `/home/app`

`CRACKZ_LOG_DIR`		`logs`
`PYTHONUNBUFFERED`		`1`

`/home/app/central_logs`

```
docker run --rm -e CRACKZ_LOG_DIR=central_logs -v "$(pwd)/project:/home/app" \
  crackz-cli detect run --project /home/app/demo-project
```

`/home/app`

```
make build        # full image
make build-slim   # slim image
make build-gpu    # gpu image (CUDA 12.8)
make init NAME=demo-project              # creates project under /home/app
make import PROJECT=demo-project DATA_DIR=./samples
make detect PROJECT=demo-project
make detect-gpu PROJECT=demo-project     # GPU detection
make train-gpu PROJECT=demo-project      # GPU training
```

`act`

```
# Install act (macOS)
brew install act

# Test GPU build with make targets
make act-gpu-dryrun          # Dry-run to validate workflow
make act-gpu                 # Actually run the GPU build locally
make act-docker-build        # Test all Docker builds (full, slim, gpu)
```

	`latest`
	`slim`
	`gpu`
	`gpu` `TORCH_INDEX_URL`

`CRACKZ_LOG_DIR`

`--gpus all`

`TORCH_INDEX_URL`

-
-
-

Azure VM (GPU)

Azure Windows VM with CUDA for Crackz

`azure_vm_cuda.sh`

```
# Deploy default GPU VM
./scripts/azure/azure_vm_cuda.sh

# Deploy with custom configuration
./scripts/azure/azure_vm_cuda.sh \
  --name ml-workstation \
  --size Standard_NC12s_v3 \
  --location westus2

# Connect via RDP and verify
# RDP to the public IP, then run:
nvidia-smi
cd C:\Crackz && .venv\Scripts\Activate.ps1
python -m crackz --version
```

`Standard_NC6s_v3`						
`Standard_NC12s_v3`						
`Standard_NC24s_v3`						

`Standard_NC6s_v2`						
`Standard_NV6`						

```
# Default deployment (V100 GPU, East US)
./scripts/azure/azure_vm_cuda.sh

# Custom name and location
./scripts/azure/azure_vm_cuda.sh --name gpu-crackz --location westeurope

# Specific GPU configuration
./scripts/azure/azure_vm_cuda.sh --size Standard_NC12s_v3 --location northeurope
```

`--name NAME`		`crackz-gpu-vm`
`--location REGION`		`eastus`
`--size SIZE`		`Standard_NC6s_v3`
`--user USERNAME`		`crackzuser`
`--password PASSWORD`		`CrackzGpu2024!`
`--mode MODE`		`standard`
`--data-disk-size SIZE`		`1024`
`--storage-type TYPE`		`Premium_LRS`
`--no-data-disk`		
`--no-cuda`		
`--no-crackz`		
`--enable-nested-virt`		
`--destroy`		
`--help`		

```
# Production inference server
./scripts/azure/azure_vm_cuda.sh \
  --name crackz-prod \
  --size Standard_NC12s_v3 \
  --location westus2 \
  --mode production

# Development environment with large storage
./scripts/azure/azure_vm_cuda.sh \
  --name crackz-dev \
  --size Standard_NV6 \
  --mode development \
  --data-disk-size 2048 \
  --storage-type Premium_LRS

# Budget setup without data disk
./scripts/azure/azure_vm_cuda.sh \
  --name crackz-simple \
  --size Standard_NV6 \
  --no-data-disk

# Manual setup (no auto-install)
./scripts/azure/azure_vm_cuda.sh \
  --name crackz-manual \
  --size Standard_NC6s_v3 \
  --no-cuda \
  --no-crackz

# Docker development (with nested virtualization)
./scripts/azure/azure_vm_cuda.sh \
  --name crackz-docker \
  --size Standard_NC6s_v3 \
  --enable-nested-virt

# Clean up resources
./scripts/azure/azure_vm_cuda.sh --name crackz-gpu-vm --destroy
```

`--data-disk-size SIZE`			
`--storage-type TYPE`			
`--no-data-disk`			

```
C:\ (OS Disk - 127 GB)
├── Windows/
├── Program Files/
├── Users/crackzuser/Desktop/Crackz/   # Crackz installation
└── ...

D:\ (Data Disk - configurable size)
├── Projects/                          # ML project workspace
│   ├── datasets/                     # Training datasets
│   ├── models/                       # Model checkpoints
│   ├── outputs/                      # Detection results
```

```
│   └── exports/                      # Exported models
└── temp/                             # Temporary processing
```

```
# Production: High-performance storage
./scripts/azure/azure_vm_cuda.sh \
  --name ml-prod \
  --data-disk-size 2048 \
  --storage-type Premium_LRS

# Development: Cost-effective storage
./scripts/azure/azure_vm_cuda.sh \
  --name ml-dev \
  --data-disk-size 512 \
  --storage-type Standard_LRS

# Simple setup: No separate disk
./scripts/azure/azure_vm_cuda.sh \
  --name ml-simple \
  --no-data-disk
```

1.
2.
3.
4.
5.
6.
7.
8. `C:\Crackz\.venv`
9.
10.
11.
12.
13.
14.
15.
16.

```
# 1. Connect via RDP to the VM
# 2. Double-click "File Upload Server" desktop shortcut
# 3. Open browser and go to: http://localhost:8080
# 4. Or access remotely: http://<VM-PUBLIC-IP>:8080
```

```
# From your local computer:
# Windows: \\<VM-PUBLIC-IP>\CrackzUploads
# Mac/Linux: smb://<VM-PUBLIC-IP>/CrackzUploads

# Credentials: crackzuser / CrackzGpu2024!
```

```
# Using any SFTP client (WinSCP, FileZilla, etc.)
sftp crackzuser@<VM-PUBLIC-IP>
# Upload to: /d/Projects/uploads/images/
```

```
D:\Projects\uploads\
├── images/          # Individual images for analysis
└── batch/          # Batch processing folders
```

```
# Use Windows built-in Remote Desktop
mstsc /v:<PUBLIC_IP>

# Or download RDP file
az vm show -d -g crackz-gpu-rg -n crackz-gpu-vm --query publicIps -o tsv
```

```
# Check GPU status
nvidia-smi

# Test CUDA availability
python -c "import torch; print(f'CUDA available: {torch.cuda.is_available()}')"
python -c "import torch; print(f'GPU count: {torch.cuda.device_count()}')"
```

```
# Activate virtual environment
cd C:\Crackz
.\.venv\Scripts\Activate.ps1

# Check Crackz version
python -m crackz --version

# Test GPU detection
python -m crackz gui  # Should detect CUDA device
```

```
# Activate virtual environment
cd C:\Crackz
.\.venv\Scripts\Activate.ps1

# Option 1: Direct analysis on uploaded images
python -m crackz detect run --images "D:\Projects\uploads\images" --device cuda

# Option 2: Create project and import uploaded images
python -m crackz init harbor-inspection
python -m crackz import-images "D:\Projects\uploads\images" harbor-inspection/images/raw
python -m crackz detect run harbor-inspection --device cuda

# Option 3: Batch processing multiple folders
python -m crackz detect run --images "D:\Projects\uploads\batch\*" --device cuda

# View results
explorer "harbor-inspection\artifacts\detection_results"
```

```
# Create test project
python -m crackz init test-project

# Import sample images
python -m crackz import-images C:\path\to\images test-project/images/raw

# Run GPU-accelerated detection
python -m crackz detect run test-project --device cuda
```

-
-
-
- `az vm deallocate`

-
-
-
-

-
-
-
-
-
-
-

`CrackzGpu2024!`

```
# Manually install CUDA
# Download from: https://developer.nvidia.com/cuda-11-8-0-download-archive
# Choose Windows x86_64 local installer
```

```
# Manual installation
cd C:\Crackz
.\.venv\Scripts\Activate.ps1
uv pip install --find-links https://download.pytorch.org/whl/cu118 torch torchvision
# Download wheel from GitHub releases and install
```

```
# Check drivers
nvidia-smi

# Verify CUDA installation
nvcc --version

# Check PyTorch CUDA
python -c "import torch; print(torch.version.cuda)"
```

- `nvidia-smi -l 1`

```
# Windows Event Logs
eventvwr.msc

# CUDA installation logs
Get-Content C:\ProgramData\NVIDIA Corporation\CUDA Installer\log.txt

# Crackz logs
Get-Content $env:TEMP\crackz-*.log
```

```
az vm list-sizes --location eastus --query "[?contains(name, 'NC')]"
```

```
# Use spot instances (60-90% discount)
az vm create --priority Spot --max-price -1 ...

# Auto-shutdown schedule
az vm auto-shutdown --name crackz-gpu-vm --resource-group crackz-gpu-rg --time 1800

# Deallocate when not in use
az vm deallocate --name crackz-gpu-vm --resource-group crackz-gpu-rg
```

`%TEMP%\crackz-*.log`

Azure Batch

Azure Batch Deployment

1.
2.
3.
4. `az login`

```
./scripts/azure/azure_batch.sh \
  --batch-account mycrackzaccount \
  --storage-account mycrackzstorage
```

`mycrackzaccount` `mycrackzstorage`

```
# Get storage account key
STORAGE_KEY=$(az storage account keys list \
  --account-name mycrackzstorage \
  --resource-group subbaltica-crackz-rg \
  --query "[0].value" -o tsv)

# Upload project directory
az storage blob upload-batch \
  --account-name mycrackzstorage \
  --account-key "$STORAGE_KEY" \
  --destination crackz-data \
  --source ./my-project
```

```
# Create job
az batch job create \
  --id crackz-detection-job \
  --pool-id crackz-pool

# Add tasks to the job
az batch task create \
  --job-id crackz-detection-job \
  --task-id detect-task-1 \
  --command-line "docker run crackz-cli:latest detect run --project /mnt/batch/tasks/fsmounts/crackz-data/my-project"
```

`--batch-account`		
`--resource-group`		`subbaltica-crackz-rg`
`--location`		`swedencentral`
`--pool-id`		`crackz-pool`
`--vm-size`		`Standard_D2s_v3`
`--node-count`		`2`
`--image`		`crackz-cli:latest`
`--storage-account`		
`--container`		`crackz-data`

`Standard_D2s_v3`				
`Standard_D4s_v3`				
`Standard_D8s_v3`				
`Standard_NC6s_v3`				
`Standard_NC4as_T4_v3`				

```
./scripts/azure/azure_batch.sh \
  --batch-account crackz-batch \
  --pool-id high-performance-pool \
  --vm-size Standard_D8s_v3 \
  --node-count 10 \
  --storage-account crackzstore
```

```
./scripts/azure/azure_batch.sh \
  --batch-account crackz-batch-gpu \
  --pool-id gpu-pool \
  --vm-size Standard_NC4as_T4_v3 \
  --node-count 2 \
  --image crackz-cli:gpu
```

```
# Enable auto-scaling
az batch pool autoscale enable \
  --pool-id crackz-pool \
  --account-name mycrackzaccount \
  --auto-scale-formula '
    startingNumberOfVMs = 2;
    maxNumberofVMs = 10;
    pendingTaskSamplePercent = $PendingTasks.GetSamplePercent(180 * TimeInterval_Second);
    pendingTaskSamples = pendingTaskSamplePercent < 70 ? startingNumberOfVMs : avg($PendingTasks.GetSample(180 * TimeInterval_Second));
    $TargetDedicatedNodes = min(maxNumberofVMs, pendingTaskSamples);
  '
```

```
#!/bin/bash
# Create job
az batch job create --id multi-project-job --pool-id crackz-pool

# Submit tasks for each project
for project in project1 project2 project3; do
  az batch task create \
    --job-id multi-project-job \
    --task-id "detect-$project" \
    --command-line "docker run crackz-cli:latest detect run --project /data/$project"
done
```

```
# Training task
az batch task create \
  --job-id pipeline-job \
  --task-id train \
  --command-line "docker run crackz-cli:latest detect train --project /data/my-project --epochs 50"

# Detection task (depends on training)
az batch task create \
  --job-id pipeline-job \
  --task-id detect \
  --depends-on train \
  --command-line "docker run crackz-cli:latest detect run --project /data/my-project"
```

```
az batch task create \
  --job-id output-job \
  --task-id detect-with-output \
  --command-line "docker run -v /mnt/output:/output crackz-cli:latest detect run --project /data/project" \
  --resource-files '[{"httpUrl":"https://mystorage.blob.core.windows.net/input/data.zip","filePath":"data.zip"}]' \
  --output-files '[{"filePattern":"/mnt/output/**/*","destination":{"container":{"containerUrl":"https://mystorage.blob.core.windows.net/results"}},"uploadOptions":{"uploadCo
```

```
az batch pool show \
  --pool-id crackz-pool \
  --account-name mycrackzaccount \
  --query "{State:allocationState, DedicatedNodes:currentDedicatedNodes}"
```

```
az batch job list \
  --account-name mycrackzaccount \
  -o table
```

```
az batch task list \
  --job-id crackz-detection-job \
  --account-name mycrackzaccount \
  -o table
```

```
az batch task file download \
  --job-id crackz-detection-job \
  --task-id detect-task-1 \
  --file-path stdout.txt \
  --destination ./task-stdout.txt \
  --account-name mycrackzaccount
```

```
# After pool creation, modify to use low-priority nodes
az batch pool resize \
```

```
  --pool-id crackz-pool \
  --account-name mycrackzaccount \
  --target-dedicated-nodes 0 \
  --target-low-priority-nodes 4
```

```
./scripts/azure/azure_batch.sh \
  --batch-account mycrackzaccount \
  --pool-id crackz-pool \
  --destroy
```

	`az batch location quotas show --location <region>`
	`az batch task file download`
	`az batch account login`

```
# Get task details
az batch task show \
  --job-id crackz-detection-job \
  --task-id detect-task-1 \
  --account-name mycrackzaccount

# Download stderr
az batch task file download \
  --job-id crackz-detection-job \
  --task-id detect-task-1 \
  --file-path stderr.txt \
  --destination ./task-stderr.txt \
  --account-name mycrackzaccount
```

```
# Check current quotas
az batch location quotas show --location swedencentral

# Submit quota increase request through Azure Portal
# Support > New support request > Issue type: Service and subscription limits
```

```
name: Azure Batch Processing
on:
  push:
    branches: [main]

jobs:
  batch-process:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v5

      - name: Azure Login
        uses: azure/login@v1
        with:
          creds: ${{ secrets.AZURE_CREDENTIALS }}

      - name: Create Batch Pool
        run: |
          ./scripts/azure/azure_batch.sh \
            --batch-account ${{ secrets.BATCH_ACCOUNT }} \
            --storage-account ${{ secrets.STORAGE_ACCOUNT }}

      - name: Submit Batch Job
        run: |
```

```
        az batch job create --id ci-job-${{ github.run_number }} --pool-id crackz-pool
        az batch task create \
          --job-id ci-job-${{ github.run_number }} \
          --task-id detect \
          --command-line "docker run crackz-cli:latest detect run --project /data/project"
```

1.
2.
3.
4.
5.

`taskSlotsPerNode`

```
# Modify pool to allow multiple tasks per node
az batch pool set \
  --pool-id crackz-pool \
  --account-name mycrackzaccount \
  --json-file pool-config.json
```

`pool-config.json`

```
{
  "taskSlotsPerNode": 4
}
```


```
ai:
  detection:
    batch_size: 16  # Adjust based on VM size
```

-
-
-
-
-

-
-
-

VM vs Batch

Azure Deployment Comparison

-
-
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-
-

```
# Create Windows 11 VM
./scripts/azure/azure_vm.sh \
  --name crackz-dev \
  --size Standard_D4s_v3

# Connect via RDP and use GUI
# IP address displayed after creation
```

-
-
-

-

```
# Create batch pool with 10 nodes
./scripts/azure/azure_batch.sh \
  --batch-account mybatchaccount \
  --storage-account mystorage \
  --node-count 10 \
  --vm-size Standard_D4s_v3

# Submit 100 detection jobs
for i in {1..100}; do
  az batch task create \
    --job-id detection-job \
    --task-id "task-$i" \
    --command-line "docker run crackz-cli:latest detect run --project /data/project-$i"
done
```

```
# Phase 1: Development on VM
./scripts/azure/azure_vm.sh --name crackz-dev --size Standard_D4s_v3
# RDP in, use GUI to develop and test
# Export config: crackz export-config --project demo --out optimized.yaml

# Phase 2: Production on Batch
./scripts/azure/azure_batch.sh \
  --batch-account prod-batch \
  --pool-id production-pool \
```

```
  --node-count 20 \
  --vm-size Standard_D8s_v3

# Upload optimized config and data to storage
# Submit batch jobs with optimized parameters
```

	`azure_vm.sh`	`azure_batch.sh`

1.
2.
3.
4.

1.
2.
3.
4.
5.

1.
2.
3.
4.

```
# 1. Develop on VM
./scripts/azure/azure_vm.sh --name dev-vm

# 2. Test configuration
crackz detect train --project my-project --epochs 50
crackz detect run --project my-project

# 3. Export configuration
crackz export-config --project my-project --out batch-config.yaml

# 4. Upload to Azure Storage
az storage blob upload-batch \
  --account-name mystorage \
  --destination crackz-data \
  --source my-project

# 5. Create Batch pool
./scripts/azure/azure_batch.sh \
  --batch-account mybatch \
  --storage-account mystorage \
  --node-count 10

# 6. Submit batch jobs
az batch job create --id production-job --pool-id crackz-pool
# ... add tasks
```

- `scripts/azure/azure_vm.sh --help`

`--size Standard_NC*`

`az batch task file download`

Model Evaluation

Model Evaluation

```
# Run evaluation on test dataset
crackz evaluate run --project my_harbor

# Specific metrics
crackz evaluate run --project my_harbor --metrics accuracy,iou

# Export results
crackz evaluate run --project my_harbor --output results.json
crackz evaluate run --project my_harbor --output results.csv
```

```
from pathlib import Path
from crackz.ai.evaluation import evaluate_model, export_json

# Run evaluation
results = evaluate_model(
    project_path=Path("my_harbor"),
    metrics=["accuracy", "precision", "recall", "iou"]
)

# Display results
print(results)

# Export to JSON
export_json(results, Path("evaluation_results.json"))
```

```
Accuracy = (TP + TN) / (TP + TN + FP + FN)
```

```
Precision = TP / (TP + FP)
```

```
Recall = TP / (TP + FN)
```

```
F1 = 2 * (Precision * Recall) / (Precision + Recall)
```

```
IoU = Area of Overlap / Area of Union
```

```
# Default threshold (0.5)
crackz evaluate run --project my_harbor

# Custom threshold
crackz evaluate run --project my_harbor --threshold 0.7
```

```
# All metrics (default)
crackz evaluate run --project my_harbor

# Specific metrics
crackz evaluate run --project my_harbor --metrics accuracy,precision,recall
crackz evaluate run --project my_harbor --metrics iou  # Only IoU
```

`accuracy` `precision` `recall` `f1` `iou`

```
Evaluation Results (500 images)
==================================================
Accuracy:     92.3%
Precision:    89.1%
Recall:       94.8%
F1 Score:     91.9%
Mean IoU:     0.847
Threshold:    0.5
```

```
{
  "num_images": 500,
  "threshold": 0.5,
  "metrics": {
    "accuracy": 0.923,
    "precision": 0.891,
    "recall": 0.948,
    "f1_score": 0.919,
    "iou": 0.847
  }
}
```

```
Metric,Value
Num Images,500
Threshold,0.5
Accuracy,0.9230
Precision,0.8910
Recall,0.9480
F1 Score,0.9190
IoU,0.8470
```

```
project/
├── images/
│   └── test/          # Test images
│       ├── img1.png
│       ├── img2.png
│       └── ...
└── masks/
    └── test/          # Ground truth masks
        ├── img1.png
```

```
    ├── img2.png
    └── ...
```

```
def evaluate_model(
    project_path: Path,
    test_images: list[Path] | None = None,
    test_masks: list[Path] | None = None,
    metrics: list[str] | None = None,
    threshold: float = 0.5,
) -> EvaluationResults:
    """Evaluate crack detection model on test dataset."""
```

`project_path` `test_images`
`test_masks` `metrics`
`threshold`

`EvaluationResults`

```
@dataclass
class EvaluationResults:
    accuracy: float
    precision: float
    recall: float
    f1_score: float
    iou: float
    num_images: int
    threshold: float
```

```
# Export to JSON
export_json(results, Path("results.json"))

# Export to CSV
export_csv(results, Path("results.csv"))
```

```
ValueError: No test images found
```

`<project>/images/test/`

```
ls <project>/images/test/
```

```
ValueError: Image/mask count mismatch: 100 vs 98
```

```
# Check counts
ls <project>/images/test/ | wc -l
ls <project>/masks/test/ | wc -l

# Find missing masks
diff <(ls <project>/images/test/) <(ls <project>/masks/test/)
```

```
FileNotFoundError: No trained model checkpoint found
```

```
crackz detect train --project <project>
```

Advanced Performance

Advanced Performance & Reproducibility

`ai.training` `ai.detection` `ProjectConfig`

	`precision`	`ai.training`	
	`devices`	`ai.training`	
	`strategy`	`ai.training`	`ddp`
	`seed`	`ai.training` `ai.detection`	

`16` `16-mixed` `bf16-mixed`
`32`

```
ai:
  training:
    precision: 16-mixed
    batch_size: 16
```

```
crackz detect train --project demo-project
```

`devices` `strategy` `ddp`

```
ai:
  training:
    devices: 2
    strategy: ddp
    batch_size: 8
    precision: bf16-mixed
```

```
crackz detect train --project demo-project
```

```
docker run --gpus all --rm -v %cd%\project:/home/app crackz-cli:gpu \
  detect train --project /home/app/demo-project
```

`ddp`		
`ddp_spawn`		
`auto`		

`seed`

```
ai:
  training:
```

```
    seed: 1234
  detection:
    seed: 1234
```

`PYTHONHASHSEED` `random`

```
import torch
torch.use_deterministic_algorithms(True)
```

-
-

```
project:
  name: demo-project
  description: Perf tuned
paths:
  project_path: demo-project
  image_path: demo-project/images
  image_mask_path: demo-project/masks
  artifacts_path: demo-project/artifacts
logging:
  level: INFO
  log_dir: logs
ai:
  model:
    encoder: resnet34
    device: cuda
    input_size: [512, 512]
    threshold: 0.5
  training:
    batch_size: 16
    epochs: 20
    learning_rate: 0.0005
    precision: 16-mixed
    devices: 2
    strategy: ddp
    seed: 4242
  detection:
    batch_size: 8
    seed: 4242
```


`crackz export-config`

```
crackz export-config --project demo-project --out tuned_config.yaml
```

`tuned_config.yaml`

`ai.detection.seed`

`ddp_spawn`

	`torch.use_deterministic_algorithms(True)`

Migration Guide

Project Schema Migration

`schema_version`
`ai.training.tensorboard`

`schema_version`
`CURRENT_SCHEMA_VERSION` `crackz.project.migration`
`schema_version`

	`schema_version` `ai.training.tensorboard`	
	`schema_version: 1` `ai.training.tensorboard: false`	
	`data` `mask_threshold: 1`	
	`project.crackz_version`	

```yaml
schema_version: 3
project:
  name: my_project
  description: My crack detection project
  created: '2025-01-15T10:30:00Z'
  crackz_version: '0.50.0'  # NEW: Tracks Crackz app version
```

`project.crackz_version`
`checkpoints/` `models/` `runtime/`
`detections/visualizations/`

`checkpoints_path` `checkpoints/`
`models/` `runtime/`
`crackz_version`

`data`

```yaml
data:
  mask_threshold: 1
schema_version: 2
```

`data.mask_threshold`

`detect train` `detect run`

1.
2. `Project schema v0 < v1. Migrate now? [y/N]:`
3. `.bak`
4.
5. `CRACKZ_AUTO_MIGRATE=1`
6.

```
crackz migrate --project <path-to-project-dir>
# or
crackz migrate --cfg <path-to-project-yaml>
```

`--dry-run` `--no-backup` `<file>.bak` `--migrate-files`

```
# Using project directory
crackz migrate --project my_project --dry-run
crackz migrate --project my_project

# Using direct YAML file path
crackz migrate --cfg my_project/my_project_project.yaml
```

```
# Preview what files would be moved
crackz migrate --project my_project --migrate-files --dry-run

# Migrate both schema AND files
crackz migrate --project my_project --migrate-files
```

`checkpoints/` `models/` `training/tensorboard/` `runtime/tensorboard/` `training/lightning_logs/` `runtime/lightning_logs/` `detections/*.png` `detections/visualizations/`

`{filename}.v{version}.{timestamp}.bak`

```
myproject_project.v2.20250115_143022.bak
```

`v2`
`20250115_143022`

`myproject_project.v2.20250115_143022.bak`
`.v2.20250115_143022.bak`

`CRACKZ_AUTO_MIGRATE=1`	

```
from pathlib import Path
from crackz.project.migration import migrate_project_file

result = migrate_project_file(Path("my_project"))  # or path/to_yaml
print(result)
# Example result:
# {
#   'status': 'migrated',
#   'from_version': 2,
#   'to_version': 3,
#   'steps': ['2->3: updated to artifacts structure v3 + added crackz_version tracking'],
#   'path': 'my_project/my_project_project.yaml',
#   'backup_path': 'my_project/my_project_project.v2.20250115_143022.bak',
```

```
#   'dry_run': False
# }
```

`status` `from_version` `to_version` `steps` `path` `backup_path` `dry_run`

1.
2.

`.bak`

`.bak`

Architecture

System Architecture

```
crackz/                          # Workspace root
├── pyproject.toml               # Workspace config, dev deps, tool settings (no [project])
├── core/                        # crackz-core package
│   ├── pyproject.toml           # Package metadata and dependencies
│   ├── src/crackz/              # Core library (AI, CLI, config, logging, project)
│   └── tests/                   # Core unit + integration tests
├── gui/                         # crackz-gui package (depends on crackz-core)
│   ├── pyproject.toml           # Package metadata, dependency on crackz-core
│   ├── src/crackz_gui/          # PySide6 Qt GUI application
│   └── tests/                   # GUI unit + integration tests
└── tests/                       # Root-level tests (cross-package, legacy)
```

`core/` `gui/` `core/`
`uv sync --dev` `from crackz.ai import ...` `from crackz_gui.qt import ...`

```
┌──────────────────────────────────────────────────────────┐
│  Layer 4 — Presentation / Entry Points                    │
│  crackz.cli · crackz_gui · crackz.dashboard               │
├──────────────────────────────────────────────────────────┤
│  Layer 3 — Application Façade                             │
│  crackz.app                                               │
├──────────────────────────────────────────────────────────┤
│  Layer 2 — Domain / Business Logic                        │
│  crackz.ai · crackz.project · crackz.report               │
├──────────────────────────────────────────────────────────┤
│  Layer 1 — Infrastructure / Support                       │
│  crackz.config · crackz.log · crackz.tracing · crackz.util │
└──────────────────────────────────────────────────────────┘
```

```
graph TD
  subgraph L4["Layer 4 — Presentation"]
    CLI["crackz.cli"]
    GUI["crackz_gui"]
    DASH["crackz.dashboard"]
  end
  subgraph L3["Layer 3 — Application Façade"]
    APP["crackz.app"]
  end
  subgraph L2["Layer 2 — Domain"]
    AI["crackz.ai"]
    PROJECT["crackz.project"]
    REPORT["crackz.report"]
  end
  subgraph L1["Layer 1 — Infrastructure"]
    CONFIG["crackz.config"]
    LOG["crackz.log"]
    TRACING["crackz.tracing"]
    UTIL["crackz.util"]
  end

  CLI --> APP
  GUI --> APP
  DASH --> APP
  APP --> AI
  APP --> PROJECT
  APP --> REPORT
  AI --> CONFIG
  AI --> LOG
  AI --> UTIL
  PROJECT --> CONFIG
  PROJECT --> LOG
  PROJECT --> UTIL
  REPORT --> CONFIG
  REPORT --> LOG
  CONFIG --> LOG
  CONFIG --> UTIL
```

	crackz.cli	crackz_gui crackz.dashboard	
	crackz_gui	crackz.dashboard	crackz.cli
	crackz.dashboard		crackz.cli crackz_gui
	crackz.app		
	crackz.ai		
	crackz.project		
	crackz.report		
	crackz.config	crackz.project crackz.log crackz.util	crackz.ai crackz.report
	crackz.log		
	crackz.tracing	crackz.log	
	crackz.util	crackz.log	

.importlinter scripts/check_architecture.py uv run task pre_commit

---

pydantic-settings
raw/ train/ val/ test/ CrackzDataset
Dataset

loguru artifacts/models/ artifacts/runtime/

	crackz.db	crackz_gui.data.projects_db	
	<project>/artifacts/project.db	crackz.ai.model.registry	

artifacts/project.db crackz.db

<project>/artifacts/crackz.db
<project>/artifacts/project.db crackz.db

binary				
multiclass				

`num_classes`

- `0` `1` `255` `{0, 1, 255}`
- `ai.model.task_mode` `binary` `task_mode` `binary` `multiclass`

1.
2.
3.
4. `crackz detect train`
5.
6.
7.
8. `artifacts/models/`
9. `crackz detect run`
10.

`segmentation-models-pytorch`

`project.yaml`

```
ai:
  model:
    backbone: segformer-b2   # or segformer-b0/b4, efficientnet-b3, resnet34, etc.
annotation:
  categories:
    - id: 1
      key: crack
      name: Crack
      color: "#ef4444"
```

`crack` `crackz init --name harbor --annotation-template crack-only`

`crack` `spall` `corrosion` `joint_damage` `crackz init --name harbor --annotation-template harbor-defects` `annotation.categories`

```
C4Container
 title Crackz — Container View (Implemented)
 Person(user, "User")
 System_Boundary(crackz, "Crackz Local System") {
   Container(cli, "CLI", "Typer", "Command parsing & dispatch")
   Container(gui, "GUI", "PySide6 (Qt 6)", "Desktop inspection & invocation")
   Container(core, "Core Library", "Python Packages", "project, ai, config, logging")
   Container(fs, "Local Filesystem", "images/ + artifacts/", "Images, masks, checkpoints, metrics")
   ContainerDb(udb, "User DB", "SQLite", "Known projects + welcome aggregates")
   ContainerDb(pdb, "Project DB", "SQLite", "Per-project models, detections, training runs")
 }
 Rel(user, cli, "Runs commands")
 Rel(user, gui, "Interacts (point & click)")
 Rel(cli, core, "Invokes APIs")
 Rel(gui, core, "Invokes APIs")
 Rel(core, fs, "read/write images, masks, checkpoints, metrics")
 Rel(gui, udb, "read/write project index")
 Rel(core, pdb, "read/write project metadata")
```

```
C4Context
 title Crackz — System Context (Local)
 Person(user, "User", "Engineer / Operator")
 System(local, "Crackz Application", "CLI + GUI + Core Library")
 System_Ext(osfs, "Host Filesystem", "Images + Artifacts")
 Rel(user, local, "Invoke CLI / Launch GUI")
 Rel(local, osfs, "read / write")
```

```
C4Component
 title Core Library Components (Implemented)
 Container(core, "core", "Python")
 Component(project, "Project & Paths", "pydantic", "Project metadata & path helpers")
 Component(config, "Configuration", "pydantic-settings", "Merge defaults / env / YAML")
 Component(importer, "Image Importer", "Pillow", "Resize, quality, organize splits & masks")
 Component(dataset, "Dataset Layer", "Torch Dataset", "CrackzDataset + DataLoader factory")
 Component(train, "Training Orchestrator", "PyTorch Lightning", "Fit/validate/test + callbacks")
 Component(detect, "Detection Pipeline", "PyTorch", "Batch inference over dataset splits (raw/test)")
 Component(metrics, "Metrics & Checkpoints", "Lightning + JSON", "Save val/test metrics & best.ckpt/last.ckpt")
 Component(perf, "Performance Utilities", "Seeding", "Seed + trainer overrides")
 Component(logging, "Logging", "loguru", "Structured log sink & setup")
 Component(evaluation, "Evaluation Framework", "Metrics", "IoU, precision, recall, F1, accuracy")
 Component(reporter, "Report Generator", "python-docx", "Branded Word reports")
 Component(viz, "Visualization", "Matplotlib", "Detection overlays (3-panel)")
 Component(pipeline_orch, "Pipeline Orchestrator", "Workflow", "Train→detect + cancellation")
 Component(device_mgmt, "Device Management", "Abstraction", "CPU/GPU/MPS selection")
 Component(severity, "Severity Classifier", "Categorization", "3-tier crack severity")
 Component(suggester, "Mask Suggester", "Semi-automation", "Model-assisted annotation")
 Component(perf_est, "Performance Estimator", "RuntimeTracker", "Time estimates")
 Component(async_cb, "Async Callbacks", "Janus", "Sync/async bridging")
 Component(tracing_mod, "Tracing", "OpenTelemetry", "AI API observability")
 Component(ai_chat_svc, "AI Chat", "Anthropic", "User assistance")
 Rel(importer, project, "Uses path layout")
 Rel(dataset, project, "Resolves split directories")
 Rel(train, dataset, "Consumes (train/val/test) loaders")
 Rel(detect, dataset, "Consumes detection loader (raw or test)")
 Rel(train, metrics, "Saves checkpoints + metrics")
 Rel(detect, metrics, "Writes detection results JSON/PNGs")
 Rel(project, config, "Initializes from effective settings")
 Rel(train, perf, "Reads overrides / seed")
 Rel(detect, perf, "Sets seed (optional)")
 Rel(dataset, logging, "Reports warnings (missing masks / size mismatch)")
 Rel(detect, evaluation, "Evaluates performance")
 Rel(evaluation, metrics, "Calculates metrics")
 Rel(detect, viz, "Visualizes results")
 Rel(detect, reporter, "Generates reports")
 Rel(train, pipeline_orch, "Orchestrated by")
 Rel(detect, pipeline_orch, "Orchestrated by")
 Rel(train, device_mgmt, "Selects device")
 Rel(detect, device_mgmt, "Selects device")
 Rel(detect, severity, "Categorizes cracks")
 Rel(suggester, detect, "Uses model")
 Rel(train, perf_est, "Tracks runtime")
 Rel(detect, perf_est, "Tracks runtime")
 Rel(train, async_cb, "Reports progress")
 Rel(detect, async_cb, "Reports progress")
 Rel(ai_chat_svc, tracing_mod, "Optionally traced")
```

```
flowchart LR
 A["Test Dataset"] --> B["Evaluator"]
 B --> C["Metrics Calculator"]
 C --> D["Evaluation Results"]
 D --> E["JSON/CSV Reporter"]
 D --> F["Report Generator"]
 F --> G["Branded Word Document"]
 I["Detection Results"] --> J["Visualization"]
 J --> K["Detection Overlays"]
 I --> F
```

```
flowchart TB
 A["Training Thread (sync)"] --> B["CallbackDispatcher.sync_q"]
 B --> C["Janus Queue"]
 C --> D["CallbackDispatcher.async_q"]
 D --> E["GUI Event Loop (async)"]
 E --> F["UI Update"]
```

```
flowchart LR
 A["Import Command / GUI Action"] --> B["Image Importer"]
 B --> C["Split Directories (images/*, masks/*)"]
 C --> D["Train Command"]
 D --> E["Training Orchestrator<br/>(Trainer.fit)"]
 E --> F["Checkpoints &amp; Metrics<br/>(best.ckpt / last.ckpt / JSON)"]
 F --> G["Detect Command"]
 G --> H["Inference Loop<br/>(sigmoid → threshold)"]
 H --> I["Annotated Output<br/>(PNGs + detection_results.json)"]
```

```
sequenceDiagram
 participant User
 participant CLI as CLI / GUI
 participant Project
 participant Import as ImageImporter
 participant FS as Filesystem
 User->>CLI: import-images --src ./samples --type train
 CLI->>Project: load_project(path|yaml)
 CLI->>Import: iterate source images
 Import->>FS: resize + normalize + write split file
 Import->>FS: write mask (if provided)
 Import-->>CLI: progress callback
 CLI-->>User: summary (count, skipped)
```

```
sequenceDiagram
 participant User
 participant CLI as CLI / GUI
 participant Project
 participant DS as CrackzDataset
 participant Trainer as Lightning Trainer
 participant Model
 participant FS as Filesystem
 User->>CLI: detect train --project P
 CLI->>Project: load_project()
 CLI->>DS: build train/val/test datasets
 DS-->>CLI: DataLoaders
 CLI->>Trainer: fit(model, loaders)
 Trainer->>FS: save best.ckpt / last.ckpt
 Trainer-->>CLI: metrics (val/test)
 CLI-->>User: training summary
 User->>CLI: detect run --project P
 CLI->>Project: load_project()
 CLI->>Model: load checkpoint (best or last)
 CLI->>DS: build raw dataset (or test)
 DS-->>CLI: DataLoader
 CLI->>Model: forward (batched)
 Model-->>CLI: probabilities
```

```
 CLI->>CLI: threshold -> mask
 CLI->>FS: write annotated PNG + JSON metrics
 CLI-->>User: detection summary
```

```
flowchart TD
 Defaults[Built-in Defaults] --> Merge
 YAML[Project YAML] --> Merge
 Env[Environment] --> Merge
 Merge --> Effective[Effective Settings]
```

---

project defaults *catalog of trained models*

		`ai.model.backbone` `project.yaml`
		`ai.model.active_checkpoint`
		`artifacts/project.db`
		Start Training
		Model Selector
	`active_checkpoint`	`ai.model.backbone`
`project.yaml`	*Start Training*	*Save as new project default*

```
flowchart LR
 User([User]) --> CLI["CLI (Typer)"]
 User --> GUI["GUI (PySide6)"]
 CLI --> CORE["Core Library"]
 GUI --> CORE
 CORE --> FS["Filesystem<br/>images / artifacts/models / artifacts/runtime"]
 CORE -.optional.-> GPU[("GPU")]

 subgraph Optional_Docker [Optional Docker]
   IMG["Container Image<br/>(Python 3.13 + deps)"]
 end

 %% Show that container can host CLI/GUI
 IMG --> CLI
 IMG --> GUI
```

	`crackz.project.project`	
	`pydantic-settings`	
	`project.import_images`	
	`crackz.ai.data`	
	`crackz.ai.detection.train`	
	`crackz.ai.detection.detect`	
	`crackz.ai.model`	
	`crackz.ai.checkpoints`	
	`crackz.ai.device`	

	`crackz.util`	
	`crackz.ai.performance`	
	`crackz.log.crackz_logger`	
	`crackz_gui.*`	
	`crackz.ai.evaluation`	
	`crackz.report.generator`	
	`crackz.ai.visualization`	
	`crackz.ai.pipeline`	
	`crackz.ai.pipeline.cancellation`	
	`crackz.ai.detection.severity`	
	`crackz.ai.mask_suggester`	
	`crackz.ai.performance.runtime_estimation`	
	`crackz.ai.core.async_callbacks`	
	`crackz.tracing`	
	`crackz_gui.services.ai_chat`	

`train/` `val/`
`test/` `raw/`

- `core/` `gui/`

Last updated: synchronized with repository implementation on commit timeline of current editing session.

Architecture Guidelines

Architecture Guidelines

```
┌─────────────────────────────────────────┐
│  User Interfaces (CLI, GUI)             │  ← Entry points
├─────────────────────────────────────────┤
│  Business Logic (Commands, Operations)  │  ← Use cases
├─────────────────────────────────────────┤
│  Core Domain (Project, AI, Config)      │  ← Domain models
├─────────────────────────────────────────┤
│  Infrastructure (Logging, Storage)      │  ← Technical concerns
└─────────────────────────────────────────┘
```

```
# □ GOOD: Clear separation
# core/src/crackz/cli/detection_cli.py - CLI commands only
# core/src/crackz/ai/detection.py - Detection logic only
# core/src/crackz/project/structure.py - Project layout only

# × BAD: Mixed concerns
# core/src/crackz/everything.py - CLI, detection, project management all in one file
```

```
# □ GOOD: Depend on protocol
from typing import Protocol

class ModelProtocol(Protocol):
    def predict(self, image: torch.Tensor) -> torch.Tensor: ...

def run_detection(model: ModelProtocol, image: torch.Tensor) -> dict:
    """Works with any model implementing the protocol."""
    return {"mask": model.predict(image)}

# × BAD: Depend on concrete class
def run_detection(model: SegmentationModel, image: torch.Tensor) -> dict:
    """Tightly coupled to SegmentationModel."""
    return {"mask": model.predict(image)}
```

```
# □ GOOD: Single responsibility
class ProjectLoader:
    """Loads project configuration from disk."""
    def load(self, path: Path) -> ProjectConfig: ...

class ProjectValidator:
    """Validates project structure and configuration."""
    def validate(self, project: ProjectConfig) -> list[str]: ...

# × BAD: Multiple responsibilities
class ProjectManager:
    """Loads, validates, trains models, exports results."""
    def load(self, path: Path) -> ProjectConfig: ...
    def validate(self, project: ProjectConfig) -> list[str]: ...
    def train_model(self, project: ProjectConfig) -> None: ...
    def export_results(self, project: ProjectConfig) -> None: ...
```

```
# □ GOOD: Extensible via configuration
def create_model(config: ModelConfig) -> nn.Module:
    """Factory method - add new encoders via config."""
    if config.encoder in SUPPORTED_ENCODERS:
        return create_segmentation_model(config)
    raise ValueError(f"Unsupported encoder: {config.encoder}")

# × BAD: Requires code changes for new models
def create_model(encoder_name: str) -> nn.Module:
    if encoder_name == "resnet34":
        return ResNet34Model()
    elif encoder_name == "efficientnet":
        return EfficientNetModel()
    # Must modify this function for each new encoder!
```

```
# □ GOOD: Explicit parameters
def train_model(
    project_path: Path,
    epochs: int,
    batch_size: int,
    learning_rate: float,
) -> dict[str, float]:
    """All parameters explicit."""
    pass
```

```
# × BAD: Implicit global state
GLOBAL_CONFIG = {}  # Modified elsewhere

def train_model():
    """What parameters? Where do they come from?"""
    epochs = GLOBAL_CONFIG["epochs"]  # Implicit dependency
```

```
core/src/crackz/          # Core library (crackz-core package)
├── ai/                   # AI/ML pipeline (domain logic)
│   ├── data.py           # Dataset, data loading
│   ├── models.py         # Model definitions
│   ├── training.py       # Training pipeline
│   └── detection.py      # Inference pipeline
├── cli/                  # CLI interface (entry points)
│   ├── options.py        # Shared CLI options
│   ├── exceptions.py     # CLI error handling
│   ├── project_cli.py
│   └── detection_cli.py
├── config/               # Configuration (domain models)
│   ├── project.py        # ProjectConfig
│   ├── ai.py             # AIConfig
│   └── logging.py        # LogConfig
├── log/                  # Logging (infrastructure)
│   └── setup.py          # Loguru configuration
└── project/              # Project management (domain logic)
    └── structure.py      # Directory layout

gui/src/crackz_gui/       # GUI package (crackz-gui, depends on crackz-core)
├── qt/                   # PySide6 Qt application
│   ├── app_state.py      # QtAppState (reactive state)
│   └── async_bridge.py   # TaskSignals, run_in_background
├── services/             # External service integrations
├── state/                # State management
└── util/                 # GUI utilities
```

```
CLI (core)  ──→  Business Logic  ──→  Domain Models  ──→  Infrastructure
GUI (gui)   ──→  (ai/, project/)     (config/)          (log/)

Examples:
- core: cli/detection_cli.py → ai/detection.py → config/ai.py → log/setup.py □
- gui:  crackz_gui/qt/panels/train.py → crackz.ai.training → crackz.config □
- gui:  crackz-gui depends on crackz-core (declared in gui/pyproject.toml) □
```

```
Domain ──X→ GUI/CLI     # Domain shouldn't know about UI
Infrastructure ──X→ Domain  # Infrastructure shouldn't depend on business logic
GUI ──X→ CLI            # GUI shouldn't import CLI modules

Examples:
- ai/detection.py → cli/options.py ×  # Domain shouldn't import from CLI
- log/setup.py → config/project.py ×  # Infrastructure shouldn't import domain
- crackz_gui → crackz.cli ×  # GUI should use core APIs, not CLI
```

```
# × BAD: Domain imports from CLI
from crackz.cli.options import PROJECT_PATH_OPTION
def train_model(project_path: Path = PROJECT_PATH_OPTION): ...

# □ GOOD: CLI calls domain with explicit parameters
from crackz.cli.options import PROJECT_PATH_OPTION
from crackz.ai.training import train_model

def train_cli(project_path: Path = PROJECT_PATH_OPTION):
    train_model(project_path)  # Domain function has no CLI dependency
```

`core/src/crackz/ai/models.py`

```
def create_segmentation_model(config: ModelConfig) -> nn.Module:
    """Factory for creating segmentation models."""
    import segmentation_models_pytorch as smp

    return smp.Unet(
        encoder_name=config.encoder,
        encoder_weights=config.encoder_weights,
        in_channels=config.in_channels,
        classes=config.num_classes,
    )
```

`core/src/crackz/ai/detection.py`

```
class DetectionStrategy(Protocol):
    def detect(self, images: list[Path]) -> dict[Path, np.ndarray]: ...

class SingleImageDetection:
    def detect(self, images: list[Path]) -> dict[Path, np.ndarray]:
        """Process one at a time."""
        return {img: self._detect_one(img) for img in images}

class BatchDetection:
    def detect(self, images: list[Path]) -> dict[Path, np.ndarray]:
        """Process in batches for efficiency."""
        results = {}
        for batch in chunk_list(images, self.batch_size):
            results.update(self._detect_batch(batch))
        return results
```

`core/src/crackz/config/project.py`

```
class ProjectConfig(BaseModel):
    """Pydantic acts as builder with validation."""
    name: str
    project_path: Path
    training: TrainingConfig = Field(default_factory=TrainingConfig)
    logging: LogConfig = Field(default_factory=LogConfig)

    @classmethod
    def load(cls, path: Path) -> ProjectConfig:
        """Build from YAML with defaults."""
        with open(path) as f:
            data = yaml.safe_load(f)
        return cls(**data)  # Validation + defaults applied
```

`core/src/crackz/cli/*.py`

```
@app.command(name="train")
def train_command(
    project_path: Path = PROJECT_PATH_OPTION,
    epochs: int | None = EPOCHS_OPTION,
) -> None:
    """Each CLI command is a self-contained operation."""
    try:
        # Load configuration
        config = ProjectConfig.load(project_path / f"{project_path.name}_project.yaml")

        # Override with CLI args
        if epochs:
            config.training.epochs = epochs

        # Execute
        train_model(config)
```

```
        typer.secho("□ Training completed", fg=typer.colors.GREEN)
    except Exception as e:
        cli_failure(e, "Training")
```

`core/src/crackz/project/structure.py`

```
class ProjectRepository:
    """Abstracts project storage."""

    def load(self, path: Path) -> ProjectConfig:
        """Load project from filesystem."""
        return ProjectConfig.load(path / f"{path.name}_project.yaml")

    def save(self, project: ProjectConfig) -> None:
        """Save project to filesystem."""
        project.save(project.project_path / f"{project.name}_project.yaml")

    def exists(self, path: Path) -> bool:
        """Check if project exists."""
        return (path / f"{path.name}_project.yaml").exists()
```

`gui/src/crackz_gui/data/projects_db.py`
`core/src/crackz/ai/model/registry.py`

`crackz.db`
`<project>/artifacts/project.db` `<project>/artifacts/project.db`
`crackz.db`
`crackz.db`
`artifacts/project.db`

```
# User-level DB: known projects + cached aggregate stats.
projects_db = ProjectsDB()
projects_db.upsert_project(str(project.root), name=project.name)
projects_db.update_project_stats(
    str(project.root),
    image_count=summary.image_count,
    defect_count=summary.defect_count,
    model_count=summary.model_count,
    best_val_iou=summary.best_val_iou,
)

# Project-level DB: authoritative project operation history.
registry = ModelRegistry(project.registry_db_path, project=project)
registry.record_detection_run(detection_run)
registry.upsert_training_epoch(training_epoch)
```

`ProjectsDB(db_path=...)`
`registry_db_path` `artifacts/project.db` `crackz.db`
`project.db`

`core/src/crackz/ai/core/async_callbacks.py`

`RuntimeError: no running event loop`

`sync_q` `async_q`

```
from crackz.ai.core.async_callbacks import CallbackDispatcher

# In training setup (main thread)
dispatcher = CallbackDispatcher(maxsize=100)

# In PyTorch Lightning callback (training thread)
sync_reporter = dispatcher.create_sync_progress_reporter()
sync_reporter(0.5, "Training epoch 5/10")  # Thread-safe put

# In GUI (async event loop)
```

```
async for progress, message in dispatcher.iter_progress():
    update_progress_bar(progress, message)  # Async get
```

`AsyncProgressReporter` `AsyncMetricsReporter`

`core/src/crackz/tracing/`

`--extra tracing`

```
# In core/src/crackz/tracing/setup.py
from crackz.tracing import setup_tracing, is_tracing_enabled

# Automatic instrumentation on setup
if is_tracing_enabled():
    setup_tracing()  # Instruments Anthropic client automatically

# The instrumentation happens automatically via AnthropicInstrumentor
# No manual client instrumentation needed
```

```
# Install tracing dependencies
uv sync --extra tracing

# Enable at runtime
export CRACKZ_TRACING_ENABLED=1
```

`core/src/crackz/ai/evaluation/reporter.py` `core/src/crackz/report/generator.py`

```
from crackz.ai.evaluation.reporter import export_json, export_csv
from crackz.report.generator import ReportGenerator

# Aggregation (format-agnostic)
results = {
    "accuracy": 0.95,
    "precision": 0.92,
    "recall": 0.89,
    "f1_score": 0.90,
    "iou": 0.85
}

# Export to JSON (lightweight)
export_json(
    results,
    output_path=project_path / "evaluation_results.json"
)

# Export to CSV (analysis-friendly)
export_csv(
    results,
    output_path=project_path / "evaluation_results.csv"
)

# Export to DOCX (rich formatting)
generator = ReportGenerator(
    project_path=project_path,
    template_path=template_path
)
generator.generate_report(
    detection_results=results,
```

```
    output_path=project_path / "report.docx"
)
```

`core/src/crackz/ai/device/device.py`

`torch.device("cuda")`

```
from crackz.ai.device.device import AIDevice

# Auto-select best available device
device = AIDevice.default()  # Returns CUDA if available, else MPS, else CPU

# Check availability
available = AIDevice.available_devices()  # [AIDevice.CPU, AIDevice.CUDA]

# Test device works
success, message, details = AIDevice.test_device("cuda")
if success:
    print(f"CUDA available: {details['device_name']}")

# Use in model
model = SegmentationModel()
model.to(device.value)  # device.value is "cuda", "mps", "cpu", etc.
```

`core/src/crackz/ai/detection/severity.py`

```
from crackz.ai.detection.severity import categorize_crack_severity

# Default thresholds: 5% (high), 2% (medium)
severity = categorize_crack_severity(confidence=0.08)  # "Crack Detected"
severity = categorize_crack_severity(confidence=0.03)  # "Suspected Crack"
severity = categorize_crack_severity(confidence=0.01)  # "No Crack"

# Custom thresholds via config
severity = categorize_crack_severity(
    confidence=0.04,
    high_threshold=0.06,    # Raise "Crack Detected" threshold
    medium_threshold=0.03    # Raise "Suspected" threshold
)  # "Suspected Crack" (was "Crack Detected" with defaults)

# Override with has_crack flag (confidence + area thresholds met)
severity = categorize_crack_severity(confidence=0.01, has_crack=True)
# "Crack Detected" (flag indicates combined criteria met)
```

```
# project.yaml
ai:
  detection:
    high_confidence_threshold: 0.05   # 5% crack area
    medium_confidence_threshold: 0.02  # 2% crack area
    min_confidence: 0.005            # 0.5% minimum to save
```

```
# core/src/crackz/cli/options.py
PROJECT_PATH_OPTION = typer.Option(
    None,
    "--project",
    help="Path to project directory",
)

FORCE_OPTION = typer.Option(
    False,
    "--force",
    help="Force operation without confirmation",
)

# core/src/crackz/cli/some_command.py
from crackz.cli.options import PROJECT_PATH_OPTION, FORCE_OPTION

def my_command(
    project_path: Path | None = PROJECT_PATH_OPTION,
    force: bool = FORCE_OPTION,
) -> None:
    pass
```

`print()` `logger` `typer.echo()`

```
from crackz.log import logger
import typer

# User-facing success/error messages
typer.secho("□ Detection completed", fg=typer.colors.GREEN)
typer.secho("× Training failed", fg=typer.colors.RED)

# Technical/operational logs
logger.info("Loading {} images from {}", len(images), path)
logger.debug("Model config: {}", config.model_dump())
logger.error("Failed to load checkpoint: {}", error, exc_info=True)
```

`typer.secho()` `logger.*`
`print()` `typer.echo()`

```
from crackz.cli.exceptions import cli_failure

def train_command():
    try:
        # Business logic
        result = train_model(project_path)
        typer.secho("□ Training completed", fg=typer.colors.GREEN)
    except FileNotFoundError as e:
        cli_failure(e, "File access")
    except RuntimeError as e:
        cli_failure(e, "Training execution")
    except Exception as e:
        cli_failure(e, "Unexpected error")
```

`cli_failure` `logger.error()` `typer.secho()`

```
class AIConfig(BaseModel):
    model: ModelConfig = Field(default_factory=ModelConfig)
    training: TrainingConfig = Field(default_factory=TrainingConfig)

# Usage
config = AIConfig()  # Loads from defaults only
config_from_yaml = AIConfig(**yaml_data)  # Loads from YAML + defaults

# CLI override (doesn't persist to YAML)
if cli_epochs:
    config.training.epochs = cli_epochs  # Runtime only
```

```
logger.info("message")
    ↓
┌─────────────────┐
│  Loguru Router  │
└─────────────────┘
         │
         ├──→ Central Log: /tmp/crackz_logs/app.log (all operations)
         └──→ Project Log: <project>/logs/operations.log (project-specific)
```

```
from loguru import logger

# Configure at startup
logger.add(
    central_log_path,
    format="{time} {level} {message}",
    level="INFO",
)

# Add project-specific handler when project opens
logger.add(
    project_log_path,
    format="{time} {level} {message}",
    level="DEBUG",
    filter=lambda record: record["extra"].get("project") == project_name,
)

# Usage
logger.bind(project=project.name).info("Training started")  # Goes to both logs
```

`frozen=False`

`Field(description=...)`

`schema_version`

```
class ProjectConfig(BaseModel):
    """Project configuration with validation and versioning."""

    schema_version: int = Field(
        default=2,
        ge=1,
        description="Config schema version for migration support",
    )

    name: str = Field(
        ...,  # Required
        min_length=1,
        max_length=100,
        pattern=r"^[a-zA-Z0-9_-]+$",
        description="Project name (alphanumeric, underscore, hyphen)",
    )
```

```
    root: Path = Field(
        ...,  # Required
        description="Project root directory",
    )

    ai: AIConfig = Field(
        default_factory=AIConfig,
        description="AI/ML configuration",
    )

    @field_validator("root")
    @classmethod
    def root_must_exist(cls, v: Path) -> Path:
        """Validate project root exists."""
        if not v.exists():
            raise ValueError(f"Project root does not exist: {v}")
        return v

    @model_validator(mode="after")
    def name_matches_root(self) -> Self:
        """Validate name matches root directory."""
        if self.root.name != self.name:
            raise ValueError(f"Name '{self.name}' doesn't match directory '{self.root.name}'")
        return self
```

`schema_version: int` `crackz migrate <project>`

```
def migrate_v1_to_v2(config_data: dict) -> dict:
    """Migrate project config from v1 to v2."""
    if config_data.get("schema_version", 1) == 1:
        # v2 added 'ai.detection.batch_size'
        if "ai" not in config_data:
            config_data["ai"] = {}
        if "detection" not in config_data["ai"]:
            config_data["ai"]["detection"] = {}
        if "batch_size" not in config_data["ai"]["detection"]:
            config_data["ai"]["detection"]["batch_size"] = 8

        config_data["schema_version"] = 2

    return config_data
```

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```
class CrackzError(Exception):
    """Base exception for all Crackz errors."""
    pass

class ConfigurationError(CrackzError):
    """Invalid configuration."""
    pass

class ProjectError(CrackzError):
```

```
    """Project-related error."""
    pass

class ModelError(CrackzError):
    """Model loading/execution error."""
    pass
```

```
def command():
    try:
        # Business logic
        result = operation()
        typer.secho("✓ Success", fg=typer.colors.GREEN)

    except FileNotFoundError as e:
        # User error: file doesn't exist
        logger.error("File not found: {}", e)
        typer.secho(f"× File not found: {e}", fg=typer.colors.RED)
        raise typer.Exit(1)

    except ConfigurationError as e:
        # User error: invalid config
        logger.error("Configuration error: {}", e)
        typer.secho(f"× Configuration error: {e}", fg=typer.colors.RED)
        raise typer.Exit(1)

    except MemoryError:
        # System error: out of memory
        logger.error("Out of memory during operation")
        typer.secho("× Out of memory. Try reducing batch size.", fg=typer.colors.RED)
        raise typer.Exit(1)

    except Exception as e:
        # Programmer error: unexpected
        logger.exception("Unexpected error: {}", e)
        typer.secho("× Unexpected error. Check logs for details.", fg=typer.colors.RED)
        raise typer.Exit(1)
```

```
        /\
       /  \        E2E (Integration)         ← Few, slow, expensive
      /────\       Tests: 5% (25 tests)
     /      \
    /────────\     Integration Tests        ← Some, medium speed
   /          \    Tests: 15% (100 tests)
  /────────────\
 /______________\  Unit Tests               ← Many, fast, cheap
                   Tests: 80% (570 tests)
```

```
core/tests/
├── unit/                    # Fast, isolated, no I/O
│   └── crackz/
│       ├── ai/
│       │   ├── test_models.py      # Model creation, logic
│       │   ├── test_training.py    # Training loop logic
│       │   └── test_detection.py   # Detection logic
│       ├── cli/
│       │   ├── test_options.py     # CLI option validation
│       │   └── test_commands.py    # Command parsing
│       └── config/
│           └── test_project.py     # Config validation
└── integration/             # Slower, system-level, I/O allowed
    ├── crackz/
    │   ├── test_cli_workflows.py   # End-to-end CLI
    │   └── test_training_pipeline.py # Full training
    └── docs/
        └── test_docs_build.py      # Documentation build

gui/tests/
├── unit/                    # GUI unit tests
└── integration/             # GUI integration tests
```

```
def test_model_creation():
    """Test model factory with different configs."""
    # Arrange
    config = ModelConfig(encoder="resnet34", num_classes=1)

    # Act
    model = create_segmentation_model(config)
```

```python
    # Assert
    assert isinstance(model, nn.Module)
    assert hasattr(model, "encoder")
    assert hasattr(model, "decoder")
```

```python
@pytest.mark.integration
def test_full_training_workflow(tmp_path):
    """Test complete training workflow."""
    # Arrange
    project = create_test_project(tmp_path, num_images=10)

    # Act
    result = train_model(project, epochs=1)

    # Assert
    assert result["final_loss"] < result["initial_loss"]
    assert (project.project_path / "artifacts" / "models" / "checkpoint.ckpt").exists()
```

```python
@pytest.mark.benchmark
def test_detection_performance(benchmark, sample_images):
    """Ensure detection meets performance targets."""
    model = load_model(TEST_CHECKPOINT)

    result = benchmark(detect_cracks, model, sample_images)

    # Verify performance target: <5s per image on CPU
    assert result.mean < 5.0
```

`core/tests/conftest.py`

```python
@pytest.fixture
def tiny_model():
    """Fast model for unit tests."""
    return create_segmentation_model(ModelConfig(
        encoder="resnet18",
        encoder_weights=None,  # No pre-trained weights
        num_classes=1,
    ))

@pytest.fixture
def sample_project(tmp_path):
    """Sample project with test data."""
    project_dir = tmp_path / "test_project"
    initialize_project(project_dir, name="test_project")

    # Add minimal test data
    (project_dir / "images" / "raw").mkdir(parents=True)
    create_test_image(project_dir / "images" / "raw" / "test.png")

    return project_dir

@pytest.fixture
def mock_model(mocker):
    """Mock model for testing without actual inference."""
    mock = mocker.Mock(spec=nn.Module)
    mock.predict.return_value = torch.zeros(1, 1, 512, 512)
    return mock
```

```python
# × BAD: Load everything into memory
all_images = [Image.open(p) for p in image_paths]  # OOM for large datasets
for img in all_images:
    process(img)

# □ GOOD: Process in batches
for batch_paths in chunk_list(image_paths, batch_size=8):
    batch = [Image.open(p) for p in batch_paths]
    process_batch(batch)
    # batch garbage collected after each iteration
```

```
def train_with_cleanup():
    try:
        # Training code
        model.train()
        for batch in dataloader:
            loss = model.training_step(batch)
            loss.backward()
            optimizer.step()
    finally:
        # Always clean up GPU memory
        if torch.cuda.is_available():
            torch.cuda.empty_cache()
```

```
class Project:
    def __init__(self, root: Path):
        self.root = root
        self._config: ProjectConfig | None = None  # Not loaded yet
        self._images: list[Path] | None = None     # Not loaded yet

    @property
    def config(self) -> ProjectConfig:
        """Lazy load configuration."""
        if self._config is None:
            self._config = ProjectConfig.load(
                self.root / f"{self.root.name}_project.yaml"
            )
        return self._config

    @property
    def images(self) -> list[Path]:
        """Lazy load image list."""
        if self._images is None:
            self._images = list((self.root / "images").rglob("*.png"))
        return self._images
```

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```
from pydantic import BaseModel, Field, validator

class ProjectConfig(BaseModel):
    name: str = Field(
        ...,
        min_length=1,
        max_length=100,
        pattern=r"^[a-zA-Z0-9_-]+$",  # Prevent path traversal
    )

    @validator("name")
    def no_path_traversal(cls, v: str) -> str:
        """Prevent directory traversal attacks."""
        if ".." in v or "/" in v or "\\" in v:
            raise ValueError("Invalid project name: contains path separators")
        return v
```

```
def safe_path_join(base: Path, *parts: str) -> Path:
    """Join paths safely, preventing traversal attacks."""
    result = base
    for part in parts:
```

```
        if ".." in part or part.startswith("/") or part.startswith("\\"):
            raise ValueError(f"Unsafe path component: {part}")
        result = result / part

    # Verify result is still under base
    if not result.resolve().is_relative_to(base.resolve()):
        raise ValueError("Path traversal detected")

    return result
```

---

1. `core/src/crackz/cli/my_command.py`
2.

```
from __future__ import annotations

import typer
from crackz.cli.options import PROJECT_PATH_OPTION
from crackz.cli.exceptions import cli_failure
from crackz.log import logger

app = typer.Typer(no_args_is_help=True)

@app.command(name="my-command")
def my_command(
    project_path: Path = PROJECT_PATH_OPTION,
) -> None:
    """Description shown in --help."""
    try:
        # Business logic
        logger.info("Starting operation")
        result = do_operation(project_path)
        typer.secho("✓ Operation completed", fg=typer.colors.GREEN)
    except Exception as e:
        cli_failure(e, "Operation name")
```

3. `core/src/crackz/cli/main.py`
4. `core/tests/unit/crackz/cli/test_my_command.py`
5.

1. `core/src/crackz/config/my_feature.py`

```
from pydantic import BaseModel, Field

class MyFeatureConfig(BaseModel):
    """Configuration for my feature."""

    enabled: bool = Field(
        default=False,
        description="Enable my feature",
    )

    option: str = Field(
        default="default_value",
        description="Feature option",
    )
```

2. `core/src/crackz/config/project.py`

```
class ProjectConfig(BaseModel):
    # ... existing fields ...
    my_feature: MyFeatureConfig = Field(default_factory=MyFeatureConfig)
```

3. `core/src/crackz/project/migrations.py`
4.
5.

1. `core/src/crackz/ai/models/my_model.py`
2.

```
class MyCustomModel(nn.Module):
    def __init__(self, config: ModelConfig):
        super().__init__()
        # Model setup

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        # Forward pass
        return x

    def predict(self, x: torch.Tensor) -> torch.Tensor:
        """Inference interface."""
```

```
        self.eval()
        with torch.no_grad():
            return self(x)
```

3. `core/src/crackz/ai/models/__init__.py`

```
def create_model(config: ModelConfig) -> nn.Module:
    if config.type == "segmentation":
        return create_segmentation_model(config)
    elif config.type == "my_custom":
        return MyCustomModel(config)
    else:
        raise ValueError(f"Unknown model type: {config.type}")
```

4. `core/src/crackz/config/ai.py`
5.

1. `core/src/crackz/ai/detection/my_strategy.py`
2.

```
class MyDetectionStrategy:
    def __init__(self, model: nn.Module, config: DetectionConfig):
        self.model = model
        self.config = config

    def detect(self, images: list[Path]) -> dict[Path, np.ndarray]:
        """Run detection with my strategy."""
        results = {}
        for image_path in images:
            mask = self._detect_single(image_path)
            results[image_path] = mask
        return results
```

3. `core/src/crackz/ai/detection/__init__.py`
4. `core/src/crackz/config/ai.py`
5.

API Reference

API Reference

```
from pathlib import Path
from crackz.project.project import load_project_from_path
from crackz.ai.detection import detect

project = load_project_from_path(Path("demo-project"))
# Optional: custom progress reporter
progress = lambda p, msg=None: print(f"{p:.0%}", msg or "")

detect(project, progress_reporter=progress)
```

```
from crackz.project.project import load_project_from_path
from crackz.ai.training import train

project = load_project_from_path("demo-project")
train(project)  # writes checkpoints & metrics under project.artifacts/checkpoints
```

```
from pathlib import Path
from crackz.project.project import (
    ProjectMetadata, ProjectPaths, create_project, save_project
)

project = create_project(
    metadata=ProjectMetadata(name="demo-project", description="Example"),
    paths=ProjectPaths(
        project_path=Path("demo-project"),
        image_path=Path("demo-project/images"),
        image_mask_path=Path("demo-project/masks"),
        artifacts_path=Path("demo-project/artifacts"),
    ),
    force=False,
)
save_project(project)
```

```
from crackz.project.project import load_project_from_path
project = load_project_from_path("demo-project")
# Uses same logic as CLI import-images (see project.import_images signature)
project.import_images(
    source_path="./samples",  # directory with raw images
    masks_source_path="./samples/masks",  # optional
    image_type=project.ImagesType.RAW if hasattr(project, 'ImagesType') else None,  # or enum from crackz.ImagesType
    size=(512, 512),
    quality=90,
    overwrite=False,
)
```

`CrackzSettings` `CRACKZ_` `CRACKZ_LOG_LEVEL=DEBUG`

`BaseSettings`

▶ `core/src/crackz/config/crackz_settings.py`

`classmethod`

```
from_conf(conf_file)
```

▶ `core/src/crackz/config/crackz_settings.py`

`CrackzError`

▶ `core/src/crackz/config/exceptions.py`

[`ConfigError`]()

▶ `core/src/crackz/config/exceptions.py`

[`ConfigError`]()

▶ `core/src/crackz/config/exceptions.py`

`Project`

`project.project` `Project`

`BaseModel`

▶ `core/src/crackz/config/project_config.py`

`classmethod`

```
load(path)
```

▶ `core/src/crackz/config/project_config.py`

```
save(path)
```

▶ `core/src/crackz/config/project_config.py`

```
to_project()
```

`Project` `ProjectConfig`

▶ `core/src/crackz/config/project_config.py`

`classmethod`

```
from_project(project)
```

`ProjectConfig` `Project`

▶ `core/src/crackz/config/project_config.py`

`Project`

`BaseModel`

▶ `core/src/crackz/project/project.py`

`property`

```
name
```

`property`

```
root
```

`property`

```
config_file
```

`property`

```
detections_path
```


`Path` `Path`

`property`

```
models_path
```


`Path` `Path`

`property`

```
checkpoints_path
```


`Path` `Path`

`property`

```
runtime_path
```


`Path` `Path`

`property`

```
training_path
```

`property`

`detections_visualizations_path`

`Path`	`Path`	

`property`

`detection_results_path`

`property`

`training_metrics_path`

`property`

`tensorboard_logs_path`

`property`

`registry_db_path`

`project.db` `crackz.db`

`property`

`lightning_logs_path`

`property`

`val_path`

`Path`	`Path`	

`property`

`test_path`

`Path`	`Path`	

property

`train_path`

`Path` `Path`

property

`train_mask_path`

`Path` `Path`

property

`val_mask_path`

`Path` `Path`

property

`test_mask_path`

`Path` `Path`

property

`raw_path`

`Path` `Path`

property

`raw_mask_path`

`Path` `Path`

`property`

`project_file_path`

`report_logo_path(default_logo_path)`

▶ `core/src/crackz/project/project.py`

`resolve_task_mode()`

`ai.model.task_mode` `"binary"`

▶ `core/src/crackz/project/project.py`

`sync_derived_fields()`

▶ `core/src/crackz/project/project.py`

`map_type_to_dir(image_type)`

`raw` `train`
`test` `val` `raw`

`Path`
`None`

▶ `core/src/crackz/project/project.py`

`map_type_to_masks_dir(image_type)`

`raw` `train`
`test` `val` `raw`

`Path`
`None`

▶ `core/src/crackz/project/project.py`

`save_project()`

`model_dump`

`model_dump`

▶ `core/src/crackz/project/project.py`

```
import_image_files(
    files,
    dest_dir,
    size=None,
    quality=90,
    *,
    overwrite=False,
    progress_cb=None,
)
```

▶ `core/src/crackz/project/project.py`

```
import_images(
    source_path,
    masks_source_path=None,
    image_type=ImagesType.RAW,
    size=None,
    quality=90,
    *,
    overwrite=False,
    progress_cb=None,
    split=False,
    train_ratio=0.7,
    val_ratio=0.15,
    test_ratio=0.15,
    random_seed=None,
    mask_label_map=None,
)
```

▶ `core/src/crackz/project/project.py`

```
create_project_dir(*, force)
```

`force`

`project_path` `artifacts_path`

▶ `core/src/crackz/project/project.py`

```
clean(*, logs_only=False, keep_best_model=True)
```

▶ `core/src/crackz/project/project.py`

```
create_project_and_save(
    metadata,
    paths,
    ai=default_ai_config,
    logging=default_logging_config,
    *,
    force=False,
)
```

▶ `core/src/crackz/project/project.py`

```
create_project(
    metadata, paths, ai=None, logging=None, *, force=False
)
```

▶ `core/src/crackz/project/project.py`

```
save_project(project)
```

`Project.save_project`

▶ `core/src/crackz/project/project.py`

```
load_project(project_file_path)
```

`Project`

`project_file_path`	`Path`		*required*

[`Project`]()	`Project`
[`Project`]()	`None`

▶ `core/src/crackz/project/project.py`

```
load_project_from_path(project_path)
```

▶ `core/src/crackz/project/project.py`

```
load_from_project_path_or_cfg(cfg, project_path)
```

▶ `core/src/crackz/project/project.py`

`StrEnum`

`RAW`		
`TRAIN`		
`TEST`		
`VAL`		

▶ `core/src/crackz/project/types.py`

`CrackzError`

▶ `core/src/crackz/project/exceptions.py`

`ProjectError`

▶ `core/src/crackz/project/exceptions.py`

`ProjectError`

▶ `core/src/crackz/project/exceptions.py`

`ProjectError`

▶ `core/src/crackz/project/exceptions.py`

`ProjectError`

▶ `core/src/crackz/project/exceptions.py`

`ProjectError`

▶ `core/src/crackz/project/exceptions.py`

`BaseModel`

▶ `core/src/crackz/ai/detection/results.py`

```
categorize_crack_severity(
    confidence,
    *,
    high_threshold=DEFAULT_HIGH_CONFIDENCE,
    medium_threshold=DEFAULT_MEDIUM_CONFIDENCE,
    has_crack=False,
)
```

`confidence`	`float`		*required*
`high_threshold`	`float`		`DEFAULT_HIGH_CONFIDENCE`
`medium_threshold`	`float`		`DEFAULT_MEDIUM_CONFIDENCE`
`has_crack`	`bool`		`False`

`str`	

```
>>> categorize_crack_severity(0.08)  # 8% crack area
'Crack Detected'
>>> categorize_crack_severity(0.03)  # 3% crack area
'Suspected Crack'
>>> categorize_crack_severity(0.01)  # 1% crack area
'No Crack'
>>> categorize_crack_severity(0.01, has_crack=True)
'Crack Detected'  # has_crack flag indicates both confidence and area thresholds met
```

▶ `core/src/crackz/ai/detection/severity.py`

```
create_tensorboard_logger(project)
```

`project`	[`Project`]()		*required*

`TensorBoardLogger | None`	

▶ `core/src/crackz/ai/training/tensorboard.py`

```
train(
    project,
    progress_reporter=None,
    cancellation_token=None,
    metrics_callback=None,
    ckpt_path=None,
    backbone_override=None,
)
```

`project`	[`Project`]()		*required*
`progress_reporter`	`ProgressReporter | None`		`None`
`cancellation_token`	[`CancellationToken`]() `| None`		`None`
`metrics_callback`	`MetricsReporter | None`		`None`
`ckpt_path`	`Path | None`		`None`
`backbone_override`	`str | None`	`ckpt_path` `project.ai.model.backbone`	`None`

`CancellationError`	

▶ `core/src/crackz/ai/training/training_runner.py`

```
detect(
    project,
    progress_reporter=None,
    cancellation_token=None,
    checkpoint_name=None,
    result_callback=None,
    *,
    on_incompatible="original",
)
```

`project`	[`Project`]()		*required*
`progress_reporter`	`ProgressReporter | None`		`None`
`cancellation_token`	[`CancellationToken`]() `| None`		`None`

checkpoint_name	str | None		None
result_callback	Callable[[dict[str, Any]], None] | None		None
on_incompatible	IncompatibleMode	"untrained" "original" "raise"	'original'

None	
None	

CancellationError	
CheckpointIncompatibleError	*on_incompatible* "raise"

▶ core/src/crackz/ai/detection/detection_runner.py

`__getattr__(name)`

▶ core/src/crackz/ai/detection/__init__.py

▶ core/src/crackz/ai/data/dataset.py

`apply(im)`

▶ core/src/crackz/ai/data/dataset.py

Dataset[Sample]

▶ core/src/crackz/ai/data/dataset.py

```
load_image_with_crackz_image(image_path)
```

`image_path`	`Path`		*required*

`Image`	

[`ImageLoadingError`]()	

▶ `core/src/crackz/ai/data/dataset.py`

```
CrackzDetectionDataset(
    image_dir, transform=None, normalization_config=None
)
```

`image_dir`	`Path | str`		*required*
`transform`	`ImageTransform | None`		`None`
`normalization_config`	`NormalizationConfig | None`		`None`

[`CrackzDataset`]()	

▶ `core/src/crackz/ai/data/dataset.py`

```
create_training_datasets(
    train_path,
    train_mask_path,
    val_path,
    val_mask_path,
    test_path,
    test_mask_path,
)
```

`train_path`	`Path`		*required*
`train_mask_path`	`Path`		*required*
`val_path`	`Path`		*required*
`val_mask_path`	`Path`		*required*
`test_path`	`Path`		*required*
`test_mask_path`	`Path`		*required*

`tuple[CrackzDataset, CrackzDataset, CrackzDataset]`	

▼

▶ `core/src/crackz/ai/data/dataset.py`

```
create_dataset(
    project,
    transforms,
    mask_transforms,
    image_dir=None,
    image_masks_dir=None,
)
```

`project`	`Project`		*required*
`transforms`	`ImageTransform | None`		*required*
`mask_transforms`	`MaskTransform | None`		*required*
`image_dir`	`Path | None`		`None`
`image_masks_dir`	`Path | None`		`None`

`CrackzDataset`	

▶ `core/src/crackz/ai/data/dataset.py`

```
collate(batch)
```

▶ `core/src/crackz/ai/data/dataloader.py`

```
make_crackz_loader(
    dataset, batch_size=32, num_workers=8, *, shuffle=True
)
```

▶ `core/src/crackz/ai/data/dataloader.py`

```
create_training_dataloaders(
    train_ds,
    val_ds,
    test_ds,
    batch_size,
    num_workers,
    *,
    shuffle_train=True,
)
```

`train_ds`	[CrackzDataset]()		*required*
`val_ds`	[CrackzDataset]()		*required*
`test_ds`	[CrackzDataset]()		*required*
`batch_size`	`int`		*required*
`num_workers`	`int`		*required*
`shuffle_train`	`bool`		`True`

`tuple[DataLoader, DataLoader, DataLoader]`	

▶ `core/src/crackz/ai/data/dataloader.py`

`CrackzModel`

`CrackzModel` `TaskStrategy` `task_mode` `crackz.ai.model.segmentation.tasks`

`Module`

- `aux_weight`
- `aux_weight`

`focal_weight` `~crackz.ai.model.segmentation.model.CrackzConfig` `aux_weight`

▶ `core/src/crackz/ai/model/segmentation/loss.py`

`forward(y_pred, y_true)`

`dice_weight * Dice + aux_weight * (focal | cross_entropy)`

▶ `core/src/crackz/ai/model/segmentation/loss.py`

dataclass

`CrackzModel`

`lr`	`float`	
`backbone`	`str`	
`in_channels`	`int`	
`num_classes`	`int`	
`threshold`	`float`	
`dice_weight`	`float`	
`focal_weight`	`float`	
`task_mode`	`TaskMode`	
`category_keys`	`tuple[str, ...]`	

▶ `core/src/crackz/ai/model/segmentation/model.py`

staticmethod

`from_project(project, lr)`

`CrackzConfig` `Project`

▶ `core/src/crackz/ai/model/segmentation/model.py`

`LightningModule`

▶ `core/src/crackz/ai/model/segmentation/model.py`

`training_step(batch, _batch_idx)`

▶ `core/src/crackz/ai/model/segmentation/model.py`

`validation_step(batch, _batch_idx)`

▶ `core/src/crackz/ai/model/segmentation/model.py`

`test_step(batch, _batch_idx)`

▶ `core/src/crackz/ai/model/segmentation/model.py`

`on_train_epoch_end()`

▶ `core/src/crackz/ai/model/segmentation/model.py`

`on_validation_epoch_end()`

▶ `core/src/crackz/ai/model/segmentation/model.py`

`on_test_epoch_end()`

▶ `core/src/crackz/ai/model/segmentation/model.py`

`configure_optimizers()`

▶ `core/src/crackz/ai/model/segmentation/model.py`

`get_loss_function(config)`

`loss_name == "combo"` `ComboLoss`

▶ `core/src/crackz/ai/model/segmentation/model.py`

`Callback`

▶ `core/src/crackz/ai/training/callbacks.py`

```
on_train_start(trainer, pl_module)
```

▶ `core/src/crackz/ai/training/callbacks.py`

```
on_train_batch_end(
    trainer, pl_module, outputs, batch, batch_idx
)
```

▶ `core/src/crackz/ai/training/callbacks.py`

```
on_train_epoch_end(trainer, pl_module)
```

▶ `core/src/crackz/ai/training/callbacks.py`

```
on_validation_epoch_end(trainer, pl_module)
```

▶ `core/src/crackz/ai/training/callbacks.py`

```
on_test_epoch_end(trainer, pl_module)
```

▶ `core/src/crackz/ai/training/callbacks.py`

```
on_fit_end(trainer, pl_module)
```

▶ `core/src/crackz/ai/training/callbacks.py`

`Callback`

▶ `core/src/crackz/ai/training/callbacks.py`

```
__call__(progress, message=None)
```

▶ `core/src/crackz/ai/training/callbacks.py`

```
on_train_start(trainer, pl_module)
```

▶ `core/src/crackz/ai/training/callbacks.py`

```
on_train_epoch_start(trainer, pl_module)
```

▶ `core/src/crackz/ai/training/callbacks.py`

```
on_train_batch_end(
    trainer, pl_module, outputs, batch, batch_idx
)
```

▶ `core/src/crackz/ai/training/callbacks.py`

```
on_train_epoch_end(trainer, pl_module)
```

▶ `core/src/crackz/ai/training/callbacks.py`

```
on_fit_end(trainer, pl_module)
```

▶ `core/src/crackz/ai/training/callbacks.py`

```
on_test_end(trainer, pl_module)
```

▶ `core/src/crackz/ai/training/callbacks.py`

`Callback`

▶ `core/src/crackz/ai/training/callbacks.py`

```
on_validation_epoch_end(trainer, pl_module)
```

▶ `core/src/crackz/ai/training/callbacks.py`

```
make_reporter(ui_callback)
```

▶ `core/src/crackz/ai/training/callbacks.py`

```
create_training_callbacks(
    project,
    val_ds,
    tb_logger,
    progress_reporter,
    runtime_tracker,
    cancellation_token,
)
```

`project`	[`Project`]()		*required*
`val_ds`	[`CrackzDataset`]()		*required*
`tb_logger`	`TensorBoardLogger | None`		*required*
`progress_reporter`	`ProgressReporter | None`		*required*
`runtime_tracker`	`RuntimeTracker`		*required*
`cancellation_token`	`CancellationToken | None`		*required*

`list[Any]`	

▶ `core/src/crackz/ai/training/callbacks.py`

```
get_available_checkpoints(checkpoints_dir)
```

`checkpoints_dir`	`Path`		*required*

`list[Path]`	

▶ `core/src/crackz/ai/model/checkpoints.py`

```
has_trained_model(project)
```

`project`	[`Project`]()		*required*

`bool`	

▶ `core/src/crackz/ai/model/checkpoints.py`

```
select_checkpoint(checkpoints_dir)
```

`checkpoints_dir`	`Path`		*required*

`Path | None`	

▶ `core/src/crackz/ai/model/checkpoints.py`

```
peek_checkpoint_backbone(checkpoint_path)
```

▶ `core/src/crackz/ai/model/checkpoints.py`

```
get_checkpoint_compatibility(project)
```

▶ `core/src/crackz/ai/model/checkpoints.py`

```
load_model_from_checkpoint(
    project,
    lr,
    checkpoint_name=None,
    *,
    on_incompatible="original",
)
```

`project`	[`Project`]()		*required*
`lr`	`float`		*required*
`checkpoint_name`	`str | None`		`None`
`on_incompatible`	`IncompatibleMode`	`"raise"` `"untrained"` `"original"` `CheckpointIncompatibleError`	`'original'`

[`CrackzModel`]()	

[`CheckpointIncompatibleError`]()	*on_incompatible* `"raise"`

▶ `core/src/crackz/ai/model/checkpoints.py`

```
create_checkpoint_callback(project)
```

`project`	[`Project`]()		*required*

`ModelCheckpoint`	

▶ `core/src/crackz/ai/model/checkpoints.py`

```
prune_checkpoints(project, keep)
```

`project`	[`Project`]()		*required*
`keep`	`int`		*required*

`list[Path]`	

▶ `core/src/crackz/ai/model/checkpoints.py`

```
plot_image(img_path, out_file, pred_mask)
```

▶ `core/src/crackz/ai/visualization/visualization.py`

```
save_detection_figure(fig, out_file)
```

▶ `core/src/crackz/ai/visualization/visualization.py`

```
save_metrics_json(results, output_dir)
```

▶ `core/src/crackz/ai/visualization/visualization.py`

```
save_training_metrics_json(results, output_dir)
```

▶ `core/src/crackz/ai/visualization/visualization.py`

```
show_image(ax, index, title, img, alpha=None, cmap='gray')
```

▶ `core/src/crackz/ai/visualization/visualization.py`

```
to_numpy_img(tensor)
```

▶ `core/src/crackz/ai/visualization/visualization.py`

`Enum`

▶ `core/src/crackz/ai/device/device.py`

`staticmethod`

```
available_devices()
```

▶ `core/src/crackz/ai/device/device.py`

`staticmethod`

```
default()
```

▶ `core/src/crackz/ai/device/device.py`

`staticmethod`

```
test_device(device_name)
```

`device_name`	`str`		*required*

`bool`	
`str`	•
`dict[str, str | None]`	•
`tuple[bool, str, dict[str, str | None]]`	•

▶ `core/src/crackz/ai/device/device.py`

```
get_accelerator()
```

▶ `core/src/crackz/ai/device/utils.py`

```
resolve_device_and_accelerator(project)
```

`project`	[`Project`]()		*required*

`str`	

▶ `core/src/crackz/ai/device/utils.py`

`AIPipelineError`	
`ModelNotFoundError`	
`DataLoadingError`	

`CrackzError`

▶ `core/src/crackz/ai/exceptions.py`

`CrackzError`

▶ `core/src/crackz/ai/exceptions.py`

`AIPipelineError`

▶ `core/src/crackz/ai/exceptions.py`

`AIPipelineError`

▶ `core/src/crackz/ai/exceptions.py`

`AIPipelineError`

▶ `core/src/crackz/ai/exceptions.py`

[`AIPipelineError`]

▶ `core/src/crackz/ai/exceptions.py`

[`AIPipelineError`]

▶ `core/src/crackz/ai/exceptions.py`

`detection_results.json`

▼

▼

`crackz.ai.eval`

`BaseModel`

▶ `core/src/crackz/ai/detection/results.py`

```
load_detection_results(project_or_path, *, strict=False)
```

▶ `core/src/crackz/ai/detection/results.py`

```
result_has_finding(result)
```

▶ `core/src/crackz/ai/detection/results.py`

`result_present_categories(result)`

▶ `core/src/crackz/ai/detection/results.py`

`result_primary_category(result)`

▶ `core/src/crackz/ai/detection/results.py`

`result_area_pct(result)`

▶ `core/src/crackz/ai/detection/results.py`

`result_category_label(category)`

▶ `core/src/crackz/ai/detection/results.py`

`iter_iou(results)`

▶ `core/src/crackz/ai/detection/results.py`

`DetectionResult`

`aggregate_iou(results)`

▶ `core/src/crackz/ai/detection/eval.py` `basic_summary(results)`

▶ `core/src/crackz/ai/detection/eval.py`

`get_available_checkpoints(checkpoints_dir)`

`checkpoints_dir`	`Path`		*required*

`list[Path]`	

▶ `core/src/crackz/ai/model/checkpoints.py`

`has_trained_model(project)`

`project`	[`Project`]()		*required*

`bool`	

▶ `core/src/crackz/ai/model/checkpoints.py`

`select_checkpoint(checkpoints_dir)`

`checkpoints_dir`	`Path`		*required*

`Path | None`	

▶ `core/src/crackz/ai/model/checkpoints.py`

```
peek_checkpoint_backbone(checkpoint_path)
```

▶ `core/src/crackz/ai/model/checkpoints.py`

```
get_checkpoint_compatibility(project)
```

▶ `core/src/crackz/ai/model/checkpoints.py`

```
load_model_from_checkpoint(
    project,
    lr,
    checkpoint_name=None,
    *,
    on_incompatible="original",
)
```

`project`	[`Project`]()		*required*
`lr`	`float`		*required*
`checkpoint_name`	`str | None`		`None`
`on_incompatible`	`IncompatibleMode`	`"original"` `"untrained"` `"raise"` `CheckpointIncompatibleError`	`'original'`

[`CrackzModel`]()	

`CheckpointIncompatibleError`	*on_incompatible* `"raise"`

▶ `core/src/crackz/ai/model/checkpoints.py`

```
create_checkpoint_callback(project)
```

`project`	`Project`		*required*

`ModelCheckpoint`	

▶ `core/src/crackz/ai/model/checkpoints.py`

```
prune_checkpoints(project, keep)
```

`project`	`Project`		*required*
`keep`	`int`		*required*

`list[Path]`	

▶ `core/src/crackz/ai/model/checkpoints.py`

▶ `core/src/crackz/ai/pipeline/cancellation.py`

```
__init__()
```

▶ `core/src/crackz/ai/pipeline/cancellation.py`

```
cancel()
```

▶ `core/src/crackz/ai/pipeline/cancellation.py`

```
is_cancelled()
```


`bool`

▶ `core/src/crackz/ai/pipeline/cancellation.py`

```
check_cancelled()
```


`CancellationError`

▶ `core/src/crackz/ai/pipeline/cancellation.py`

`Handler`

▶ `core/src/crackz/log/crackz_logger.py` `emit(record)`

▶ `core/src/crackz/log/crackz_logger.py`

`use_rich_console()`

`bool`	

▶ `core/src/crackz/log/crackz_logger.py`

`setup_logging(level=None, log_dir=None, *, use_rich=None)`

`level`	`str | None`		`None`
`log_dir`	`Path | None`		`None`
`use_rich`	`bool | None`		`None`

`Logger`	

▶ `core/src/crackz/log/crackz_logger.py`

`get_logger(name, filename=None, log_dir='logs')`

`logging` `filename` `InterceptHandler`

▶ `core/src/crackz/log/crackz_logger.py`

`TyperGroup`

▶ `core/src/crackz/cli/crackz_cli.py`

```
main(
    args=None,
    prog_name=None,
    complete_var=None,
    standalone_mode=True,
    windows_expand_args=True,
    **extra,
)
```

▶ `core/src/crackz/cli/crackz_cli.py`

```
list_commands(ctx)
```

▶ `core/src/crackz/cli/crackz_cli.py`

```
get_command(ctx, cmd_name)
```

▶ `core/src/crackz/cli/crackz_cli.py`

```
format_commands(ctx, formatter)
```

▶ `core/src/crackz/cli/crackz_cli.py`

```
root_callback(
    ctx,
    version=False,
    debug=False,
    quiet=False,
    log_dir=None,
    telemetry=False,
)
```

▶ `core/src/crackz/cli/crackz_cli.py`

```
main()
```

▶ `core/src/crackz/cli/crackz_cli.py`

```
init_project(
    name=NAME_OPTION,
    image_path=IMAGE_PATH_OPTION,
    image_mask_path=IMAGE_MASK_PATH_OPTION,
    artifacts_path=ARTIFACTS_PATH_OPTION,
    description=DESCRIPTION_OPTION,
    category_template=CATEGORY_TEMPLATE_OPTION,
    task_mode=TASK_MODE_OPTION,
    log_dir=LOG_DIR_OPTION,
    *,
    force=FORCE_OPTION,
)
```

▶ `core/src/crackz/cli/project_cli.py`

```
export_project_config(
    cfg=CFG_OPTION,
    project_path=PROJECT_PATH_OPTION,
    out=EXPORT_OUTPUT_OPTION,
)
```

▶ `core/src/crackz/cli/project_cli.py`

```
migrate_project_config(
    cfg=CFG_OPTION,
    project_path=PROJECT_PATH_OPTION,
    *,
    dry_run=DRY_RUN_OPTION,
    no_backup=NO_BACKUP_OPTION,
    migrate_files=typer.Option(
        default=False,
        help="Also migrate artifacts directory structure (checkpoints to models, etc.)",
    ),
)
```

▶ `core/src/crackz/cli/project_cli.py`

```
clean_project(
    project_path=PROJECT_PATH_OPTION,
    *,
    dry_run=DRY_RUN_OPTION,
    logs_only=LOGS_ONLY_OPTION,
    keep_best_model=KEEP_BEST_MODEL_OPTION,
)
```

▶ `core/src/crackz/cli/project_cli.py`

```
import_images(
    cfg=CFG_OPTION,
    project_path=PROJECT_PATH_OPTION,
    source_path=SOURCE_PATH_OPTION,
    masks_source_path=MASKS_SOURCE_PATH_OPTION,
    image_type=TYPE_OPTION,
    image_size=SIZE_OPTION,
    quality=QUALITY_OPTION,
    target_size=TARGET_SIZE_OPTION,
    *,
    overwrite=OVERWRITE_OPTION,
    split=SPLIT_OPTION,
    train_ratio=TRAIN_RATIO_OPTION,
    val_ratio=VAL_RATIO_OPTION,
    test_ratio=TEST_RATIO_OPTION,
    random_seed=RANDOM_SEED_OPTION,
    mask_label_map=MASK_LABEL_MAP_OPTION,
)
```

`cfg`	`Path | None`		`CFG_OPTION`
`project_path`	`Path | None`		`PROJECT_PATH_OPTION`
`source_path`	`Path`		`SOURCE_PATH_OPTION`
`masks_source_path`	`Path | None`	`source_path`	`MASKS_SOURCE_PATH_OPTION`
`image_type`	[`ImagesType`]()		`TYPE_OPTION`
`image_size`	`SizeType`		`SIZE_OPTION`
`quality`	`int`		`QUALITY_OPTION`
`target_size`	`int | None`		`TARGET_SIZE_OPTION`
`overwrite`	`bool`		`OVERWRITE_OPTION`
`split`	`bool`		`SPLIT_OPTION`
`train_ratio`	`float`		`TRAIN_RATIO_OPTION`

`val_ratio`	`float`		`VAL_RATIO_OPTION`
`test_ratio`	`float`		`TEST_RATIO_OPTION`
`random_seed`	`int | None`		`RANDOM_SEED_OPTION`
`mask_label_map`	`str | None`		`MASK_LABEL_MAP_OPTION`

`BadParameter`	
`Exit`	

`None`	

▶ `core/src/crackz/cli/image_import_cli.py`

```
run_detection(
    cfg=CFG_OPTION,
    project_path=PROJECT_PATH_OPTION,
    model=CHECKPOINT_NAME_OPTION,
)
```

▶ `core/src/crackz/cli/detection_cli.py`

```
detection_train(
    cfg=CFG_OPTION,
    project_path=PROJECT_PATH_OPTION,
    epochs=EPOCHS_OPTION,
    batch_size=BATCH_SIZE_OPTION,
    resume=RESUME_FROM_OPTION,
)
```

`cfg`	`Path | None`	`project_path`	`CFG_OPTION`
`project_path`	`Path | None`	`cfg`	`PROJECT_PATH_OPTION`

`epochs`	`int | None`		`EPOCHS_OPTION`
`batch_size`	`int | None`		`BATCH_SIZE_OPTION`
`resume`	`str | None`		`RESUME_FROM_OPTION`

`BadParameter`	`cfg` `project_path`
`Exit`	

▶ `core/src/crackz/cli/detection_cli.py`

```
prune_checkpoints_cmd(
    keep=KEEP_CHECKPOINTS_OPTION,
    cfg=CFG_OPTION,
    project_path=PROJECT_PATH_OPTION,
)
```

▶ `core/src/crackz/cli/detection_cli.py`

	`CrackzModel` `CrackzDataset`	
	`MetricsLoggerCallback` `resolve_device_and_accelerator` `DetectionResult` `Pipeline`	

1.
2.
3. `DetectionResult`
4.
5. `evaluation.py`
6.
7.
8.
9.
10.
11.
12.
13.
14.
15.
16.
17.
18. `ProjectConfig`

19.
20.
21.
22. `Pipeline`

```
Project -> load_model_from_checkpoint -> CrackzModel.eval() -> DataLoader -> forward -> sigmoid -> threshold -> mask -> metrics(json)
```

`requirements.txt`

Async Callbacks

Async Callback Architecture

```
from crackz.ai.core.async_callbacks import CallbackDispatcher

dispatcher = CallbackDispatcher(maxsize=100)

# Create sync reporter for PyTorch Lightning (training thread)
progress_reporter = dispatcher.create_sync_progress_reporter()

# Use in training (synchronous context)
train(project, progress_reporter=progress_reporter)

# Consume updates asynchronously (main thread)
async for progress, message in dispatcher.iter_progress():
    update_progress_bar(progress, message)
```

```
from crackz.ai.core.async_callbacks import (
    create_async_progress_consumer,
    create_async_metrics_consumer,
)

# Define callback
def update_ui(progress: float, message: str | None = None) -> None:
    progress_bar.set_value(progress)

# Create async consumer
consumer = create_async_progress_consumer(dispatcher, update_ui)

# Run in async context
await consumer()
```

```
from crackz_gui.qt.async_bridge import (
    start_gui_callback_consumers,
    stop_gui_callback_consumers,
)

# Start consumers (manages event loop)
loop, tasks = start_gui_callback_consumers(
    dispatcher,
    progress_callback=update_progress,
    metrics_callback=update_metrics,
)

# ... run training ...
```

```
# Cleanup
stop_gui_callback_consumers(dispatcher, loop)
```

`TaskSignals` `async_bridge.py`

```
from crackz_gui.qt.async_bridge import TaskSignals, run_in_background

def _do_work(signals: TaskSignals) -> None:
    signals.progress.emit(0.5)   # Safe from any thread
    signals.status.emit("Processing...")
    result = expensive_operation()
    signals.data.emit(result)
    signals.finished.emit()

signals = run_in_background(_do_work)
signals.progress.connect(self._on_progress)
signals.finished.connect(self._on_complete)
signals.error.connect(self._on_error)
```

`threading.Lock` `after()`

`crackz_gui.qt.async_bridge`

```
def progress_callback(progress: float, message: str | None = None) -> None:
    print(f"Progress: {progress:.2%}")

train(project, progress_reporter=progress_callback)
```

```
dispatcher = CallbackDispatcher(maxsize=100)

# Create sync reporters
progress_reporter = dispatcher.create_sync_progress_reporter()
metrics_reporter = dispatcher.create_sync_metrics_reporter()

# Start training in background
training_thread = threading.Thread(
    target=train,
    args=(project,),
    kwargs={
        "progress_reporter": progress_reporter,
        "metrics_callback": metrics_reporter,
    }
)
training_thread.start()

# Consume updates asynchronously
async def consume_updates():
    async for progress, message in dispatcher.iter_progress():
        print(f"Progress: {progress:.2%} - {message}")

asyncio.run(consume_updates())
```

```
from crackz.ai.core.async_callbacks import CallbackDispatcher
from crackz_gui.qt.async_bridge import (
    start_gui_callback_consumers,
    stop_gui_callback_consumers,
)

dispatcher = CallbackDispatcher(maxsize=100)

# Start async consumers with Qt-thread-safe GUI updates
loop, tasks, emitters = start_gui_callback_consumers(
    dispatcher,
    progress_callback=self.update_progress_bar,
```

```
    metrics_callback=self.update_metrics_display,
)
self._qt_emitters = emitters  # Keep signal emitters alive

# Create reporters
progress_reporter = dispatcher.create_sync_progress_reporter()
metrics_reporter = dispatcher.create_sync_metrics_reporter()

try:
    # Run training in background thread
    training_thread = threading.Thread(
        target=train,
        args=(project,),
        kwargs={
            "progress_reporter": progress_reporter,
            "metrics_callback": metrics_reporter,
        }
    )
    training_thread.start()
    training_thread.join()
finally:
    stop_gui_callback_consumers(dispatcher, loop)
```

```
Training Loop:
  Process batch → Update metrics → **Wait for UI** → Next batch
                                      ↑ 1-5ms overhead
```

```
Training Loop:
  Process batch → Put in queue → Next batch (no wait!)
                       ↓ 0.1ms
  Async Consumer: Takes from queue → Updates UI
```

```
dispatcher = CallbackDispatcher(maxsize=100)

# ... use reporters ...

# Check progress queue stats
stats = dispatcher.progress_stats
print(f"Total items put: {stats.total_put}")
print(f"Total items retrieved: {stats.total_get}")
print(f"Total items dropped: {stats.total_dropped}")
print(f"Current queue size: {stats.current_size}")
print(f"Max queue size: {stats.max_size}")

# Check metrics queue stats
metrics_stats = dispatcher.metrics_stats
```

`total_dropped`

- `sync_q` `async_q`
- `threading.Event`
-
-

```
Training Thread          Async Event Loop Thread     GUI Main Thread
     |                            |                        |
     ├─ sync_q.put() ────────────>|                        |
     |  (thread-safe)             |                        |
     |                            ├─ async_q.get()         |
     |                            |                        |
```

```
    │                            ├─ Qt signal emit ─────>│
    │                            │                        │─ Update UI
```

```
# Before (still works)
train(project, progress_reporter=my_callback)

# After (optional upgrade for better performance)
dispatcher = CallbackDispatcher()
train(project, progress_reporter=dispatcher.create_sync_progress_reporter())
```

```
from crackz.ai.core.async_callbacks import CallbackDispatcher
from crackz_gui.qt.async_bridge import start_gui_callback_consumers

dispatcher = CallbackDispatcher(maxsize=100)
loop, tasks = start_gui_callback_consumers(
    dispatcher,
    progress_callback=update_progress,
    metrics_callback=update_metrics,
)

# Use dispatcher's sync reporters in training
train(
    project,
    progress_reporter=dispatcher.create_sync_progress_reporter(),
    metrics_callback=dispatcher.create_sync_metrics_reporter(),
)

stop_gui_callback_consumers(dispatcher, loop)
```

`crackz.ai.core.async_callbacks`

- `CallbackDispatcher`
- `QueueStats`
- `create_async_progress_consumer()`
- `create_async_metrics_consumer()`

`crackz_gui.qt.async_bridge`

- `AsyncEventLoop`
- `create_threadsafe_gui_callback()`
- `create_threadsafe_metrics_callback()`
- `start_gui_callback_consumers()`
- `stop_gui_callback_consumers()`

`help()`

```
from crackz.ai.core.async_callbacks import CallbackDispatcher
help(CallbackDispatcher)
```

docs/examples/async_callbacks_example.py

MCP Integration

MCP (Model Context Protocol) Integration

`.github/copilot-instructions.md`

```
┌──────────────────────────┬───────────────────┬───────────────────┐
│ Development Agents       │ GitHub Copilot    │ End-User GUI      │
├──────────────────────────┼───────────────────┼───────────────────┤
│ ✅ 4 MCP Servers         │ ✗ No MCP          │ ✗ No MCP          │
│   • filesystem           │                   │                   │
│   • github               │ IDE context       │ Simple API        │
│   • python               │ Workspace files   │ Project info only │
│   • uv                   │                   │                   │
├──────────────────────────┼───────────────────┼───────────────────┤
│ Tools: Claude Desktop,   │ Tools: VS Code,   │ Tools: Crackz GUI │
│        Cline             │        IDE        │        Chat Panel │
└──────────────────────────┴───────────────────┴───────────────────┘
```

```
TOOL CALL: filesystem.readDirectory("src/")
→ returns 50 files in context

TOOL CALL: filesystem.readFile("src/file1.py")
→ returns full file in context

TOOL CALL: filesystem.readFile("src/file2.py")
→ returns full file in context
```

```
# Discover and filter in code
files = await filesystem.readDirectory("src/")
python_files = [f for f in files if f.endswith(".py")]

# Load only what's needed
for filename in python_files[:5]:  # Top 5 only
    content = await filesystem.readFile(f"src/{filename}")
    # Process in code, log summary only
    print(f"{filename}: {len(content)} lines")
```

1.
2.
3.
4.

`<type>/<scope>-<short-slug>`

```
feat/gui-live-preview       # New GUI feature
fix/pdf-rendering-crash     # Bug fix in PDF generation
perf/dataloader-cache       # Performance improvement
docs/api-reference-update   # Documentation update
refactor/config-cleanup     # Code refactoring
```

`<type>(<scope>): <description>`

```
feat(gui): add real-time detection preview
fix(cli): correct path handling in Windows
perf(dataloader): implement batch caching
docs(api): update training configuration reference
```

`<type>(<scope>): <description>`

```
feat(gui): add export panel for detection results
fix(pdf): handle missing hero image gracefully
perf(training): optimize data loading pipeline
```

```
Fix GUI bugs and add new feature
Update documentation
Add export functionality
```

`feat:` `feat(<scope>):` `fix:` `fix(<scope>):`
`perf:` `perf(<scope>):`

`docs:` `refactor:`
`test:` `chore:` `ci:` `build:`

`typer.secho()` `logger` `crackz.log`
`crackz.cli.options` `print()`

`uv run task tests`

`uv run task mypy`

```
crackz mcp init
```

`.mcp/config.json`

```
crackz mcp show
```

```
crackz mcp validate
```

`.mcp/config.json`

```
{
  "mcpServers": {
    "filesystem": {
      "type": "stdio",
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-filesystem", "."],
      "description": "Filesystem operations with progressive disclosure"
    },
    "github": {
      "type": "stdio",
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-github"],
      "env": {
        "GITHUB_PERSONAL_ACCESS_TOKEN": "${GITHUB_TOKEN}"
      },
      "description": "GitHub operations (issues, PRs, releases)"
    },
    "python": {
      "type": "stdio",
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-python"],
      "description": "Python code execution environment"
    },
    "uv": {
      "type": "stdio",
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-uv"],
      "description": "UV package manager operations"
    }
  }
}
```

```json
{
  "metadata": {
    "project": "Crackz",
    "language": "python",
    "version": "3.13",
    "package_manager": "uv",
    "framework": "pytorch-lightning",
    "cli": "typer",
    "gui": "pyside6",
    "testing": "pytest",
    "ci": "github-actions",
    "release": "semantic-release",
    "code_execution": true,
    "progressive_disclosure": true
  }
}
```



```json
{
  "patterns": {
    "code_execution": {
      "enabled": true,
      "description": "Use code execution to interact with MCP servers efficiently",
      "examples": [
        "Filter large datasets in code before returning results",
        "Loop over operations without alternating tool calls",
        "Compose multiple MCP operations in a single execution"
      ]
    },
    "progressive_disclosure": {
      "enabled": true,
      "description": "Load tool definitions on-demand",
      "workflow": [
        "List directories to discover modules",
        "Read specific files as needed",
        "Load documentation when relevant"
      ]
    }
  }
}
```


`--output` `-o` `.mcp/config.json` `--force` `-f`

```bash
# Create with defaults
crackz mcp init

# Create in custom location
crackz mcp init --output custom/mcp.json

# Overwrite existing config
crackz mcp init --force
```

`--config` `-c` `.mcp/config.json`

```bash
# Show default config
crackz mcp show

# Show custom config
crackz mcp show --config custom/mcp.json
```

`--config` `-c` `.mcp/config.json`

```
# Validate default config
crackz mcp validate

# Validate custom config
crackz mcp validate --config custom/mcp.json
```

`.mcp/config.json`

```
{
  "mcpServers": {
    "filesystem": { ... },
    "custom-api": {
      "type": "stdio",
      "command": "node",
      "args": ["./custom-mcp-server.js"],
      "description": "Custom MCP server for specialized operations"
    }
  }
}
```

```
{
  "metadata": {
    "project": "Crackz",
    "language": "python",
    "team": "ML Ops",
    "repository": "https://github.com/Anaxiatech-AB/crackz",
    "custom_field": "your_value"
  }
}
```

1.
2.
3.
4.
5.

```
# List Python files in src/
files = await filesystem.readDirectory("src/crackz")
py_files = [f for f in files if f.endswith(".py")]

# Read only the first 3 for initial analysis
for filename in py_files[:3]:
    content = await filesystem.readFile(f"core/src/crackz/{filename}")
    lines = len(content.split('\n'))
    print(f"{filename}: {lines} lines")
```

```
# Get recent PRs and filter in code
prs = await github.listPullRequests(repo="Anaxiatech-AB/crackz", state="open")
priority_prs = [pr for pr in prs if "feat:" in pr.title or "fix:" in pr.title]

print(f"Found {len(priority_prs)} priority PRs out of {len(prs)} total")
for pr in priority_prs[:5]:  # Show top 5 only
    print(f"#{pr.number}: {pr.title}")
```

```
# Install multiple packages in one operation
packages = ["pytest", "ruff", "mypy"]
for pkg in packages:
    await uv.install(package=pkg)
    print(f"✓ Installed {pkg}")
```

`logger`

`print()`

1.
2. `CRACKZ_LOG_DIR` `--log-dir`
3.
4.
5. `<project>/logs/`
6.
7.
8.

Info level - operational progress

Warning level - non-critical issues

Error level - failures with details

Debug level - detailed troubleshooting info

Tracing & Observability

Tracing with OpenTelemetry

```
uv sync --extra tracing
```

`opentelemetry-instrumentation-anthropic` `opentelemetry-sdk` `opentelemetry-exporter-otlp-proto-http`

`Ctrl+Shift+P` `Cmd+Shift+P`

```
AI Toolkit: Open Tracing
```

`ai-mlstudio.tracing.open`

```
export CRACKZ_TRACING_ENABLED=1
crackz-gui  # Tracing auto-enabled
```

```
from crackz.tracing import setup_tracing

# Initialize with default AI Toolkit endpoint
setup_tracing()

# Or custom endpoint
setup_tracing(
    endpoint="http://custom-collector:4318/v1/traces",
    service_name="crackz-production"
)
```

-
-
-
-
-

```
crackz
└── anthropic.messages.create
    ├── model: claude-sonnet-4-5-20250929
    ├── input_tokens: 1245
    ├── output_tokens: 382
    └── duration: 2.3s
```

`CRACKZ_TRACING_ENABLED`	`0`	`1` `0`

`http://localhost:4318/v1/traces` `http://localhost:4317`

```
setup_tracing(endpoint="http://your-collector:4318/v1/traces")
```

```
# Enable tracing for all CLI operations
export CRACKZ_TRACING_ENABLED=1
crackz detect run --project my_project

# Disable for specific run
export CRACKZ_TRACING_ENABLED=0
crackz detect run --project my_project
```

```
# In main.py or startup script
from crackz.tracing import setup_tracing

# Enable before starting GUI
if os.getenv("CRACKZ_TRACING_ENABLED") == "1":
    setup_tracing()

# Run GUI
from crackz_gui.qt.app import main
main()
```

```
from crackz.tracing import setup_tracing, shutdown_tracing

# Initialize tracing
setup_tracing(service_name="crackz-batch-processor")

# Your AI operations (automatically traced)
from crackz_gui.services.ai_chat import AIChatService
chat = AIChatService(...)
chat.send_message("Analyze this crack detection result")

# Clean shutdown (flushes pending spans)
shutdown_tracing()
```

```
uv sync --extra tracing
```

```
Ctrl+Shift+P → AI Toolkit: Open Tracing
```

`uv sync --extra tracing`
`CRACKZ_TRACING_ENABLED=1`

`shutdown_tracing()`

```
import atexit
from crackz.tracing import shutdown_tracing

atexit.register(shutdown_tracing)
```

`CRACKZ_TRACING_ENABLED=1`

```
from opentelemetry import trace

tracer = trace.get_tracer(__name__)

with tracer.start_as_current_span("custom-operation") as span:
    span.set_attribute("project.name", "harbor_alpha")
    span.set_attribute("image.count", 1024)
    # Your code here
```

```
from opentelemetry.sdk.trace.sampling import TraceIdRatioBased

# Sample 10% of traces
sampler = TraceIdRatioBased(0.1)

# Pass to setup_tracing (requires custom implementation)
```

Contributing

Contributing Guide

- `.github/copilot-instructions.md`
  `.github/workflows/copilot-setup-steps.yml`

`<type>/<scope>-<short-slug>`

```
# AI creates feature branch
git checkout -b feat/gui-export-panel

# AI makes changes with proper commit messages
git commit -m "feat(gui): add export panel component"
git commit -m "feat(gui): integrate export panel with main view"

# When creating PR, AI suggests proper title
# 🎯 PR Title: "feat(gui): add export panel for detection results"
# This triggers a MINOR version release (e.g., 1.2.0 → 1.3.0)
```

```
# create & activate venv
uv venv
source .venv/bin/activate  # Windows: .venv\Scripts\activate

# install with dev dependencies
uv sync --dev

# run CLI help
uv run crackz --help

# run GUI (optional)
uv run crackz-gui
```

`pyproject.toml` `[tool.taskipy.tasks]`

```
uv run task lint        # ruff lint (auto-fix)
uv run task format      # ruff format
uv run task mypy        # static type check
uv run task tests       # unit tests only
uv run task integration # integration tests
uv run task all_tests   # full test matrix
uv run task docs        # live docs (mkdocs serve)
uv run task build_docs  # strict build
```

- `tests/unit` `core/tests/unit` `gui/tests/unit`
- `tests/integration` `@pytest.mark.integration`
- `backend/backend/tests/unit/crackz/cli/test_cli_benchmark.py` `@pytest.mark.benchmark`

- `tests/test_data`
-
- `core/tests` `gui/tests`

```
uv run task tests
```

```
uv run task all_tests
```

```
uv run task benchmark
```

`pyproject.toml` `fail_under = 75`

`tests/integration/crackz/gui/`

```
# Run all integration tests (includes GUI smoke tests)
uv run task integration

# Run only GUI smoke tests
uv run pytest tests/integration/crackz/gui/test_gui_smoke.py

# Run GUI in smoke mode (auto-close after initialization)
set CRACKZ_GUI_SMOKE=1  # Windows CMD
export CRACKZ_GUI_SMOKE=1  # Linux/macOS
uv run crackz-gui
```

`CRACKZ_GUI_SMOKE=1`

`CI` `GITHUB_ACTIONS`

```
tests/integration/crackz/gui/
├── test_gui_smoke.py        # Smoke tests for GUI initialization
└── ...                       # Future GUI integration tests
```

1.

```
env = os.environ.copy()
env["CRACKZ_GUI_SMOKE"] = "1"
subprocess.run([sys.executable, "gui/src/crackz_gui/crackz_gui.py"], env=env)
```

2.

```
@pytest.mark.skipif(not _has_display() or os.environ.get("CI"),
                    reason="Requires display")
def test_component(qapp):
    component = MyComponent()
    assert component is not None
```

3.

```
def test_no_unsupported_parameters():
    # Ensure widget initialization works without errors
    browser.toggle_toolbar_buttons(visible=True)  # Should not raise
```

`benchmark`

```
# Run benchmarks - compares to last saved baseline and fails on regression
uv run task benchmark
```

```
# Update the baseline after intentional performance-impacting changes
uv run task benchmark_save
```

```
# Manual benchmark with comparison
uv run pytest -m benchmark --benchmark-only --benchmark-autosave --benchmark-compare --benchmark-compare-fail=mean:20%
```

`.benchmarks/`

- `uv run task format` `lint`
-
-
- `loguru` `print`
-

```
<type>(<optional-scope>): <description>
```

`feat:` `feat(<scope>):` `fix:` `fix(<scope>):` `perf:` `perf(<scope>):`

`docs:` `refactor:` `test:` `chore:` `ci:` `build:` `style:`

```
feat(gui): add return-to-run-view functionality
fix(pdf): correct missing hero image
perf(dataloader): optimize batch processing
```

```
Fix GUI image display and add return-to-run-view functionality
Add new feature to handle user authentication
Update documentation for new feature
```

`feat:` `fix:` `Feat:` `Fix:` `:` `feat(gui):` `fix(cli):`

```
# Install pre-commit hooks (one-time setup)
pip install pre-commit  # or: uv tool install pre-commit
pre-commit install
pre-commit install --hook-type commit-msg
```

```
# Test with a valid message
echo "feat(gui): add export panel" > temp_msg.txt
cz check --commit-msg-file temp_msg.txt

# Test with an invalid message
echo "Fix bug" > temp_msg.txt
cz check --commit-msg-file temp_msg.txt
```

```
git commit --no-verify -m "your message"
```

```
uv run task docs  # serves at http://127.0.0.1:8000
```

```
uv run task build_docs
```

`scripts/export_pdf.py`

```
uv run task pdf
```

- `main` `cz bump` `pyproject.toml` `CHANGELOG.md` `main`

`logs/crackz.log`

---

main

<type>/<scope>-<short-slug>

feat/gui-live-preview fix/pdf-rendering-crash perf/dataloader-optimization docs/api-reference-update refactor/config-module-cleanup

feat/gui-live-preview
feat(gui): add live preview panel fix/pdf-crash fix(pdf): handle null hero image

- main

<type>/<scope>-<short-slug>

```
feat/gui-live-preview
fix/pdf-hero-image
refactor/logging-bootstrap
perf/dataloader-prefetch
chore/ci-release-step
hotfix/0.17.1-pdf-null-crash
exp/vector-db-poc
```

feat fix perf
refactor docs test
chore ci build style
hotfix exp
wip

hotfix/* main fix:

```
<type>(<optional-scope>): <imperative summary>

<body — optional>

<footer — BREAKING CHANGE / issue refs>
```

```
feat(gui): add real-time detection preview panel
fix(pdf): correct missing hero when logo == cover image
refactor(logging): unify central + project logger bootstrap
chore(ci): refine release workflow version verification
```

```
feat!: remove legacy --batch flag

BREAKING CHANGE: The --batch flag was replaced by --batch-size.
```

feat fix ! BREAKING CHANGE:

-
- `feat(gui): add live preview`
- `fix(cli): correct path handling`
- `docs: update installation guide`
- `Fix GUI bugs`
- `Add new feature`
-
-
-
-

`<type>(<scope>): <description>` `feat:` `fix:` `docs:`

`Dev Build` `<base>.devYYYYMMDD+<shortsha>` `dev-<computed-dev-version>` `dev-`

1. `main`
2. `cz bump`
3.
4. `CHANGELOG.md` `vX.Y.Z` `main`
5.
6.
7. `:X.Y.Z` `:latest`
8.
9.
10.
11.
12. `v0.35.0` `latest`
13.
14.

1. `hotfix/<next-patch>-<short-desc>` `main`
2. `fix:`
3. `cz bump`

`TrainingConfig`

`tests/unit/crackz/project/test_training_config_schema.py` `.pre-commit-config.yaml`

```
pip install pre-commit  # or uv tool install pre-commit
pre-commit install
```

`test_project.py`

`docs/configuration.md` *TrainingConfig Contract & Compatibility* `CHANGELOG.md`

`ModelConfig` `DetectionConfig`

		`feat/cli-batch-export`
		`fix/train-metrics-path`
		`perf/dataloader-cache`
		`refactor/model-config`
		`docs/pdf-section`
		`test/cli-train-e2e`
		`chore/update-ruff`
		`ci/release-tag-fix`
		`build/multiarch-images`
		`style/ruff-cleanup`
		`hotfix/0.17.2-threshold-null`
		`exp/alt-model-head`
		`wip/gui-layout-pass1`

`feat(gui):`

`BREAKING CHANGE:`

-
-
- `feat/*`

`typer.secho`

```
# Success messages
typer.secho(" Operation completed successfully", fg=typer.colors.GREEN)

# Error messages
typer.secho("× Operation failed", fg=typer.colors.RED)

# Warning messages
typer.secho("⚠ Warning message", fg=typer.colors.YELLOW)

# Info messages
typer.secho("ℹ Information", fg=typer.colors.CYAN)
```

`logger`

```
from crackz.log import logger

# Detailed progress, technical info, debugging
logger.info("Detailed operational information")
logger.warning("Warning about internal state")
logger.error("Error details for troubleshooting")
```

NEVER USE

- `print()`
- `typer.echo()` `typer.secho()`
- `console.print()`

-

`core/src/crackz/cli/options.py`

```
from crackz.cli.options import PROJECT_PATH_OPTION, FORCE_OPTION

def command_function(
    project_path: Path | None = PROJECT_PATH_OPTION,
    force: bool = FORCE_OPTION,
) -> None:
```

```
# Only for options not reused elsewhere
LOCAL_OPTION = typer.Option(
    None,
    "--local-flag",
    help="Module-specific option not reused elsewhere",
)
```

NEVER

- `FORCE_OPTION`
- `typer.Option()`
-

```
"""Module docstring explaining CLI command purpose."""

from __future__ import annotations

from pathlib import Path
import typer

from crackz.cli.exceptions import cli_failure
from crackz.cli.options import PROJECT_PATH_OPTION, FORCE_OPTION
from crackz.log import logger  # Always import logger

app = typer.Typer(no_args_is_help=True)

@app.command(name="command")
def command_function(
    project_path: Path | None = PROJECT_PATH_OPTION,
    force: bool = FORCE_OPTION,
) -> None:
    """Command description."""
    try:
        # Business logic here
        logger.info("Processing started")
        typer.secho("□ Operation completed", fg=typer.colors.GREEN)
    except Exception as e:
        cli_failure(e, "Operation name")
```

-
-
-
- `loguru` `print()`
-

-
-
-
-

-
- `tests/unit/`
- `tests/integration/`
- `@pytest.mark.integration` `@pytest.mark.benchmark`

1.
2. `options.py`
3.
4. `uv run task lint && uv run task format`
5. `uv run task mypy`
6. `uv run task tests`

1. `typer.secho()`
2. `logger`
3. `options.py`
4.
5.

1. **`print()`**
2. **`typer.echo()`** `typer.secho()`
3. `options.py`
4.
5.
6.
7.

Release Flow

Release Flow

`.github/workflows/release.yml`

`main`

1.
2.
3. `CHANGELOG.md` `vX.Y.Z`
4. `main`
5.
6.

- `workflow_run` `Integration Tests` `main`
- `workflow_dispatch` `confirm=YES`

```
uv run cz bump --dry-run
```

- `0`
- `21` `NoneIncrementExit`
-

- `feat:`
- `fix:`
- `perf:`

`docs:` `chore:` `refactor:` `test:` `ci:` `build:` `style:`

`version-bump`

-
- `commitizen`
-
-
- `cz bump --yes --changelog`

-
-
- `CHANGELOG.md`

-
-

```
uv lock
```

`uv.lock`

`main`

- `HEAD` `main`
-

`version-bump`

- `CHANGELOG.md`
- `vX.Y.Z`

- `build-wheels`
-
-
-
- `build-sdist`
- `.tar.gz`

`crackz`

`release`

-
- `dist/`
- `site/crackz-docs.pdf` `.0`
- `CHANGELOG.md`
- `SHA256SUMS.txt`
-
-
- `--draft=false`

-
- `cu118` `cu128`
-
- `CHANGELOG.md`
- `SHA256SUMS.txt`
- `.0`

`dist-combined`

- `ai-release-notes.yml`
-
-
- `pyinstaller-build.yml`
-
-
- `docker-build.yml`
-
-

-
-
-
-

- `feat:` `fix:` `perf:` `main`
-

-
-
-
- `.0`

```
gh run list --limit 10
gh run view <run-id>
gh run watch
```

PR Review Workflow

Pull Request Review Workflow with Claude Code

- `gh`
-
-

```
# List open PRs
gh pr list

# View PR details
gh pr view <number>

# View PR diff
gh pr diff <number>

# Check CI status
gh pr checks <number>

# Review PR
gh pr review <number> --approve
gh pr review <number> --comment -b "your comment"
gh pr review <number> --request-changes -b "needs changes"
```

```
# List all open PRs
gh pr list

# List PRs with specific labels
gh pr list --label "ready-for-review"

# List your own PRs
gh pr list --author @me
```

```
# View PR summary
gh pr view 254

# View PR with comments
gh pr view 254 --comments

# Open PR in browser
gh pr view 254 --web
```

```
fix(docs): resolve build failures and clarify release standards #254
Open • theresiasnow wants to merge 4 commits into main from fix/docs-build-and-standards

  This PR fixes critical documentation build failures...

  View this pull request on GitHub: https://github.com/owner/repo/pull/254
```

```
# View full diff
gh pr diff 254

# View only changed filenames
gh pr diff 254 --name-only

# View diff in browser
gh pr diff 254 --web
```

```
# View all checks
gh pr checks 254
```

```
# View only required checks
gh pr checks 254 --required

# Watch checks until completion
gh pr checks 254 --watch

# Open checks in browser
gh pr checks 254 --web
```

```
✓ Build and Test (ubuntu-latest) — pass
✓ Build and Test (windows-latest) — pass
✓ Integration Tests — pass
✓ Lint and Format — pass
```

```
# Checkout PR to review code locally
gh pr checkout 254

# This creates/switches to the PR branch
# Now you can:
# - Run tests: uv run task tests
# - Run linting: uv run task lint
# - Build docs: uv run task build_docs
# - Run the code locally
```

```
Review the changes in scripts/validate_commit_msg.py for:
- Code quality and readability
- Error handling
- Edge cases
- Documentation completeness
```

```
Review the changes in docs/contributing.md for:
- Clarity and completeness
- Correct formatting
- Broken links
- Consistency with existing docs
```

```
Check if the changes need additional tests and identify any edge cases not covered.
```

```
gh pr review 254 --approve -b "LGTM! Code quality looks good, tests pass."
```

```
gh pr review 254 --comment -b "Good work! Minor suggestion: consider adding error handling for X."
```

```
gh pr review 254 --request-changes -b "Please address the linting issues in validate_commit_msg.py before merging."
```

```
gh pr review 254 --approve -b "$(cat <<'EOF'
Great PR! Here's my review:

□ Code quality: Excellent
□ Tests: All passing
□ Documentation: Clear and complete
□ Linting: All checks pass

Minor suggestions:
- Consider adding a note about pre-commit hook in README
- The error messages are very helpful

Approved for merge.
EOF
)"
```

```
# After review, switch back to main
git checkout main
```

```
# List all PRs and store numbers
gh pr list

# Review each PR in sequence
for pr in 254 255 256; do
    echo "Reviewing PR #$pr"
    gh pr view $pr
    gh pr diff $pr --name-only
    gh pr checks $pr
    # Ask Claude Code to review changes
done
```

```
# Get PR data as JSON
gh pr view 254 --json title,body,author,createdAt,additions,deletions,files

# Get check status as JSON
gh pr checks 254 --json name,state,completedAt
```

```
# Continuously monitor checks until completion
gh pr checks 254 --watch --interval 30

# This will update every 30 seconds until all checks complete
```

Always Check CI Before Reviewing

```
# Wait for checks to complete before reviewing
gh pr checks 254 --watch
```

Review Locally for Complex Changes

```
# For complex PRs, checkout and test locally
gh pr checkout 254
uv run task lint
uv run task tests
uv run task build_docs
```

Use Claude Code for Deep Review

Provide Constructive Feedback

```
# Instead of just "needs work"
gh pr review 254 --request-changes -b "$(cat <<'EOF'
Thanks for the PR! A few items to address:

1. Linting: Please run `uv run task lint` to fix PLR2004 error
2. Tests: Add test case for empty commit message
3. Docs: Update contributing.md with hook installation steps

Otherwise looks great!
EOF
)"
```

Verify Release Triggers

```
# Check PR title format
gh pr view 254 --json title

# Should match: <type>(<scope>): <description>
# Examples:
```

```
# □ feat(gui): add export panel
# □ fix(docs): resolve build failures
# × Fix bugs
```

```
# 1. View the PR
gh pr view <number>

# 2. Check what files changed
gh pr diff <number> --name-only

# 3. View the actual changes
gh pr diff <number>

# 4. Checkout and test the fix
gh pr checkout <number>
# Test the fix...

# 5. Run all pre-commit checks (lint, format, mypy, tests)
uv run task pre_commit

# 6. Approve if all looks good
gh pr review <number> --approve -b "Bug fix verified, all pre-commit checks pass."
```

```
# 1. View the PR
gh pr view <number>

# 2. Checkout the branch
gh pr checkout <number>

# 3. Build docs to check for errors
uv run task build_docs

# 4. Review locally with live server
uv run task docs
# Opens at http://127.0.0.1:8000

# 5. Ask Claude Code to review
"Review the documentation changes for clarity, completeness, and broken links"

# 6. Approve or request changes
gh pr review <number> --approve
```

```
# 1. View PR and check size
gh pr view <number> --json additions,deletions,files

# 2. Check CI status
gh pr checks <number>

# 3. Checkout locally
gh pr checkout <number>

# 4. Run full test suite
uv run task all_tests

# 5. Test the feature manually
# ... manual testing ...

# 6. Ask Claude Code for code review
"Review this feature for:
- Code quality and maintainability
- Test coverage
- Documentation completeness
- Potential edge cases"

# 7. Provide detailed review
gh pr review <number> --approve -b "Feature works well, tests comprehensive."
```

1.
2.
3.
4.

```
Release workflow completed successfully
New release published: v0.52.1
Artifacts built and uploaded
CHANGELOG updated
Git tag created
```

```
ℹ Release workflow completed - No release created
Reason: No conventional commits found that trigger releases
This is expected behavior for documentation and maintenance commits
```

```
gh auth login
```

```
# Make sure you're in the right repository
gh repo view

# Or specify repository explicitly
gh pr view 254 -R Anaxiatech-AB/crackz
```

```
# Stash local changes first
git stash

# Then checkout PR
gh pr checkout <number>

# Restore changes later
git stash pop
```

Distribution

Distribution & Commercial Delivery

`.github/workflows/docker-build.yml`

`.github/workflows/docker-build.yml`

`uv build --wheel`
`python scripts/build_cuda_wheel.py`
`uv build --sdist`
`uv run task build_exe`
`docker build -f Dockerfile -t crackz-cli:latest .`
`docker build -f Dockerfile.slim -t crackz-cli:slim .`
`docker build -f Dockerfile.gpu -t crackz-cli:gpu .`

```
# Build with CUDA 11.8 (default, most compatible for Windows)
python scripts/build_cuda_wheel.py

# Build with CUDA 12.1
python scripts/build_cuda_wheel.py --cuda-version cu121

# Build with CUDA 12.8 (default for Linux)
python scripts/build_cuda_wheel.py --cuda-version cu128

# Build without cleaning artifacts first
python scripts/build_cuda_wheel.py --no-clean
```

```
# Method 1: Install with specific CUDA index
pip install dist/crackz-*.whl --extra-index-url https://download.pytorch.org/whl/cu118

# Method 2: Install with CUDA extra (uses pyproject.toml config)
pip install dist/crackz-*.whl[cu128]  # For CUDA 12.8 on Linux

# Method 3: Install CPU-only version
pip install dist/crackz-*.whl[cpu]
```

`cu118` `cu121`
`cu128`

`--extra-index-url` `pyproject.toml`
`python -c "import torch; print(f'CUDA available: {torch.cuda.is_available()}')"`

```
# Build both CLI and GUI executables (CPU-only)
uv run task build_exe

# Build only CLI
uv run task build_exe_cli

# Build only GUI
uv run task build_exe_gui

# Clean and rebuild
uv run task build_exe_clean
```

```
# Build with CUDA 11.8 (default, recommended for Windows)
python scripts/build_cuda_exe.py
```

```
# Build with CUDA 12.8 (for newer GPUs/Linux)
python scripts/build_cuda_exe.py --cuda-version cu128

# Build only GUI with CUDA
python scripts/build_cuda_exe.py --gui-only

# Skip CUDA check if already installed
python scripts/build_cuda_exe.py --skip-cuda-check
```

`dist/crackz` `dist/crackz-gui`

```
# Use the build script directly for more control
python scripts/build_exe.py --help
python scripts/build_exe.py --clean  # Clean then build both
python scripts/build_exe.py --no-build  # Only clean, don't build

# CUDA build script
python scripts/build_cuda_exe.py --help
python scripts/build_cuda_exe.py --clean --cuda-version cu128
```

-
- `dist/` `v0.14.0`
- `upload_to_pypi = false`

1.
2.
3.
4.

```
# Install specific version from GitHub release
pip install https://github.com/OWNER/REPO/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl

# Or download then install
wget https://github.com/OWNER/REPO/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl
pip install crackz-0.30.2-py3-none-any.whl
```

```
# Set GitHub token
export GITHUB_TOKEN="ghp_xxxxxxxxxxxx" # gitleaks:allow
```

```
# Download using gh CLI
gh release download v0.30.2 --pattern '*.whl' --repo OWNER/REPO
pip install crackz-*.whl

# Or use curl with authentication
curl -L \
  -H "Authorization: token $GITHUB_TOKEN" \
  -o crackz.whl \
  https://github.com/OWNER/REPO/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl
pip install crackz.whl
```

`PRIVATE_INDEX_URL`
`PRIVATE_INDEX_USER` `__token__` `PRIVATE_INDEX_PASSWORD`

```
pip install --index-url https://pypi.company.com/simple/ crackz
```

`.github/workflows/release.yml`

```
name: release
on:
  push:
    tags:
      - 'v*.*.*'
jobs:
  build-release:
    runs-on: ubuntu-latest
    permissions:
      contents: write   # allow releasing
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-python@v5
        with:
          python-version: '3.13'
      - name: Install uv
        run: pip install uv
      - name: Sync deps (locked)
        run: uv sync --frozen
      - name: Build wheel + sdist
        run: uv build --wheel --sdist
      - name: Build documentation PDF
        run: |
          uv run mkdocs build --clean --strict
          uv run playwright install chromium
          uv run python scripts/export_pdf.py --outline \
            --cover-image 'assets/crackz-logo.png' \
            --logo 'assets/crackz-logo.png' \
            --out site/crackz-docs.pdf
          cp site/crackz-docs.pdf dist/
      - name: Upload to private index (twine)
        if: env.PRIVATE_INDEX_URL != ''
        env:
          TWINE_USERNAME: ${{ secrets.PRIVATE_INDEX_USER }}
          TWINE_PASSWORD: ${{ secrets.PRIVATE_INDEX_PASSWORD }}
        run: |
          pip install twine
          twine upload --repository-url ${{ secrets.PRIVATE_INDEX_URL }} dist/*.whl dist/*.tar.gz
      - name: Attach artifacts to Release
        uses: softprops/action-gh-release@v2
        with:
          files: dist/*
```

`publish-release.yml`

`crackz-gui`

`v0.35.0` `latest`

`PRIVATE_INDEX_URL`
`https://pypi.company.com/simple/` `PRIVATE_INDEX_USER` `__token__`

`PRIVATE_INDEX_PASSWORD`

`.github/workflows/docker-build.yml`

`.github/workflows/docker-build.yml`

```
docker login ghcr.io -u USER -p TOKEN
# Build variants
docker build -f Dockerfile       -t ghcr.io/anaxiatech-ab/crackz:latest .
docker build -f Dockerfile.slim  -t ghcr.io/anaxiatech-ab/crackz:slim   .
docker build -f Dockerfile.gpu   -t ghcr.io/anaxiatech-ab/crackz:gpu    .
# Push
for tag in latest slim gpu; do docker push ghcr.io/anaxiatech-ab/crackz:$tag; done
```

```
docker pull ghcr.io/anaxiatech-ab/crackz:slim
```

	`pip install --index-url https://pypi.company.com/simple/ crackz`
	`pip install https://github.com/OWNER/REPO/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl`
	`pip install torch torchvision --index-url <TORCH_URL> && pip install <crackz-wheel-url>`
	`pip install --no-index --find-links wheelhouse crackz-VERSION-py3-none-any.whl`
	`pip install --require-hashes -r requirements.txt`

1.

```
# Download Crackz wheel from GitHub Release first
wget https://github.com/OWNER/REPO/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl

# Create wheelhouse directory and download all dependencies
mkdir wheelhouse
pip download crackz-VERSION-py3-none-any.whl -d wheelhouse
```

2.

```
(cd wheelhouse && shasum -a 256 * > SHA256SUMS.txt)
```

3.
4.

```
pip install --no-index --find-links wheelhouse crackz-VERSION-py3-none-any.whl
```

- `.asc`
- `pip install pip-audit` `cyclonedx-bom`
-

`~/.config/pip/pip.conf`

```
[global]
extra-index-url = https://TOKEN@pkg.company.com/simple
```

`%APPDATA%/pip/pip.ini`

-
- `escrow-*`

-
-
-

```
# Build wheel
uv build --wheel
# Build + attach release (manual example)
git tag v0.14.0 && git push origin v0.14.0
# Install from private index
pip install --index-url https://pypi.company.com/simple/ crackz
# Install from GitHub Release
pip install https://github.com/OWNER/REPO/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl
# Offline install
pip install --no-index --find-links wheelhouse crackz-VERSION-py3-none-any.whl
```

Customer Guide

Customer Guide: Crackz for Infrastructure Management

What is a Mask?

```
 Original Image          Mask Image           Result
[photo of concrete] → [white lines on    = Trained model learns
                       black background]    to detect cracks
```

When Do You Need Masks?

Recommended Tools for Creating Masks

1. CVAT (Computer Vision Annotation Tool) - Best for Teams

```
# Self-hosted option (Docker)
docker run -d -p 8080:8080 cvat/server
# Or use cloud-hosted version at app.cvat.ai
```

2. Labelme - Best for Individual Users

```
pip install labelme
labelme
```

`labelme_json_to_dataset <json_file>` `label.png`

3. Photoshop / GIMP - Best for Fine Control

`masks/train/`
`masks/val/` `masks/test/`

4. Python Scripting - Best for Automation

```
# Example: Convert polygon annotations to masks
from PIL import Image, ImageDraw
import json

def polygon_to_mask(image_path, polygon_coords, output_path):
    # Load original image to get dimensions
    img = Image.open(image_path)
    width, height = img.size

    # Create black background
    mask = Image.new('L', (width, height), 0)
    draw = ImageDraw.Draw(mask)

    # Draw white polygon for crack
    draw.polygon(polygon_coords, fill=255)

    # Save mask
    mask.save(output_path)

# Usage
polygon_coords = [(100, 200), (150, 250), (200, 230), ...]
polygon_to_mask('image_01.jpg', polygon_coords, 'masks/train/image_01.png')
```

Best Practices for Mask Creation

Quality Guidelines

-
-
-
-
- `image_01.jpg` `image_01.png`

File Naming Convention

```
✓ CORRECT:
images/train/harbor_crack_001.jpg
masks/train/harbor_crack_001.png

✗ INCORRECT:
```

```
images/train/harbor_crack_001.jpg
masks/train/harbor_crack_001_mask.png  ← Extra suffix
masks/train/HarborCrack001.png         ← Different name
```

Organizational Tips

1.
2.
3.
4.
5.

Train/Val/Test Split

```
masks/
  train/    # 70% of labeled data (200-350 images)
  val/      # 15% of labeled data (30-50 images)
  test/     # 15% of labeled data (30-50 images)
```

`images/`

```
images/
  train/    # Same filenames as masks/train/
  val/      # Same filenames as masks/val/
  test/     # Same filenames as masks/test/
```

Using Crackz with Your Masks

`images/train/`

```
# Automatic mask detection (if masks are in source_folder/masks/)
crackz import-images \
  --project my-project \
  --src ./labeled-data \
  --type train \
  --size 512 512

# Explicit mask source
crackz import-images \
  --project my-project \
  --src ./images \
  --masks_src ./masks \
  --type train \
  --size 512 512
```

```
crackz import-images \
  --project my-project \
  --src ./labeled-data \
  --masks_src ./labeled-data/masks \
  --split \
  --train-ratio 0.7 \
  --val-ratio 0.15 \
  --test-ratio 0.15 \
  --size 512 512 \
  --random-seed 42
```

Troubleshooting Mask Issues

`masks/train/` `masks/`

Advanced: Semi-Automated Mask Generation

Time Estimates

Professional Labeling Services

```
inspection-campaign-2025/
  images/          # Organized by train/val/test/raw splits
  masks/           # Ground truth for training (if available)
  artifacts/       # Detection results, metrics, checkpoints
  logs/            # Complete audit trail
  project.yaml     # Configuration snapshot
```

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Customer Access

1.
2.
3.
4.

1.
2. `templates/welcome-email.md`
3. `templates/access-revocation-email.md`
4. `templates/renewal-reminder-email.md`
5.
6.
7.
8.

1.
2.
3.

`docs/customer-access.md`

Customer Access Management

1.
2.
3. `pip install crackz`

`pip install crackz`

1.

```
# Via GitHub web UI:
# Settings > Teams > New Team
# Team name: "Customer-Acme-Corp"
# Privacy: Secret
# Repository access: crackz (Read)
```

2.

3.
4.
5.

```
Subject: Crackz Access - Welcome to [Customer Name]

Hi [Contact Name],

Thank you for your Crackz license purchase!

To access Crackz packages and Docker images, please:

1. Create a GitHub account (if you don't have one): https://github.com/signup
2. Reply with your GitHub username
3. We'll add you to the private repository

Once added, you can access:
- Releases: https://github.com/Anaxiatech-AB/crackz/releases
- Docker images: https://github.com/anaxiatech?tab=packages&repo_name=crackz
- Documentation: https://github.com/Anaxiatech-AB/crackz/tree/main/docs

License: [LICENSE-KEY or CONTRACT-REF]
Now the `release.yml` workflow will automatically publish to Azure Artifacts when configured!

Best regards,
Theresia
```

```
Subject: Crackz Access Confirmed

Hi [Contact Name],

You now have access to the Crackz repository!

Installation instructions:
https://github.com/Anaxiatech-AB/crackz/blob/main/docs/installation.md

Getting started:
https://github.com/Anaxiatech-AB/crackz/blob/main/docs/getting-started.md

Docker registry:
ghcr.io/anaxiatech-ab/crackz:latest

Need help? Email theresia.sofia.snow@gmail.com

Best regards,
Theresia
```

```
# Find latest version at: https://github.com/Anaxiatech-AB/crackz/releases
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl
```

```
# Download from GitHub Releases page
wget https://github.com/Anaxiatech-AB/crackz/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl

# Install locally
pip install crackz-0.30.2-py3-none-any.whl
```

```
# Install GitHub CLI: https://cli.github.com/
gh auth login

# Download latest release
gh release download --repo anaxiatech/crackz --pattern '*.whl'

# Install
pip install crackz-*.whl
```

```
# Generate GitHub Personal Access Token

1. Go to: https://github.com/settings/tokens/new
```

```
2. Note: "Crackz Docker Access"
3. Expiration: 1 year (or custom)
4. Scopes: Select ONLY:
   - □ read:packages
5. Click "Generate token"
6. Copy the token (starts with ghp_)
```

```
# Customer runs this once
export CR_PAT=ghp_xxxxxxxxxxxxxxxxxxxx  # Their PAT # gitleaks:allow
echo $CR_PAT | docker login ghcr.io -u GITHUB_USERNAME --password-stdin

# Now they can pull
docker pull ghcr.io/anaxiatech-ab/crackz:latest
docker pull ghcr.io/anaxiatech-ab/crackz:slim
docker pull ghcr.io/anaxiatech-ab/crackz:gpu
```

```
# Use as normal
docker run --rm ghcr.io/anaxiatech-ab/crackz:latest --version
```

1.
2.
3.
4.

   ```
   # Via Azure DevOps UI:
   # Artifacts > Create Feed
   # Name: crackz-packages
   # Visibility: Private (organization)
   # Upstream sources: Enable PyPI (optional)
   ```

5.
6.
7.
   -
   -
8.

   ```
   https://pkgs.dev.azure.com/YOUR-ORG/_packaging/crackz-packages/pypi/simple/
   ```

```
# Settings > Secrets and variables > Actions > New repository secret

PRIVATE_INDEX_URL: https://pkgs.dev.azure.com/YOUR-ORG/_packaging/crackz-packages/pypi/upload
PRIVATE_INDEX_USER: azure
PRIVATE_INDEX_PASSWORD: <Azure-PAT-with-Packaging-Write-scope> # gitleaks:allow
```

`publish-release.yml`

```
# Option 1: Install with explicit index
pip install crackz \
  --index-url https://pkgs.dev.azure.com/YOUR-ORG/_packaging/crackz-packages/pypi/simple/

# Option 2: Configure pip globally
# Linux/macOS: ~/.config/pip/pip.conf
# Windows: %APPDATA%\pip\pip.ini
```

```
[global]
index-url = https://YOUR-PAT@pkgs.dev.azure.com/YOUR-ORG/_packaging/crackz-packages/pypi/simple/
```

```
pip install crackz
crackz --version
```

```
# Install devpi
pip install devpi-server devpi-web

# Initialize server
devpi-init

# Start server
devpi-server --start --host=0.0.0.0 --port=3141

# Create user for yourself
devpi use http://localhost:3141
devpi user -c admin password=SECRET # gitleaks:allow

# Create index
devpi index -c crackz bases=root/pypi
devpi use crackz
```

```
# Install devpi client
pip install devpi-client

# Login
devpi use http://YOUR-SERVER:3141
devpi login admin --password=SECRET # gitleaks:allow
devpi use admin/crackz

# Upload wheel
devpi upload dist/crackz-*.whl
```

```
# Customer configures pip
pip install --index-url http://YOUR-SERVER:3141/admin/crackz/+simple/ crackz
```

`docker run -p 3141:3141 muccg/devpi`

1.

2.
3.
4.
5.
6.
7.
8.
9.

```
Subject: Crackz License Expired - Access Revoked

Hi [Contact],

Your Crackz license expired on [DATE].
Repository and package access has been revoked.

To renew, please contact theresia.sofia.snow@gmail.com

Thank you for using Crackz.
```

1.
2.
3.

1.
2.
3.
4.

```
# Ask customer to verify:
1. Go to: https://github.com/Anaxiatech-AB/crackz
2. If you see "404 Not Found", you don't have access yet
3. Reply with your GitHub username for access
```

`read:packages`

```
# Ask customer to:
1. Generate new PAT: https://github.com/settings/tokens/new
2. Select ONLY: read:packages
3. Run: echo PAT | docker login ghcr.io -u USERNAME --password-stdin
4. Try again: docker pull ghcr.io/anaxiatech-ab/crackz:latest
```

```
# Recommend:
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl
```

```
# You prepare:
pip download crackz -d wheelhouse
tar -czf crackz-offline-bundle.tar.gz wheelhouse/

# Customer installs:
tar -xzf crackz-offline-bundle.tar.gz
pip install --no-index --find-links wheelhouse crackz
```

1.

   ```
   # Customer pulls and retags
   docker pull ghcr.io/anaxiatech-ab/crackz:latest
   docker tag ghcr.io/anaxiatech-ab/crackz:latest registry.company.com/crackz:latest
   docker push registry.company.com/crackz:latest
   ```

2.

```
# Get release download counts
gh api repos/anaxiatech/crackz/releases \
  --jq '.[] | {name: .name, downloads: [.assets[].download_count] | add}'
```

```
{"name":"v0.30.2","downloads":47}
{"name":"v0.30.1","downloads":23}
```

```
# Requires GitHub GraphQL API (complex)
# Easier to use web UI for now
```

```
# Crackz Usage Report - October 2025

## Active Customers: 5

| Customer | License | Status | Last Access |
|----------|---------|--------|-------------|
| Acme Corp | LIC-001 | Active | 2025-10-08 |
| Beta Inc | LIC-002 | Active | 2025-10-05 |
| Gamma LLC | LIC-003 | Expired | 2025-09-30 |

## Download Statistics

- Total releases: 12
- Total downloads (wheels): 156
- Docker pulls (latest): 89
- Docker pulls (gpu): 34

## Top Versions
1. v0.30.2: 47 downloads
2. v0.30.1: 23 downloads
```

```
3. v0.30.0: 18 downloads

## Action Items
- [ ] Contact Gamma LLC for renewal
- [ ] Send v0.31.0 release announcement
```

1.
2. `PRIVATE_INDEX_*`
3.
4.
5.
6.
7.
8.
9.
10.
11.
12.

1.
2.
3. `read:packages`
4.
5.
6.
7.
8.
9.
10.
11.
12.

```
#!/bin/bash
# scripts/customer-access/onboard-customer.sh

set -euo pipefail

CUSTOMER_NAME="$1"
GITHUB_USERNAME="$2"
LICENSE_KEY="$3"

echo "□ Onboarding customer: $CUSTOMER_NAME"
echo "GitHub username: $GITHUB_USERNAME"
echo "License: $LICENSE_KEY"

# Create GitHub team (requires gh CLI with admin access)
TEAM_SLUG="customer-$(echo $CUSTOMER_NAME | tr '[:upper:]' '[:lower:]' | tr ' ' '-')"
gh api orgs/YOUR-ORG/teams -f name="Customer-$CUSTOMER_NAME" -f privacy=secret

# Add member to team
gh api orgs/YOUR-ORG/teams/$TEAM_SLUG/memberships/$GITHUB_USERNAME -X PUT

# Add team to repository
gh api orgs/YOUR-ORG/teams/$TEAM_SLUG/repos/YOUR-ORG/crackz -X PUT -f permission=pull

echo "□ Customer onboarded successfully!"
echo ""
echo "Next steps:"
echo "1. Send welcome email to customer"
echo "2. Add to tracking spreadsheet"
echo "3. Schedule renewal reminder for 1 year from now"
```

```
#!/bin/bash
# scripts/customer-access/revoke-customer.sh

set -euo pipefail

CUSTOMER_NAME="$1"
REASON="${2:-License expired}"

echo "□ Revoking access for: $CUSTOMER_NAME"
echo "Reason: $REASON"

TEAM_SLUG="customer-$(echo $CUSTOMER_NAME | tr '[:upper:]' '[:lower:]' | tr ' ' '-')"

# Remove team from repository
gh api orgs/YOUR-ORG/teams/$TEAM_SLUG/repos/YOUR-ORG/crackz -X DELETE

# Optionally delete team entirely
read -p "Delete team entirely? (y/n) " -n 1 -r
echo
if [[ $REPLY =~ ^[Yy]$ ]]; then
    gh api orgs/YOUR-ORG/teams/$TEAM_SLUG -X DELETE
    echo "□ Team deleted"
else
    echo "ℹ Team kept (access removed from repository only)"
fi

echo "□ Access revoked successfully!"
echo ""
echo "Next steps:"
echo "1. Send termination notice to customer"
echo "2. Update tracking spreadsheet"
echo "3. Archive customer records"
```

`PRIVATE_INDEX_*`

`pip install crackz`

License

License


```
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```